

CHAPTER 20

ELECTROMAGNETISM

4.5 Electromagnetic effects

4.5.1 Electromagnetic induction

- 1 Describe an experiment to demonstrate electromagnetic induction
- 2 State that the magnitude of an induced e.m.f. is affected by:
 - (a) the rate of change of the magnetic field or the rate of cutting of magnetic field lines
 - (b) the number of turns in a coil
- 3 State and use the fact that the effect of the current produced by an induced e.m.f. is to oppose the change producing it (Lenz's law) and describe how this law may be demonstrated

4.5.2 The a.c. generator

- 1 Describe a simple form of a.c. generator (rotating coil or rotating magnet) and the use of slip rings and brushes where needed
- 2 Sketch and interpret graphs of e.m.f. against time for simple a.c. generators and relate the position of the generator coil to the peaks, troughs and zeros of the e.m.f.

4.5.3 Magnetic effect of a current

- 1 Describe the pattern and direction of the magnetic field due to currents in straight wires and in solenoids and state the effect on the magnetic field of changing the magnitude and direction of the current
- 2 Describe how the magnetic effect of a current is used in relays and loudspeakers and give examples of their application

4.5 Electromagnetic effects continued

4.5.4 Forces on a current-carrying conductor

- 1 Describe an experiment to show that a force acts on a current-carrying conductor in a magnetic field, including the effect of reversing:
 - (a) the current
 - (b) the direction of the field
- 2 Recall and use the relative directions of force, magnetic field and current
- 3 Describe the magnetic field patterns between currents in parallel conductors and relate these to the forces on the conductors (excluding the Earth's field)

4.5.5 The d.c. motor

- 1 Know that a current-carrying coil in a magnetic field may experience a turning effect and that the turning effect is increased by increasing:
 - (a) the number of turns on the coil
 - (b) the current
 - (c) the strength of the magnetic field
- 2 Describe the operation of an electric motor, including the action of a split-ring commutator and brushes

4.5.6 The transformer

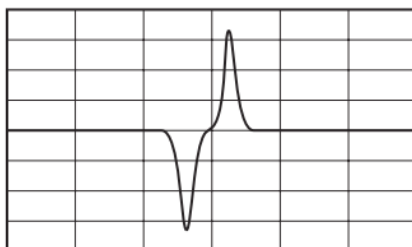
- 1 Describe the structure and explain the principle of operation of a simple iron-cored transformer
- 2 Use the terms primary, secondary, step-up and step-down
- 3 Recall and use the equation
$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$
where P and S refer to primary and secondary
- 4 State the advantages of high-voltage transmission and explain why power losses in cables are smaller when the voltage is greater

4.6 Uses of an oscilloscope

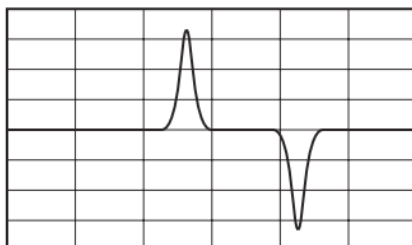
- 1 Describe the use of an oscilloscope to display waveforms (the structure of an oscilloscope is **not** required)
- 2 Describe how to measure p.d. and short intervals of time with an oscilloscope using the Y-gain and timebase

1. O/N/23/11/Q29

A magnet moves at constant speed through a coil of wire. The coil is connected to an oscilloscope. The diagram shows the trace produced on the screen.



The experiment is changed and a new trace is produced.

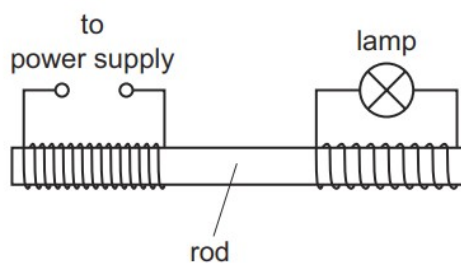


Which statement explains why the new trace is different from the original trace?

- A** The coil is longer and the other pole of the magnet enters the coil first.
- B** The coil is longer and the same pole of the magnet enters the coil first.
- C** The coil is shorter and the other pole of the magnet enters the coil first.
- D** The coil is shorter and the same pole of the magnet enters the coil first.

2. O/N/23/11/Q34

A girl makes a simple transformer to illuminate a low voltage lamp. The transformer is shown.

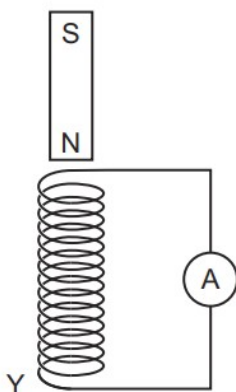


What is the material of the rod and what is the type of power supply?

	material of rod	power supply
A	plastic	d.c. supply
B	plastic	a.c. supply
C	iron	d.c. supply
D	iron	a.c. supply

3. O/N/23/11/Q30

A bar magnet is dropped through a vertical solenoid that is connected to a sensitive ammeter.



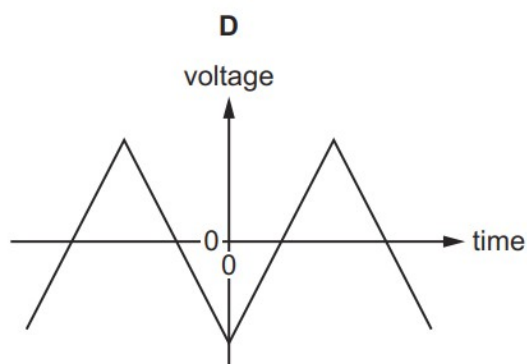
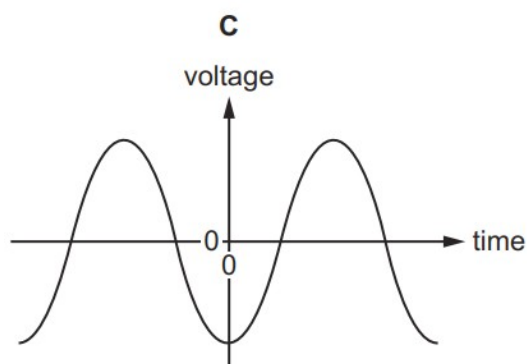
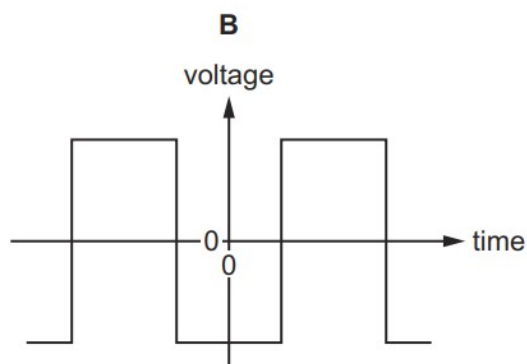
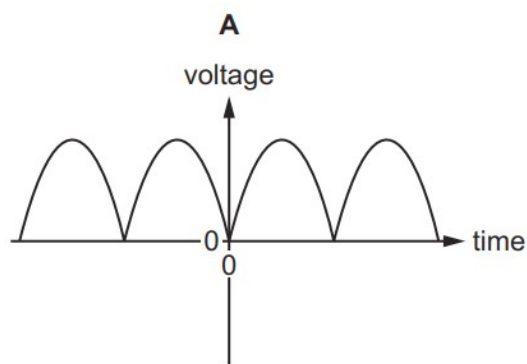
A current is induced in the solenoid and the solenoid becomes an electromagnet.

Which row describes the polarity of end Y of the solenoid and the force between the solenoid and the bar magnet when the bar magnet leaves the solenoid?

	polarity of end Y of the solenoid	force between solenoid and bar magnet
A	north	attraction
B	north	repulsion
C	south	attraction
D	south	repulsion

4. O/N/23/11/Q31

Which graph shows the voltage output against time for a simple a.c. generator?



5. O/N/23/11/Q32

A charger for a mobile phone (cellphone) uses a transformer which steps the voltage down from 230 V to 5.0 V.

The primary coil in the transformer has 2500 turns.

How many turns are on the secondary coil?

A 11

B 54

C 500

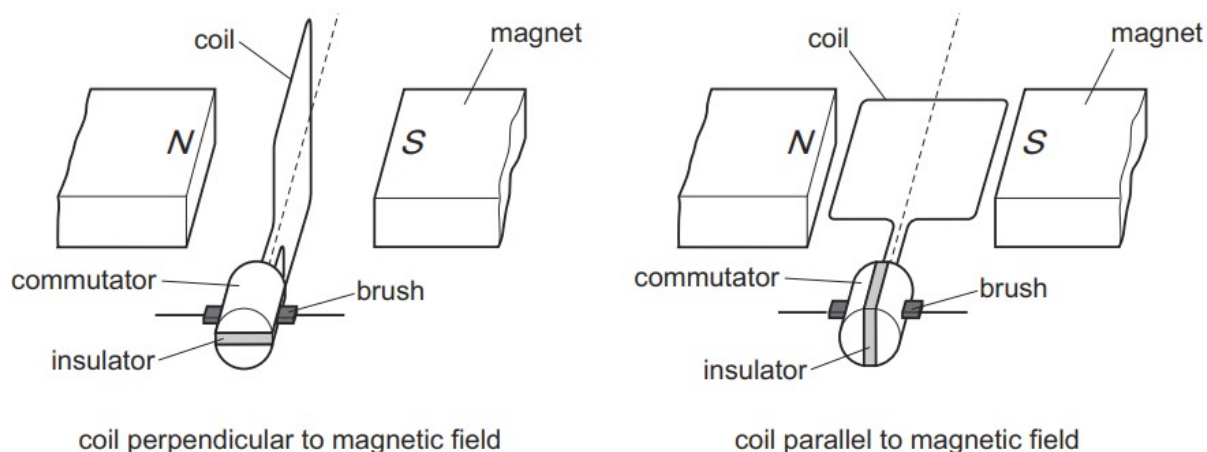
D 11 500

6. O/N/23/11/Q33

A commutator is connected to the coil of a d.c. motor.

The commutator rotates and connects the coil to the power supply.

Once every half-turn, the current in the coil reverses.



What is the name of the commutator and what is the orientation of the plane of the coil as the current reverses?

	name	orientation
A	slip ring	parallel to the magnetic field
B	slip ring	perpendicular to the magnetic field
C	split ring	parallel to the magnetic field
D	split ring	perpendicular to the magnetic field

7. O/N/23/12/Q35

The output of an a.c. generator is connected to a lamp.

The coil of the generator is rotated at a constant rate which makes the lamp flash twice every second.

The coil is then rotated at half the rate of rotation.

What happens to the maximum brightness of the lamp and how many times does the lamp flash every second?

	brightness	number of flashes every second
A	decreases	1
B	decreases	4
C	does not change	1
D	does not change	4

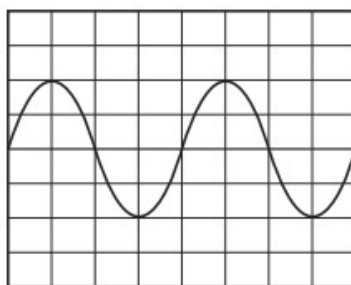
8. O/N/23/12/Q36

Which device uses the magnetic effect of an electric current?

- A** compass
- B** fuse
- C** loudspeaker
- D** magnifying glass

9. O/N/23/12/Q38

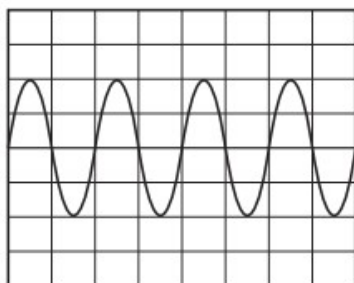
The diagram shows the trace produced on the screen of an oscilloscope due to an alternating voltage input.



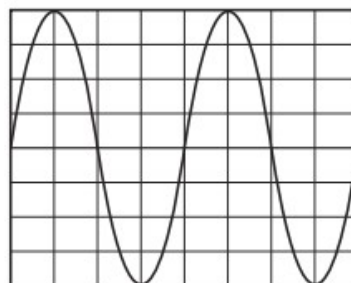
The time-base on the oscilloscope is set to 4 ms/cm.

What does the trace look like when the time base is set to 2 ms/cm?

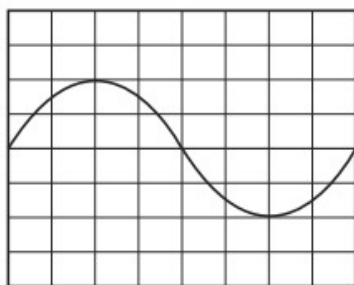
A



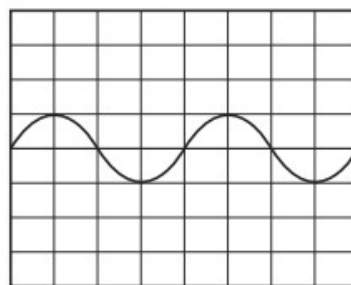
B



C

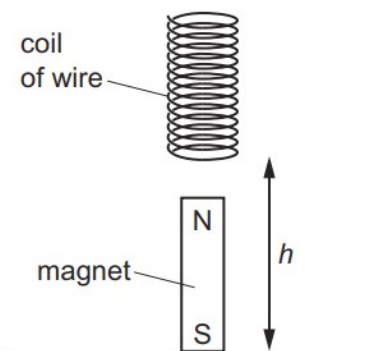


D



10. M/J/23/12/Q32

A coil of wire above a magnet is dropped. As the coil falls over the magnet, an e.m.f. is induced across the coil.

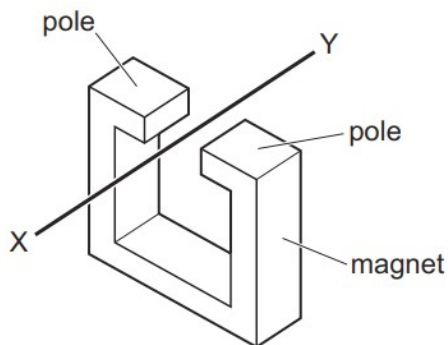


What does **not** affect the magnitude of the e.m.f. induced across the coil?

- A changing the number of turns of wire in the coil
- B changing the height h from which the coil is dropped
- C changing the direction of the magnet, so the north pole is at the bottom
- D changing the strength of the magnetic field from the magnet

11. M/J/23/11/Q32

When wire XY is moved downwards between the poles of a stationary magnet, an e.m.f. is produced across X and Y.

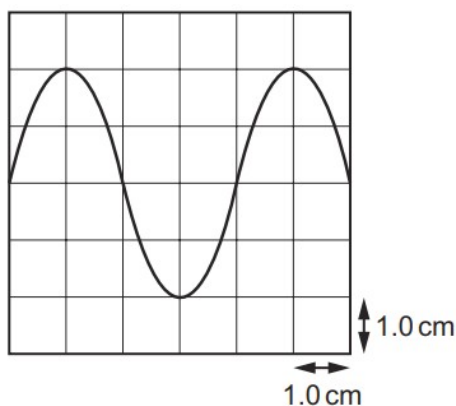


Which action produces an e.m.f. across X and Y in the opposite direction?

- A Move both the wire and the magnet upwards at the same speed.
- B The wire is kept stationary and the magnet is moved upwards.
- C The wire is moved downwards and the magnet is moved upwards.
- D The wire is moved upwards and the magnet is kept stationary.

12. M/J/23/11/Q33

An a.c. voltage is displayed on an oscilloscope screen. The Y-gain is set at 2.0 V/cm .



What is the maximum value of the voltage?

- A** 2.0 V **B** 4.0 V **C** 8.0 V **D** 12 V

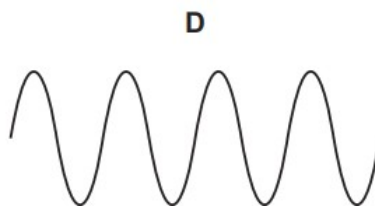
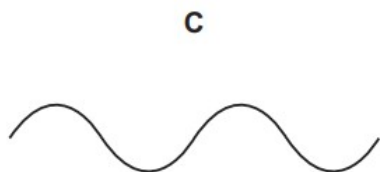
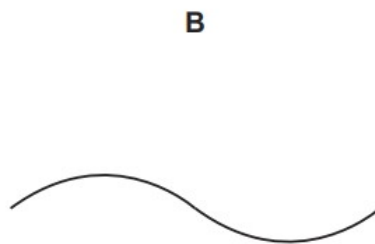
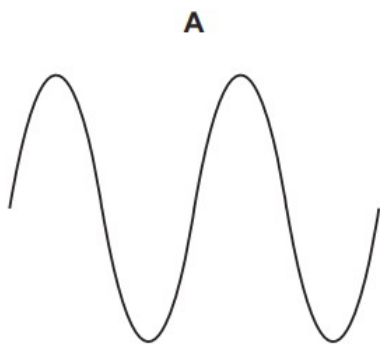
13. O/N/22/11/Q34

The diagram shows the output from an a.c. generator displayed on an oscilloscope screen.



The generator is turned at a slower rate. The settings on the oscilloscope controls are not changed.

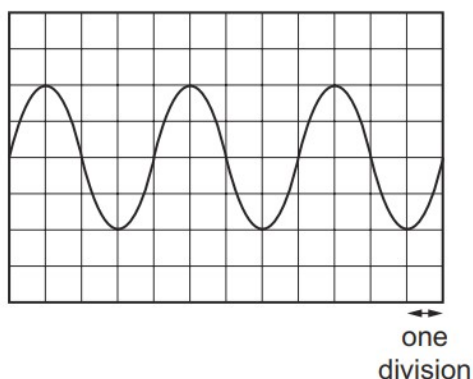
Which diagram shows the new output?



14. O/N/22/11/Q35

The sound wave produced by a note is displayed on the screen of a cathode-ray oscilloscope.

The time-base is set at 5 ms for one division.



What is the frequency of the note?

- A** 20 Hz **B** 50 Hz **C** 60 Hz **D** 150 Hz

15. O/N/22/11/Q33

Which statement about a transformer is correct?

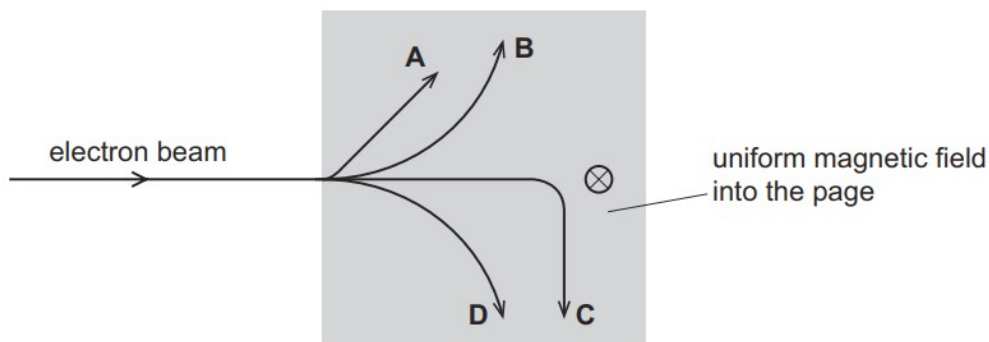
- A** The changing magnetic field in the transformer induces an e.m.f. in the secondary coil.
B The core of the transformer is made of iron because iron is a good electrical conductor.
C The transformer converts alternating current to direct current.
D The transformer converts direct current to alternating current.

16. O/N/22/11/Q37

A beam of electrons is fired into a uniform magnetic field as shown.

The direction of the magnetic field is into the page.

Which path do the electrons follow?

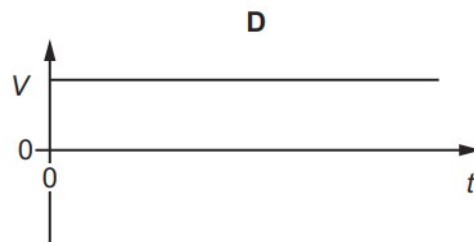
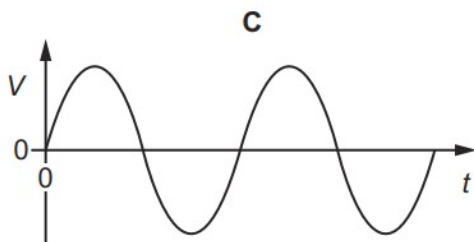
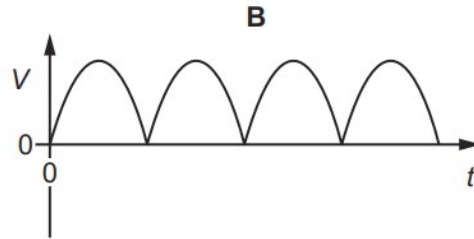
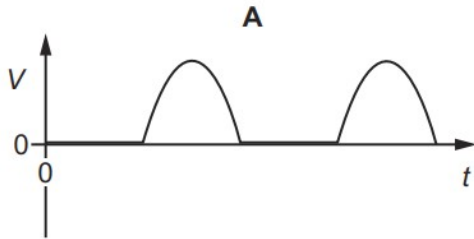


17. O/N/22/12/Q35

A coil of wire is rotated at a constant rate between the poles of a U-shaped magnet.

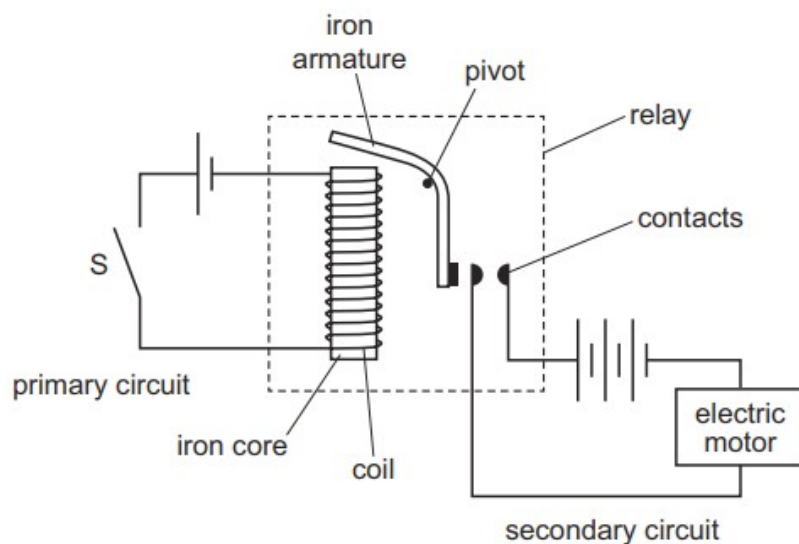
The two ends of the coil are connected to different slip rings.

Which graph shows how the voltage V between the slip rings varies with time t ?



18. O/N/22/12/Q38

The diagram shows a relay used to switch on an electric motor.



A student makes five statements to explain how the relay switches on the electric motor. The statements are **not** in the correct order.

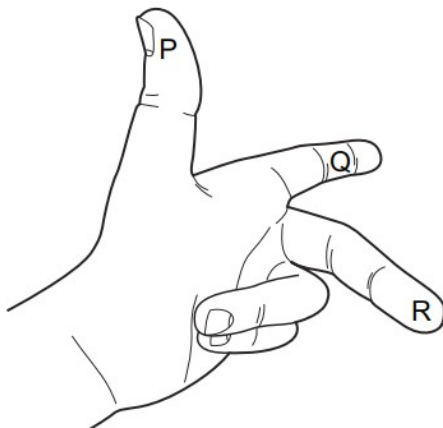
- 1 The current in the coil magnetises the electromagnet.
- 2 The armature closes the contacts.
- 3 The current in the secondary circuit makes the motor turn.
- 4 The electromagnet attracts the iron armature.
- 5 The switch S in the primary circuit is closed.

What is the correct order of the statements?

- A** 1 → 5 → 2 → 4 → 3
- B** 3 → 2 → 4 → 1 → 5
- C** 3 → 4 → 1 → 5 → 2
- D** 5 → 1 → 4 → 2 → 3

19. M/J/22/12/Q38

A left hand can be used to determine the direction of the force when a current-carrying conductor is perpendicular to a magnetic field.

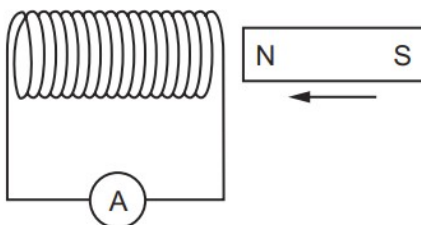


Which quantities are represented by the direction of fingers P, Q and R?

	P	Q	R
A	current	field	force
B	field	force	current
C	force	current	field
D	force	field	current

20. M/J/22/12/Q39

As a magnet is moved into the coil of wire as shown, there is a small positive reading on the sensitive ammeter.



Which change must increase the size of the reading?

- A** moving the opposite pole into the coil
- B** pulling the magnet out of the coil
- C** pushing the magnet in faster
- D** unwinding some of the turns of wire

21. M/J/22/12/Q40

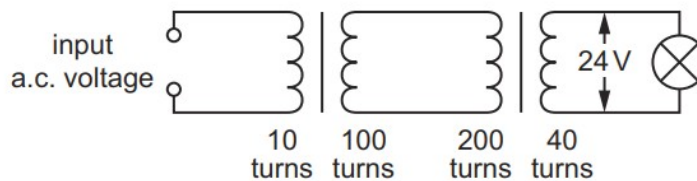
Electrical power is transmitted by cables over long distances at very high voltages.

What are the effects of using a high voltage transmission system?

	power loss in the cables	current in the cables
A	high	high
B	high	low
C	low	high
D	low	low

22. M/J/22/11/Q37

The diagram shows the circuit for a model power line with two transformers and a lamp at the output.



What is the input voltage?

- A** 0.12 V **B** 6.0 V **C** 12 V **D** 96 V

23. M/J/22/11/Q39

The diagram shows a beam of electrons about to enter a magnetic field. The magnetic field is directed into the page.



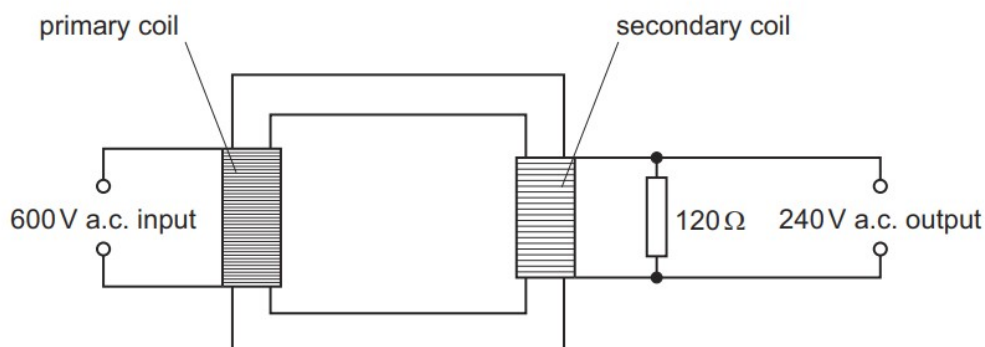
What is the direction of the deflection of the electrons as they enter the magnetic field?

- A** into the page
B out of the page
C up the page
D down the page

24. O/N/21/11/Q34

A transformer with an efficiency of 100% has a primary voltage input of 600 V and a secondary voltage output of 240 V.

The secondary coil is attached to a resistor of resistance $120\ \Omega$.

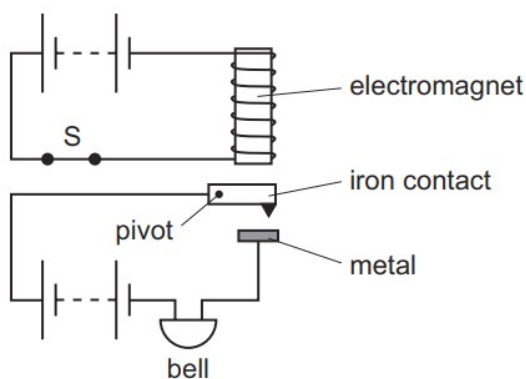


What is the power dissipated in the resistor and what is the current in the primary coil?

	power / W	current / A
A	120	0.20
B	120	5.0
C	480	0.80
D	480	1.3

25. O/N/21/11/Q35

The diagram shows an alarm system in which the switch S is shown closed. The top circuit is arranged so that the electromagnet is positioned over the soft iron contact.

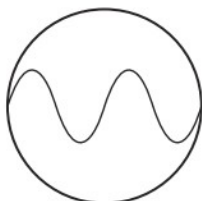


What happens when the switch S is opened?

	iron contact	bell
A	drops	rings
B	drops	stops ringing
C	moves up	rings
D	moves up	stops ringing

26. O/N/21/11/Q36

A microphone detects a musical sound and the signal is fed into an oscilloscope. The diagram shows the trace which appears on the screen.



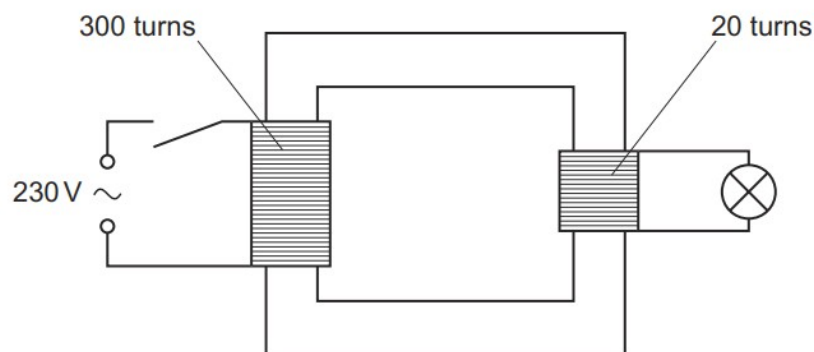
The spot of light on the oscilloscope screen takes 1.2 ms to travel across the screen.

What is the frequency of the musical sound?

- A** 170 Hz **B** 330 Hz **C** 1700 Hz **D** 3300 Hz

27. M/J/21/12/Q38

A student uses a transformer to light a filament lamp using a 230 V a.c. supply. The lamp has a maximum voltage rating of 6.0 V.

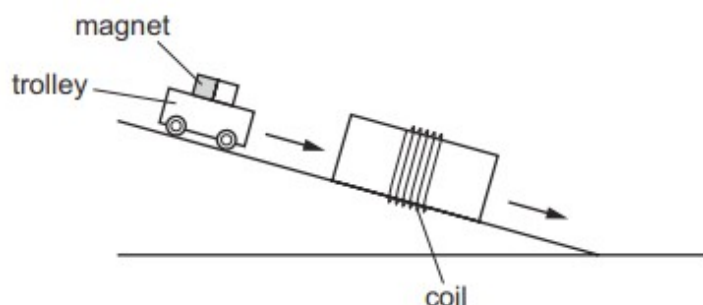


What happens when the circuit is switched on?

- A** The lamp does not light at all.
B The lamp lights at normal brightness.
C The lamp lights dimly.
D The lamp lights up brightly and then goes out.

28. M/J/21/12/Q37

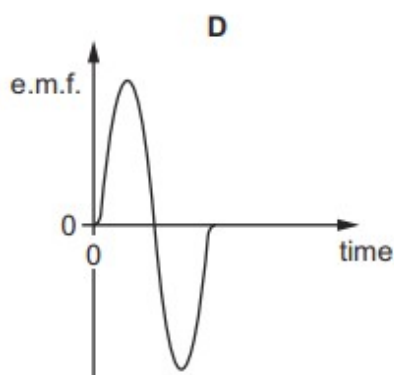
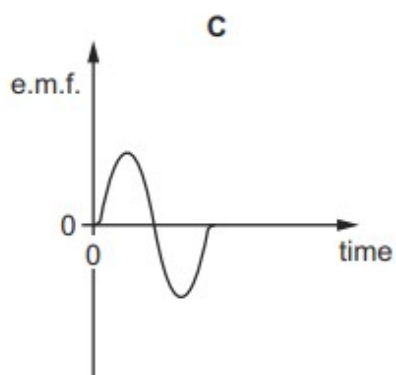
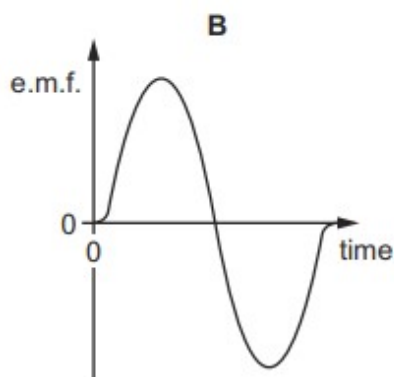
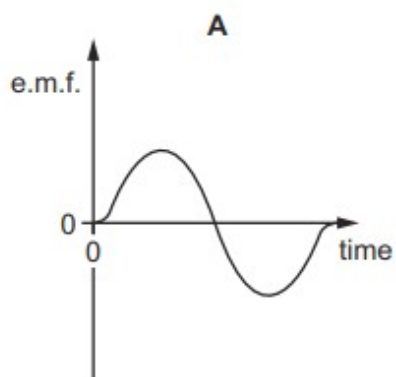
A trolley carrying a strong magnet rolls down a ramp at constant speed. It passes through a coil as shown.



An electromotive force (e.m.f.) is induced in the coil. A graph of the e.m.f. against time is plotted.

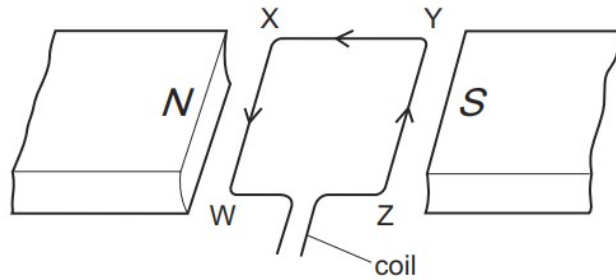
The experiment is repeated with different coils and with a steeper ramp. The trolley moves at a greater constant speed on the steeper ramp.

Which graph is produced using the coil with the least number of turns and the steepest ramp? All graphs are drawn to the same scale.



29. M/J/21/11/Q32

The diagram shows a horizontal rectangular wire coil WXYZ between the poles of a magnet.



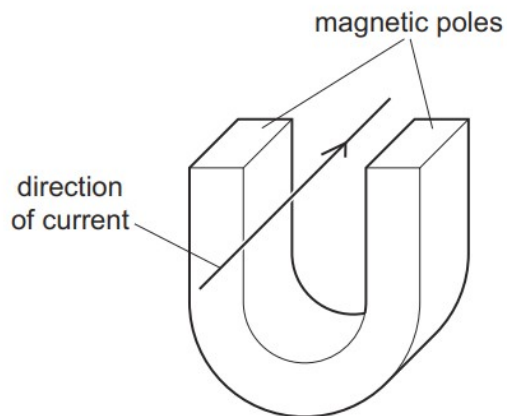
There is a current in the coil in the direction shown.

Which statement is correct?

- A** The side WX experiences an upward force.
- B** The side XY experiences an outward force.
- C** The side YZ experiences an inward force.
- D** The side ZW experiences a downward force.

30. M/J/21/11/Q34

The diagram shows a current-carrying conductor between the poles of a magnet. The force on the wire acts downwards.



Four changes are possible.

- 1 The current is increased.
- 2 A stronger magnet is used.
- 3 The current is reversed.
- 4 The poles exchange positions.

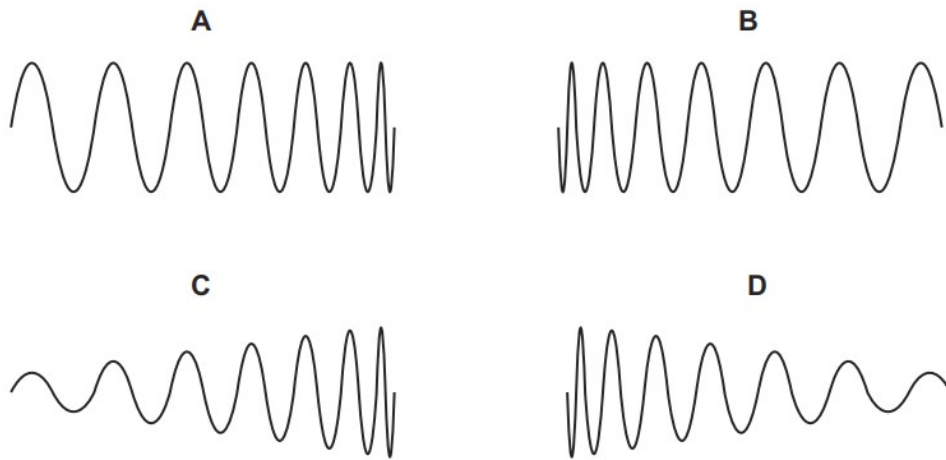
Which two changes made together keep the force acting downwards?

- A** 1 and 3 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

31. M/J/21/11/Q35

In an alternating current (a.c.) generator, a magnet rotates near a coil of wire. The induced electromotive force (e.m.f.) in the coil is displayed on an oscilloscope screen.

Which trace is produced as the magnet slows down?



32. O/N/20/12/Q34

What is used with a magnet to create an induced electromotive force (e.m.f.) in a simple a.c. generator?

- A** a battery
- B** a coil of wire
- C** a voltmeter
- D** a relay

33. O/N/20/12/Q35

What changes kinetic (movement) energy into electrical energy?

- A** a car battery
- B** an electric fire
- C** an electric motor
- D** a wind turbine

34. O/N/20/12/Q36

A transformer has 4800 turns on its primary coil and 480 turns on its secondary coil.

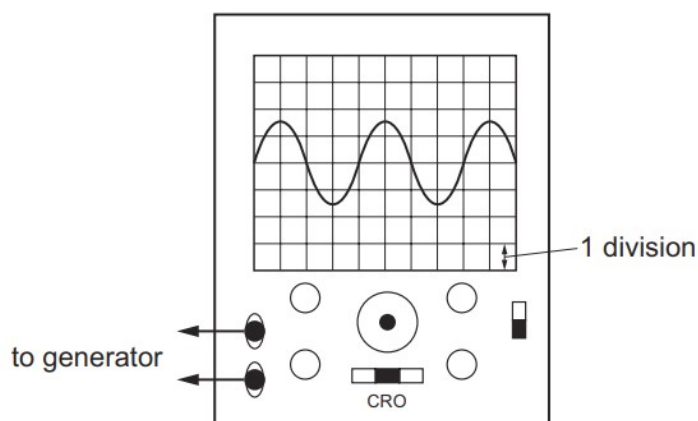
The primary coil is connected to a 240 V a.c. supply. The secondary coil is connected to a lamp.

How does the output current in the lamp compare with the input current?

- A** higher frequency a.c.
- B** lower frequency a.c.
- C** current in one direction only
- D** the same frequency a.c.

35. O/N/20/12/Q37

A cathode-ray oscilloscope with the Y-gain set at 2 V/division is connected to a generator.



Which row describes the signal from the generator and its amplitude?

	signal	amplitude / V
A	a.c.	3
B	a.c.	6
C	d.c.	3
D	d.c.	6

36. M/J/20/12/Q39

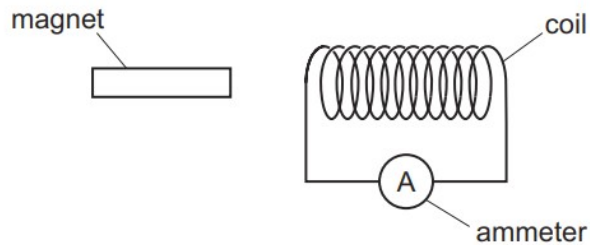
A rectangular current-carrying coil is pivoted between the poles of an electromagnet.

Which action does **not**, on its own, increase the size of the turning effect exerted on the coil?

- A** increasing the current in the coils of the electromagnet
- B** increasing the current in the rectangular coil
- C** reversing the current in the electromagnet
- D** increasing the number of turns on the rectangular coil

37. M/J/20/12/Q40

The diagram shows a coil connected to a very sensitive ammeter. A magnet is next to the coil.

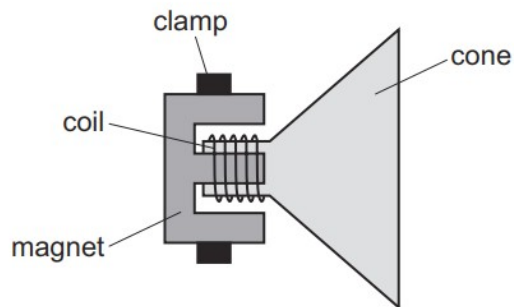


Which action results in a zero reading on the ammeter?

- A** moving the coil and the magnet at the same speed in opposite directions
- B** moving the coil and the magnet at the same speed in the same direction
- C** moving the coil away from the stationary magnet
- D** moving the magnet towards the stationary coil

38. M/J/20/11/Q33

There is varying current in the coil of the loudspeaker shown. The loudspeaker is producing a sound. The magnet is clamped.



What is vibrating to produce the sound?

- A** coil only
- B** cone only
- C** magnet only
- D** coil and cone

39. O/N/19/11/Q37

The input voltage to a transformer is 24 V a.c. and the output voltage is 6.0 V.

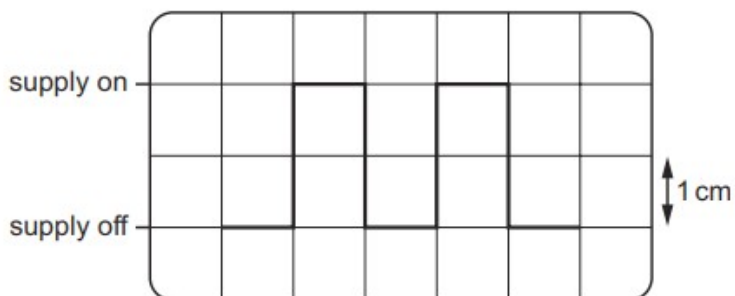
The input coil has 720 turns.

How many turns are on the output coil?

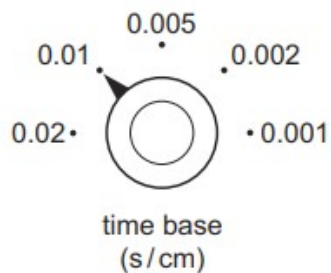
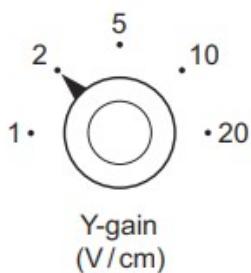
- A** 5 **B** 180 **C** 2900 **D** 100 000

40. O/N/19/11/Q39

The diagram shows the trace on an oscilloscope screen when connected to a d.c. supply which is being switched on and off.



The settings of the Y-gain control and the time base control are shown.



The voltage of the supply is v .

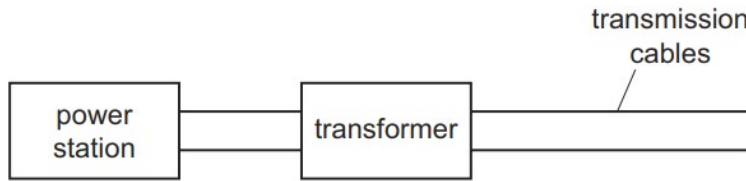
The supply is switched on f times per second.

What are the values of v and of f ?

	v/V	f
A	2	50
B	2	100
C	4	50
D	4	100

41. O/N/19/12/Q37

Transformers are used to transmit electrical energy between power stations and transmission cables, as shown.

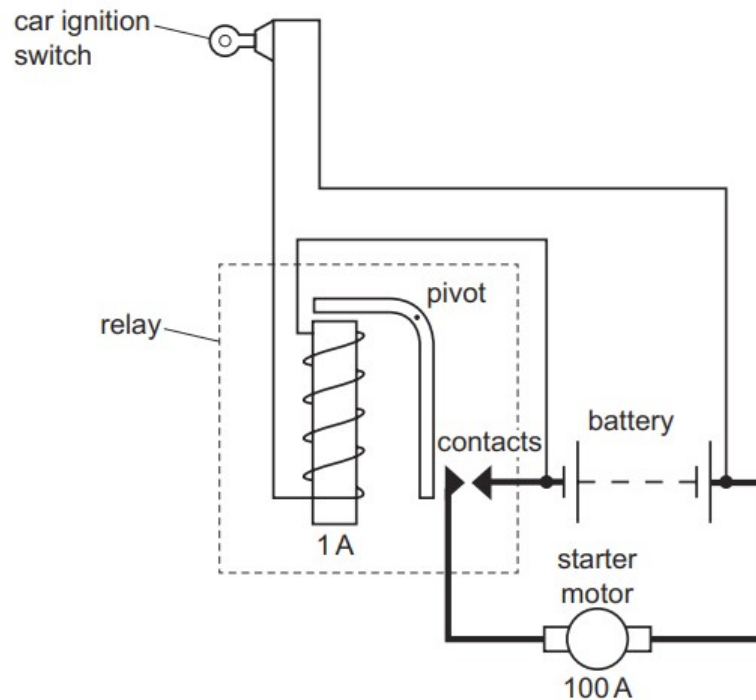


What is the purpose of the transformer in the diagram?

- A** to decrease the current and the potential difference from the power station
- B** to decrease the current and increase the potential difference from the power station
- C** to increase the current and the potential difference from the power station
- D** to increase the current and decrease the potential difference from the power station

42. M/J/19/12/Q32

The diagram shows a car ignition switch and starter motor.



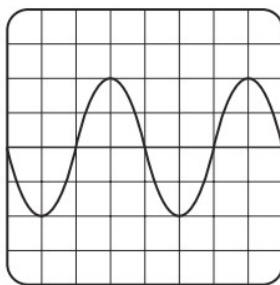
The ignition switch is in a circuit with long, thin wires. The starter motor is in a circuit with short, thick wires.

What is the explanation for the choice of wires?

- A** Each circuit needs to contain the same total mass of wire.
- B** Thicker wires heat up more quickly when the relay is switched on.
- C** Thin wires have lower resistances.
- D** The ignition switch circuit carries a smaller current than the starter motor circuit.

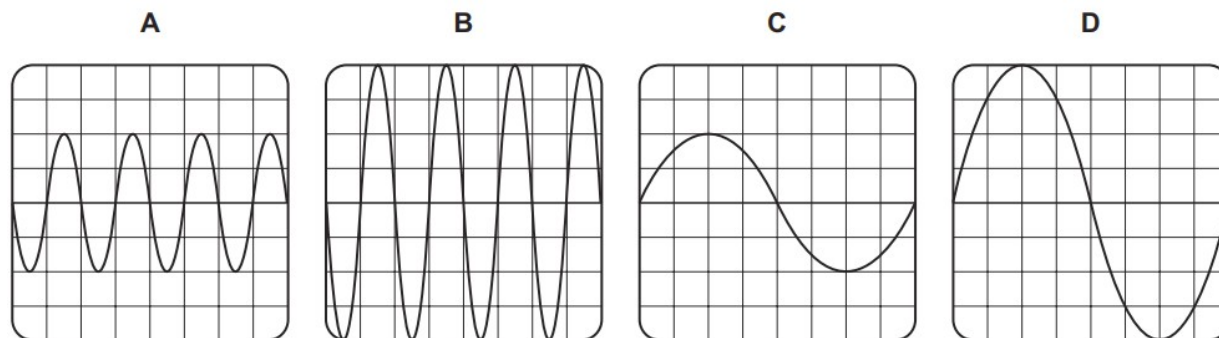
43. M/J/19/12/Q37

A coil is rotating in a magnetic field. The coil is connected to an oscilloscope. The diagram shows the trace on the screen of the oscilloscope.



The coil now speeds up and rotates twice as fast.

Which diagram shows the new trace?



44. M/J/19/12/Q38

A teacher uses the circuit shown.



The identical lamps X and Y are connected to a low voltage a.c. power supply by high resistance transmission wires. Both lamps are switched on.

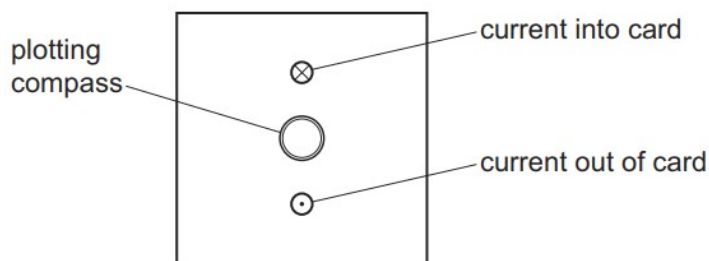
Lamp X is then switched off. Lamp Y stays switched on.

What happens to the voltage and the power supplied to lamp Y?

	the voltage supplied to lamp Y	the power supplied to lamp Y
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

45. M/J/19/11/Q29

Two vertical wires pass at right-angles through a piece of card. There is a large current in each wire in the direction shown.



A plotting compass is placed on the card.

Which diagram shows the direction in which the needle of the plotting compass points?



46. M/J/19/11/Q36

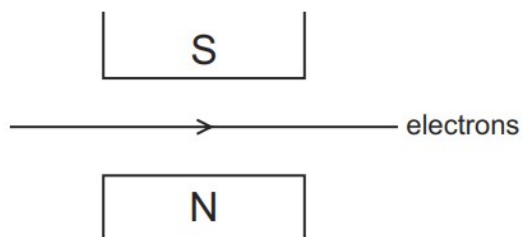
The coil of a simple motor lies between the poles of a permanent magnet. The coil rotates about its axis when there is a current in it.

What decreases the frequency of rotation of the coil?

- A** increasing the number of turns in the coil
- B** reversing the current
- C** using a lower voltage supply
- D** using a stronger magnet

47. M/J/19/11/Q37

A beam of electrons travels through a vacuum. The beam passes between the poles of a magnet as shown.

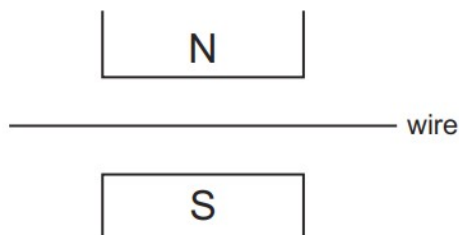


What is the direction of the conventional current and what is the direction of the magnetic field?

	direction of the conventional current	direction of the magnetic field
A	→	↓
B	→	↑
C	←	↓
D	←	↑

48. O/N/18/11/Q33

A wire is placed in the magnetic field between the poles of a magnet, as shown.



When there is a current in the wire, it experiences a force.

Which change increases the size of this force?

- A** reverse both the direction of the current and the poles of the magnet
- B** reverse just the direction of the current
- C** use a stronger magnet
- D** use a thicker wire with the same current

49. O/N/18/11/Q34

A 12 V lamp glows at normal brightness when connected to the secondary coil of a mains transformer. The mains voltage is 240 V.

The transformer is 100% efficient and the primary coil has 200 turns.

How many turns are there on the secondary coil?

- A** 10 **B** 14 **C** 4000 **D** 580 000

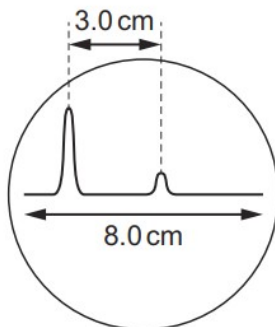
50. O/N/18/11/Q35

Which row shows how the electrical energy produced by a power station is transmitted to distant towns?

	current	voltage
A	alternating	low
B	alternating	very high
C	direct	low
D	direct	very high

51. O/N/18/11/Q36

The diagram shows the trace on an oscilloscope screen. The first peak occurs when a pulse of sound is emitted. The second, smaller peak is the echo received from a wall.



The trace moves across the screen in a time of 24 ms.

How long does it take for the sound to travel to the wall?

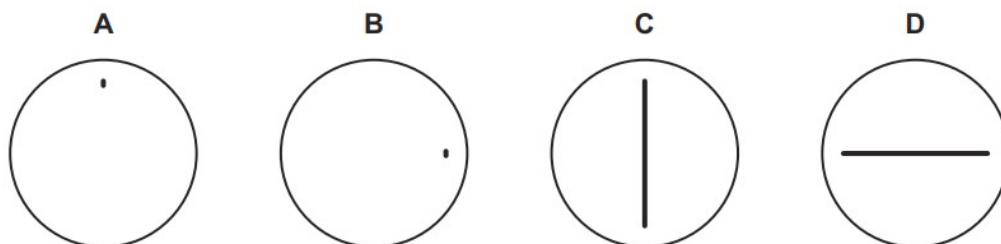
- A** 4.5 ms **B** 9.0 ms **C** 12 ms **D** 32 ms

52. O/N/18/12/Q36

An oscilloscope is connected to a d.c. power supply.

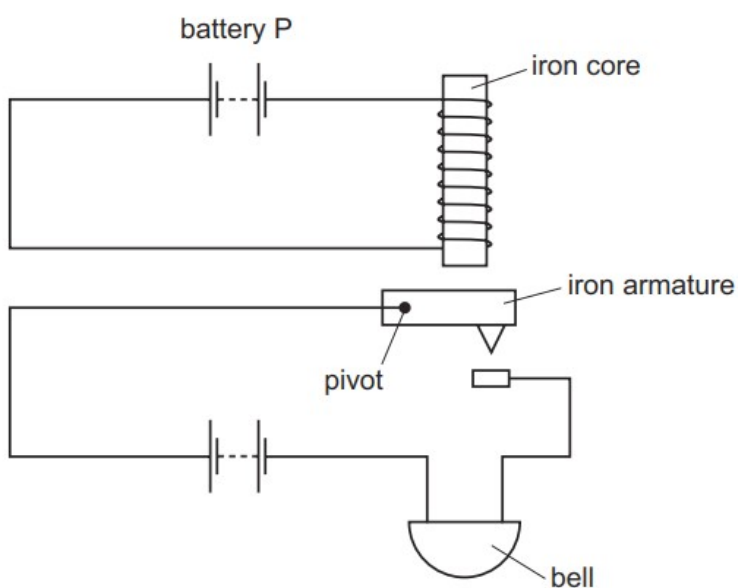
The time-base on the oscilloscope is switched off.

What is seen on the screen of the oscilloscope?



53. M/J/18/12/Q34

The diagram shows an alarm system.



What happens when battery P is disconnected?

	iron armature	bell
A	falls	rings
B	falls	stops ringing
C	moves up	rings
D	moves up	stops ringing

54. M/J/18/12/Q36

A step-down transformer has a primary coil and a secondary coil wound on a soft-iron core.

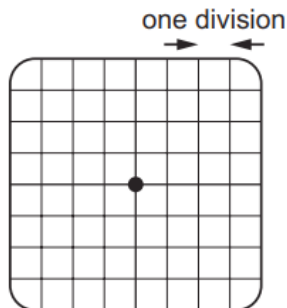
The primary coil is connected to a 6.0 V direct current (d.c.) supply.

Which statement about the transformer is correct?

- A** The output voltage is equal to 6.0 V.
- B** The output voltage is greater than 6.0 V.
- C** The output voltage is less than 6.0 V but more than zero.
- D** There is no output voltage.

55. M/J/18/12/Q39

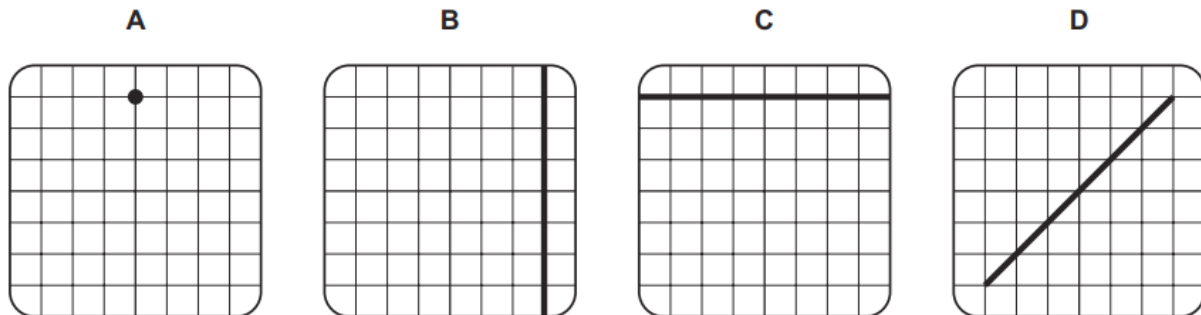
An oscilloscope is used to measure potential difference (p.d.). The trace with no input connected is shown.



A 1.5 V d.c. supply is connected to the oscilloscope.

The Y-gain is set at 0.5 V/div. The time-base is set at 0.5 ms/div.

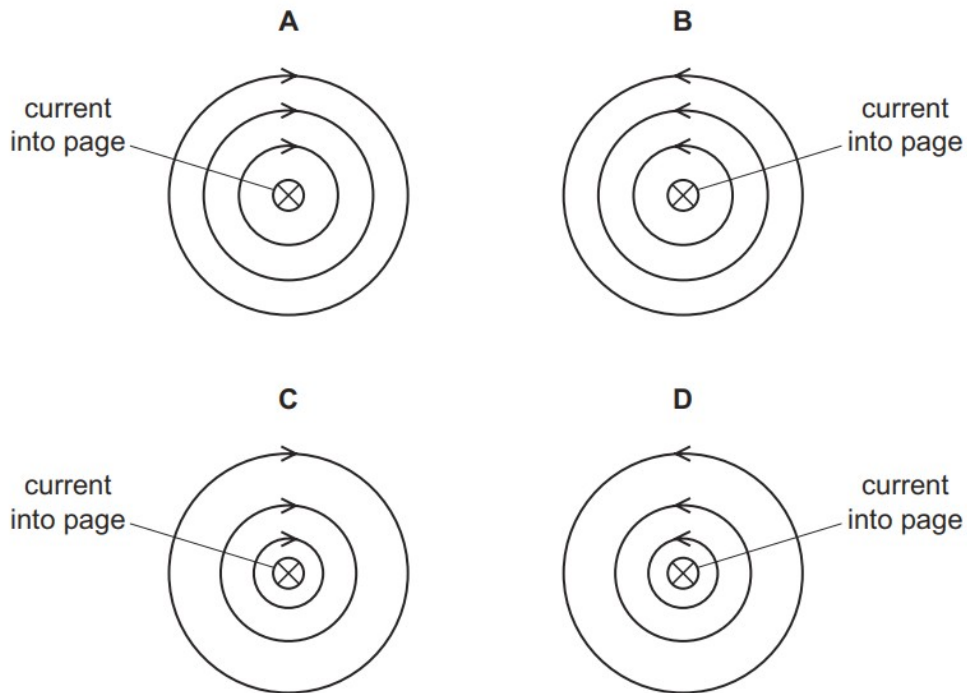
Which trace shows a supply of 1.5 V d.c.?



56. M/J/18/11/Q28

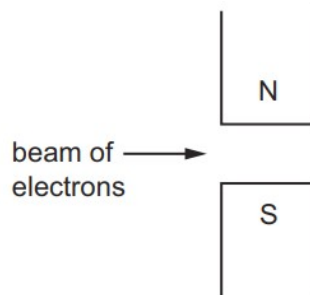
An electric current in a wire is into the page.

Which diagram shows the shape and direction of the magnetic field around the wire?



57. M/J/18/11/Q32

A horizontal beam of electrons passes between the two poles of a magnet.

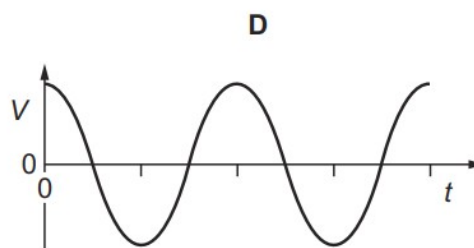
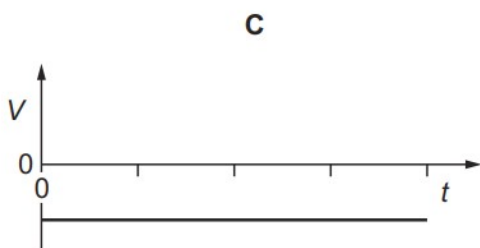
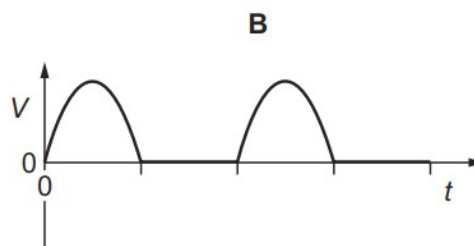
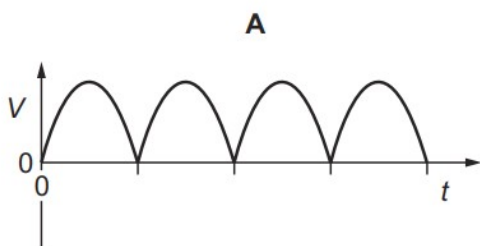


In which direction is the beam deflected?

- A** into the page
- B** out of the page
- C** towards the north pole
- D** towards the south pole

58. M/J/18/11/Q33

Which graph shows the voltage output V against time t for an a.c. generator?

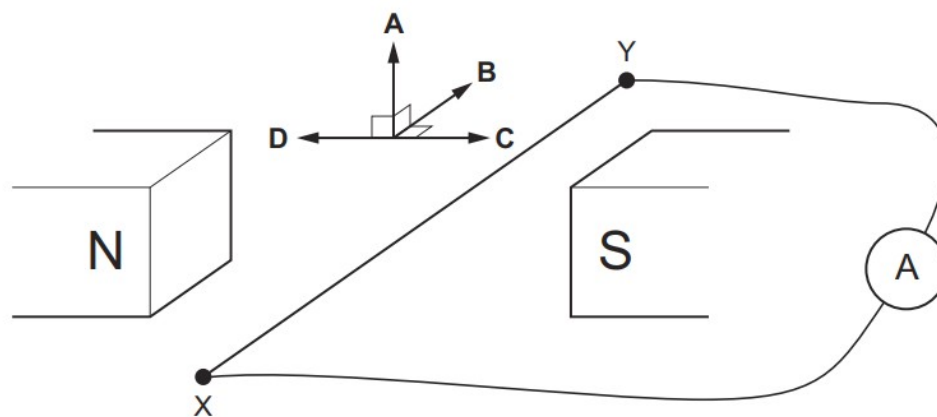


59. O/N/17/11/Q34

The diagram shows a wire XY lying between the poles of a magnet.

The ends of the wire are connected to a sensitive ammeter. The wire is moved and a reading is registered.

In which direction is the wire moved?



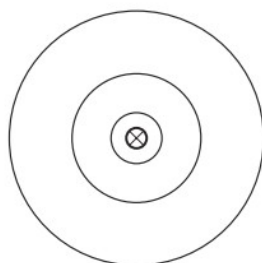
60. O/N/17/12/Q33

Why is the coil of an electric motor wound on a soft-iron cylinder?

- A** to decrease the electric current
- B** to increase the electric current
- C** to reverse the magnetic field
- D** to strengthen the magnetic field

61. O/N/17/12/Q34

A straight wire carries a current into the paper. The diagram shows three magnetic field lines around the wire.



key

⊗ current into paper

The current in the wire increases.

What is the direction of the field lines and which change occurs to the field lines as the current increases?

	direction	change
A	clockwise	the lines move closer together
B	clockwise	the lines move further apart
C	anticlockwise	the lines move closer together
D	anticlockwise	the lines move further apart

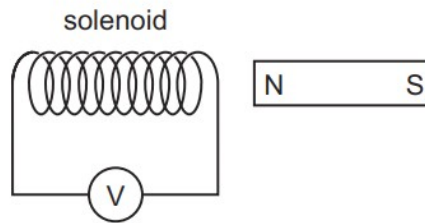
62. M/J/17/12/Q36

What is the effect of using a split-ring commutator?

- A** it ensures that the current is the same in all parts of a series circuit
- B** it generates an alternating electric current
- C** it produces a force on a current-carrying coil
- D** it reverses the direction of the current in the coil of a motor

63. M/J/17/12/Q37

The N-pole of a magnet is moved into a solenoid and an e.m.f. is induced.



What causes an increase in the induced e.m.f.?

- A moving the magnet more quickly
- B moving the magnet more slowly
- C pulling the magnet out instead of pushing it in
- D using the S-pole of the magnet instead of the N-pole

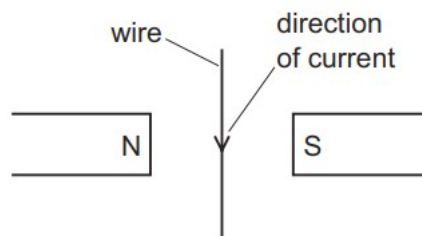
64. M/J/17/12/Q38

Which material is used for the core of a transformer?

- A copper
- B iron
- C plastic
- D steel

65. O/N/16/11/Q32

A current-carrying wire lies between the poles of two magnets, as shown.

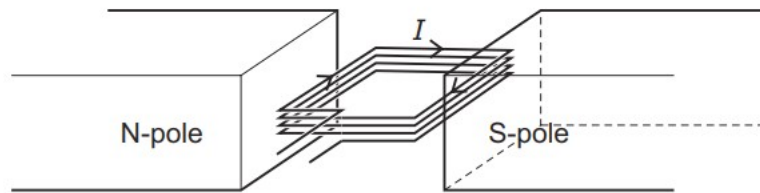


What is the direction of the force on the wire?

- A into the plane of the paper
- B out of the plane of the paper
- C towards the left
- D towards the right

66. O/N/16/11/Q33

A coil P of N turns is made from a length L of wire. The coil carries a current I when between two magnetic poles.



A similar coil Q of $2N$ turns is made from a length $2L$ of identical wire. It also carries a current I when between the two magnetic poles.

Which coil has the greater resistance and which coil experiences the greater turning effect?

	greater resistance	greater turning effect
A	P	P
B	P	Q
C	Q	P
D	Q	Q

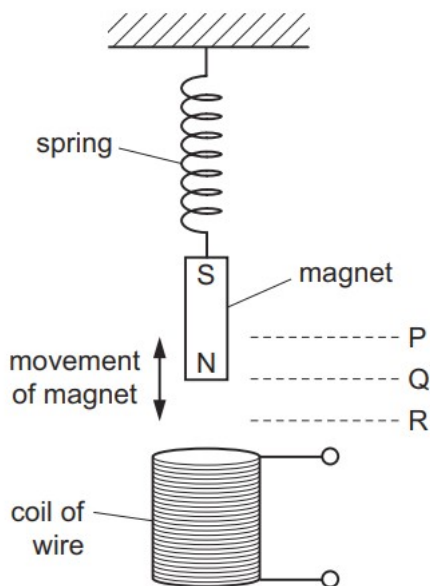
67. O/N/16/11/Q35

Which material is most suitable for the core of a transformer and which material is most suitable for the coils in the transformer?

	material for core	material for coils
A	iron	copper
B	iron	steel
C	steel	copper
D	steel	iron

68. O/N/16/11/Q34

A magnet moves up and down above a coil of wire.



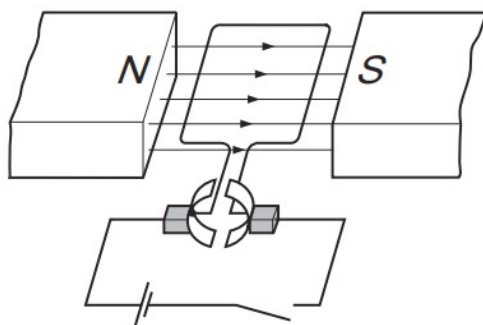
The bottom of the magnet moves up and down between P and R.

Where is the bottom of the magnet when there is no induced electromotive force (e.m.f.) in the coil?

- A at P and at Q
- B at P and at R
- C at Q only
- D at R only

69. M/J/16/12/Q32

The diagram shows a simple d.c. motor.



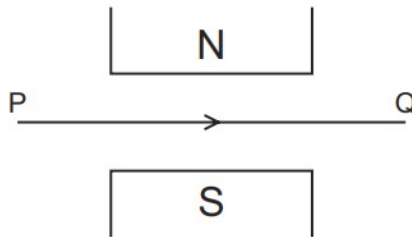
The switch is closed and the coil rotates.

Which change makes the coil rotate in the opposite direction **and** at a faster rate?

- A increase the current in the coil and increase the number of turns in the coil
- B reverse both the magnetic field and the current in the coil
- C reverse the magnetic field and decrease the current in the coil
- D reverse the magnetic field and increase the current in the coil

70. M/J/16/12/Q33

The diagram shows a wire PQ between the N-pole and the S-pole of a magnet. There is a current in the wire in the direction of the arrow.



What is the direction of the force on the wire PQ?

- A** into the page
- B** out of the page
- C** towards the N-pole
- D** towards the S-pole

71. M/J/16/12/Q34

Electrical power is transmitted by cables over long distances at very high voltages.

What are the effects of using a high voltage transmission system?

	power loss in the cables	current in the cables
A	high	high
B	high	low
C	low	high
D	low	low

72. M/J/16/11/Q35

A split-ring commutator is used in a simple d.c. motor. It reverses the current in the coil.

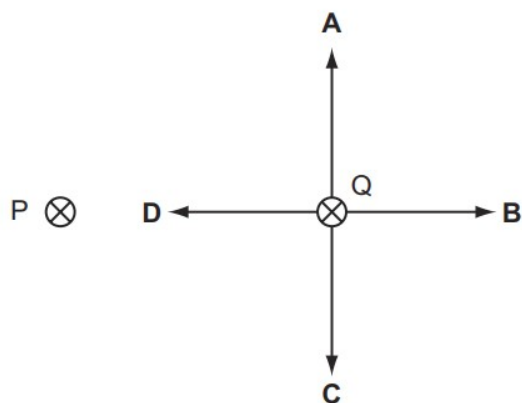
How often does it reverse the current?

- A** every quarter turn
- B** every half turn
- C** every full turn
- D** every two turns

73. O/N/15/11/Q32

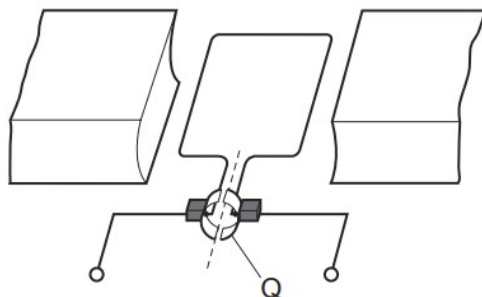
P and Q represent two, parallel, straight wires carrying currents into the plane of the paper. P and Q exert a force on each other.

Which arrow shows the force on Q?



74. O/N/15/11/Q33

The diagram shows a simple d.c. motor.



What is the part labelled Q?

- A** a coil
- B** a commutator
- C** a magnet
- D** a slip ring

75. O/N/15/11/Q35

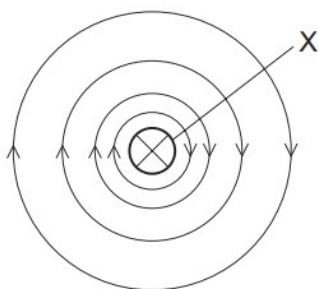
Electric power cables transmit electrical energy over large distances using high-voltage, alternating current.

What are the advantages of using a high voltage and of using an alternating current?

	advantage of using a high voltage	advantage of using an alternating current
A	high current is produced in the cable	the resistance of the cable is reduced
B	high current is produced in the cable	the voltage can be changed using a transformer
C	less energy is wasted in the cable	the resistance of the cable is reduced
D	less energy is wasted in the cable	the voltage can be changed using a transformer

76. O/N/15/12/Q31

The diagram shows the magnetic field around wire X which carries a current into the paper.

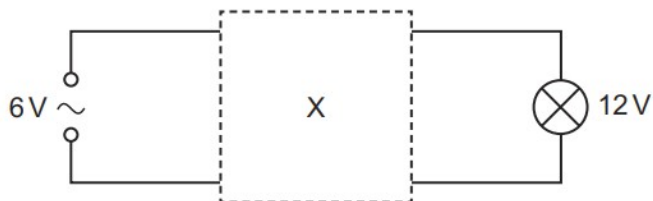


The arrows on the field lines show the direction of the force on

- A** a N-pole.
- B** a S-pole.
- C** a small negative charge.
- D** a small positive charge.

77. O/N/15/12/Q37

The diagram shows an electrical device X connected between a 6V a.c. supply and a 12V lamp



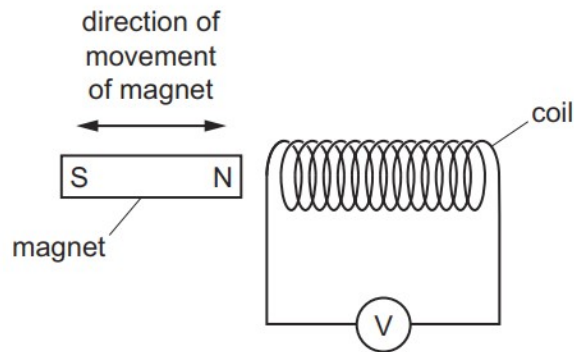
The lamp is seen to glow with normal brightness.

What is X?

- A** a capacitor
- B** a potential divider
- C** a relay
- D** a transformer

78. M/J/15/12/Q36

A teacher moves a magnet into and out of a coil of wire, as shown, in order to demonstrate electromagnetic induction.



Which statement is correct?

- A** As the magnet is moved into the coil the left-hand end of the coil becomes a S-pole.
- B** As the magnet is taken out of the coil the left-hand end of the coil becomes a N-pole.
- C** Increasing the speed at which the magnet enters the coil, increases the induced voltage.
- D** Increasing the speed at which the magnet leaves the coil decreases the induced voltage.

79. M/J/15/12/Q37

A transformer consists of two coils which are wound on to a metallic core.

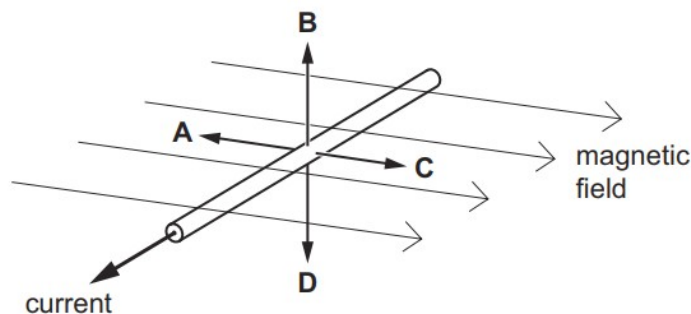
Which type of voltage is supplied to the transformer and which metal is used to make the core?

	supply voltage	metal
A	alternating	iron
B	alternating	steel
C	direct	iron
D	direct	steel

80. O/N/14/11/Q30

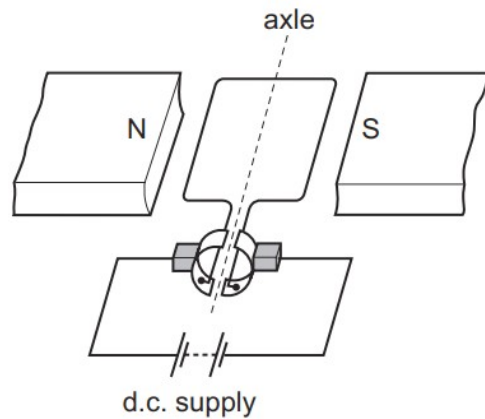
The diagram shows a current-carrying wire in a horizontal magnetic field.

Which arrow shows the direction of the force experienced by the wire?



81. O/N/14/11/Q31

The diagram shows a d.c. motor with its coil horizontal.

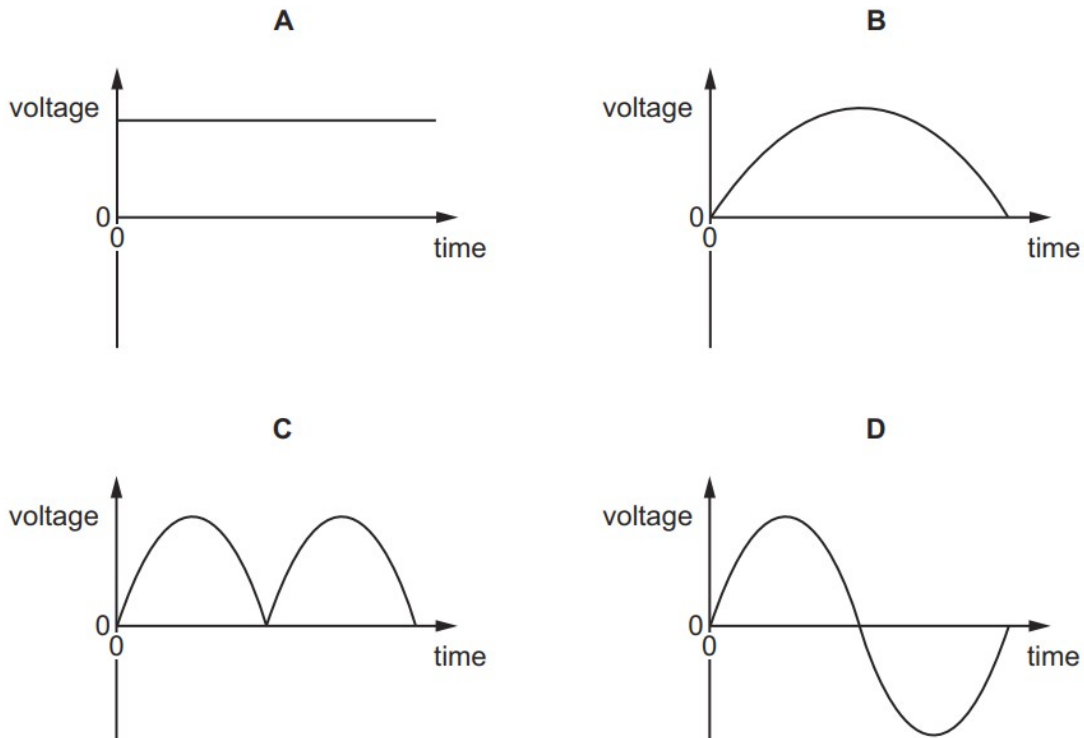


Why is a split-ring commutator used?

- A to change the current direction in the coil as the coil passes the horizontal position
- B to change the current direction in the coil as the coil passes the vertical position
- C to change the current direction in the d.c. supply as the coil passes the horizontal position
- D to change the current direction in the d.c. supply as the coil passes the vertical position

82. O/N/14/11/Q32

Which graph shows the voltage output of an a.c. generator when the coil makes **one** complete revolution?



83. O/N/14/11/Q36

Why is a reed relay used in a switching circuit?

- A to switch on a small current using a large current
- B to switch on a small voltage using a large voltage
- C to switch on a large current using a small current
- D to switch on a large voltage using a large current

84. O/N/14/12/Q31

Two long, parallel conductors carrying current lie in a horizontal plane.

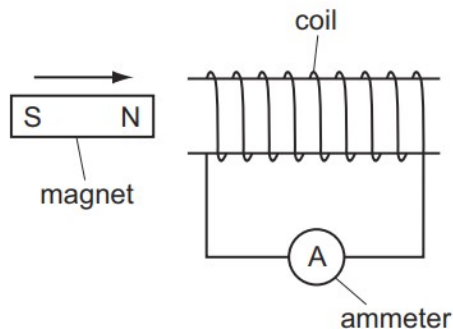
The two conductors attract one another.

The two currents **must**

- A be in the same direction.
- B be in opposite directions.
- C be parallel to the Earth's magnetic field.
- D be at 90° to the Earth's magnetic field.

85. O/N/14/12/Q33

A student moves a magnet into a coil of wire as shown in the diagram. The coil of wire is connected to a sensitive ammeter.



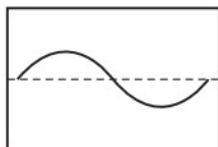
Which change does **not** produce an increase in the reading?

- A increasing the number of turns on the coil
- B increasing the resistance of the ammeter
- C increasing the speed of the magnet
- D increasing the strength of the magnet

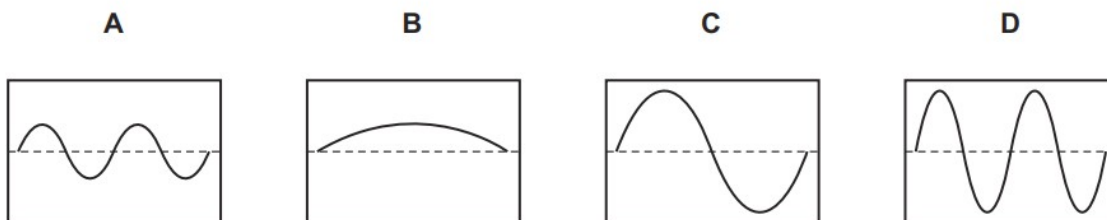
86. O/N/14/12/Q34

The coil of an a.c. generator is rotated and the output is displayed on the screen of a cathode-ray oscilloscope (c.r.o.).

The diagram shows the trace on the screen.



Which trace appears on the screen when the speed of rotation of the coil is doubled but the settings on the c.r.o. are unaltered?



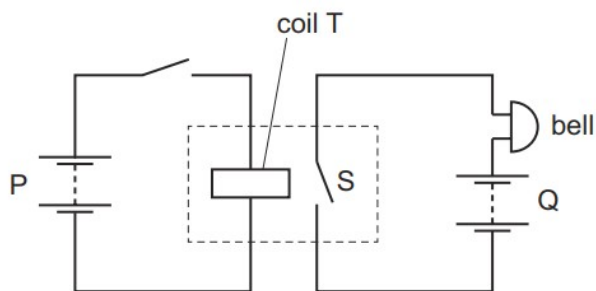
87. M/J/14/12/Q34

Which device uses the force experienced by a current in a magnetic field when in normal use?

- A** cathode-ray oscilloscope
- B** electrostatic precipitator
- C** loudspeaker
- D** transformer

88. M/J/14/12/Q35

A relay is used in a circuit containing a bell.

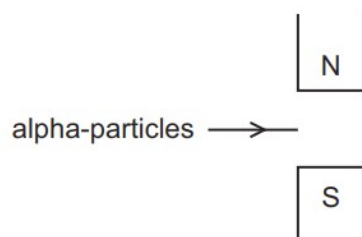


How can the apparatus be altered to make the sound of the bell louder?

- A** increase the number of turns on coil T
- B** increase the voltage of battery P
- C** increase the voltage of battery Q
- D** move the coil closer to switch S

89. M/J/14/11/Q34

A beam of alpha-particles enters the magnetic field between the poles of a magnet.



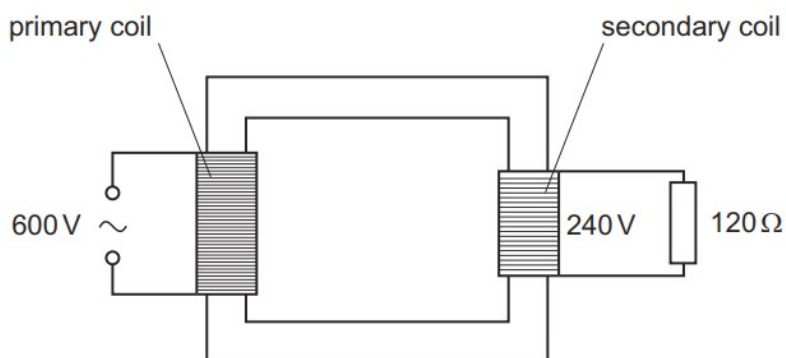
In which direction is the magnetic force on the beam?

- A** down the page
- B** into the page
- C** out of the page
- D** up the page

90. M/J/14/11/Q36

An ideal transformer has a primary voltage of 600 V and a secondary voltage of 240 V.

The secondary coil is attached to a resistor of resistance $120\ \Omega$.

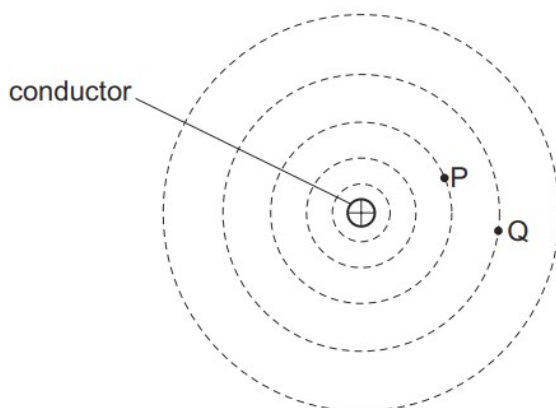


What is the power dissipated in the resistor and the current in the primary coil?

	power / W	current / A
A	120	0.20
B	120	5.0
C	480	0.80
D	480	1.3

91. O/N/13/11/Q25

The diagram shows the shape of the magnetic field lines near a current-carrying conductor.



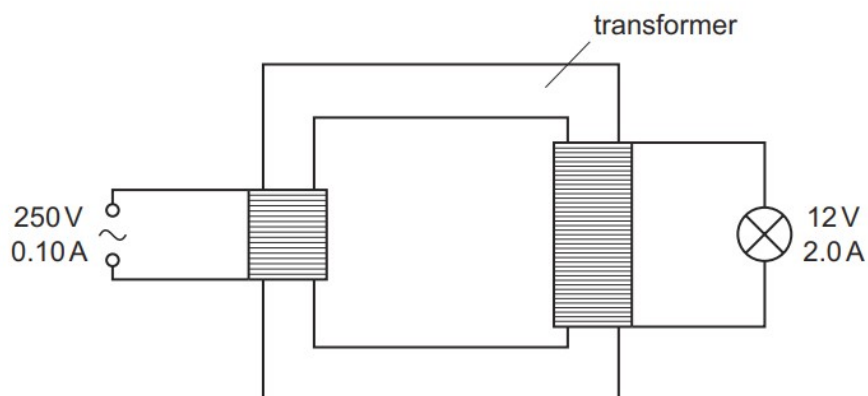
The current in the conductor is into the plane of the diagram.

Which row correctly states the direction of the field lines and compares the strengths of the field at points P and Q?

	direction of field lines	the field is stronger at
A	clockwise	P
B	clockwise	Q
C	anticlockwise	P
D	anticlockwise	Q

92. O/N/13/11/Q31

A transformer is used to operate a 12 V lamp from a 250 V mains supply.



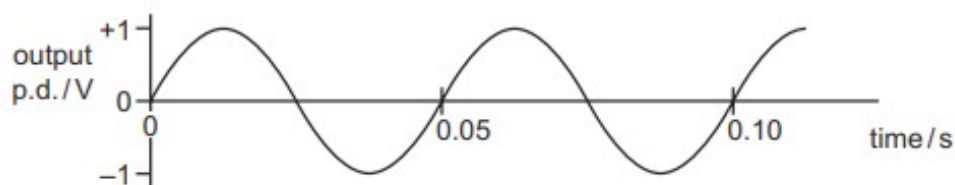
The mains current is 0.10 A. The current in the lamp is 2.0 A.

What is the efficiency of the transformer?

- A** 0.048 **B** 0.050 **C** 0.96 **D** 1.04

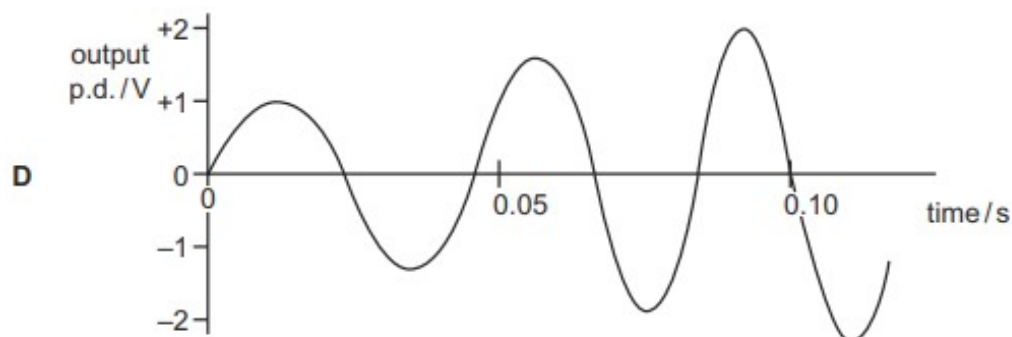
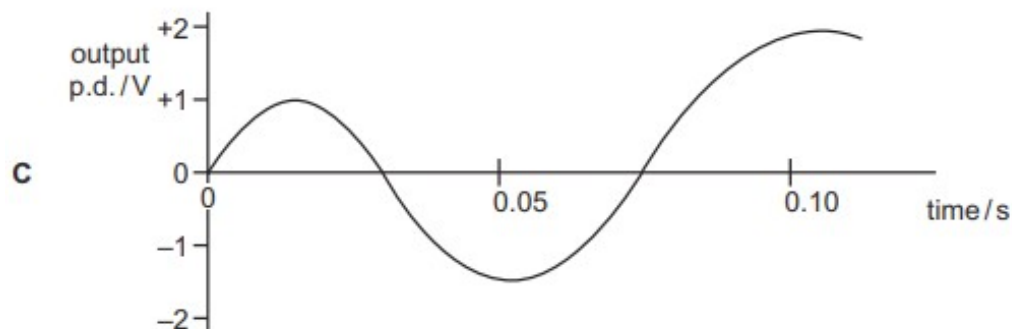
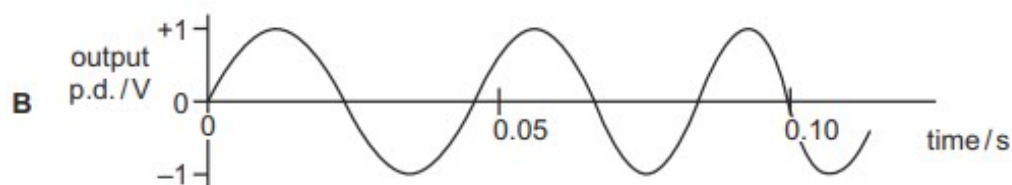
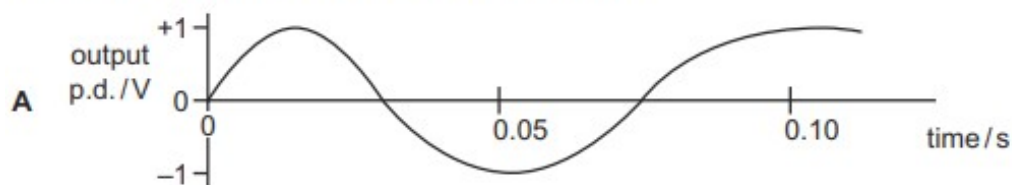
93. O/N/13/11/Q33

The graph shows the output of an a.c. generator. The coil in the generator rotates 20 times in one second.



The speed of rotation of the coil steadily increases.

Which graph best shows how the output changes?



94. O/N/13/11/Q34

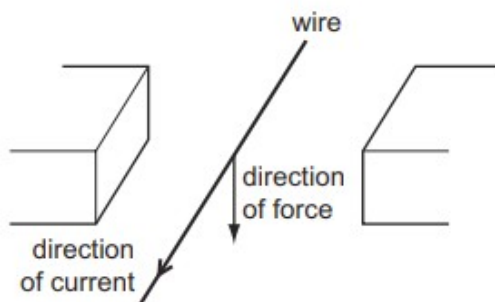
Which material is used for the core of a transformer and why?

	material	reason
A	copper	good conductor of electricity
B	copper	easy to magnetise and demagnetise
C	iron	good conductor of electricity
D	iron	easy to magnetise and demagnetise

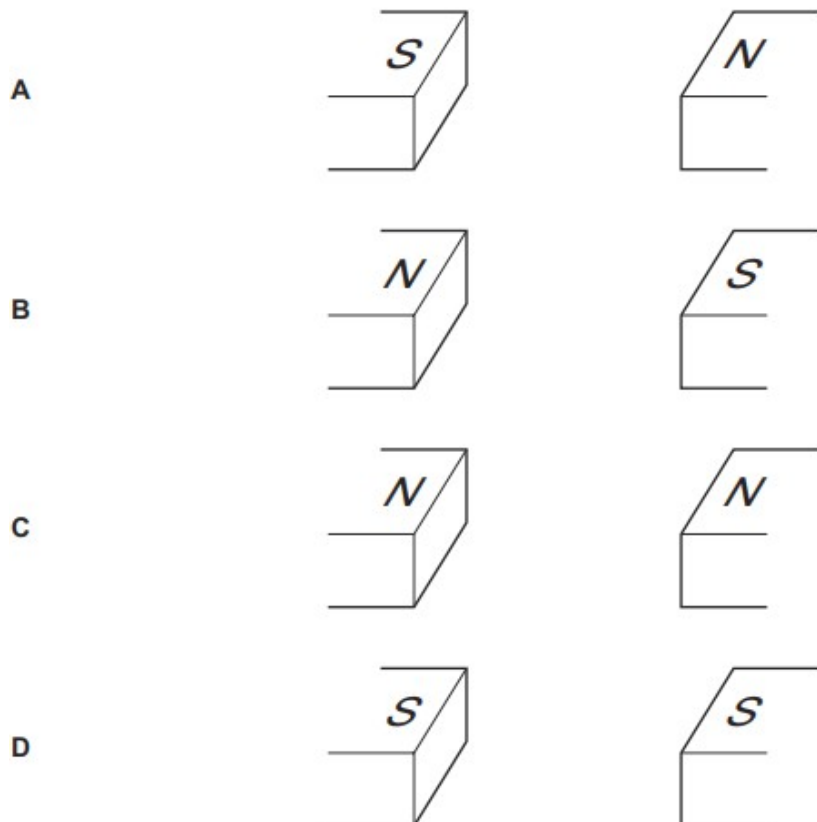
95. O/N/13/12/Q32

The diagram shows a wire placed between two magnetic poles of equal strength.

A current passes through the wire in the direction shown. The current causes a downward force on the wire.



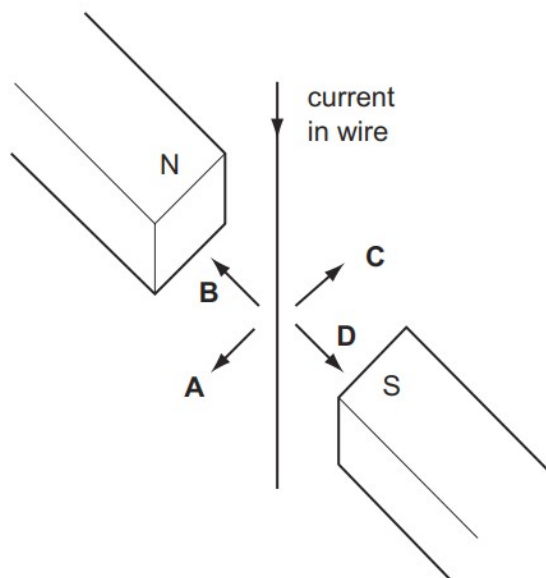
What is the arrangement of the magnetic poles?



96. M/J/13/12/Q37

A wire hangs between the poles of a magnet.

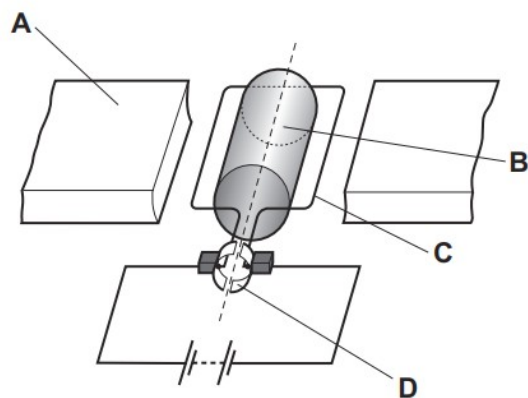
When there is a current in the wire, in which direction does the wire move?



97. M/J/13/11/Q35

The diagram shows a simple d.c. motor.

Which labelled part is the commutator?



98. M/J/13/11/Q36

Two leads emerging from a box are connected to a sensitive ammeter.



When a bar magnet moves towards the open end of the box, the needle of the ammeter deflects to the right. When the bar magnet stops, the needle returns to zero.

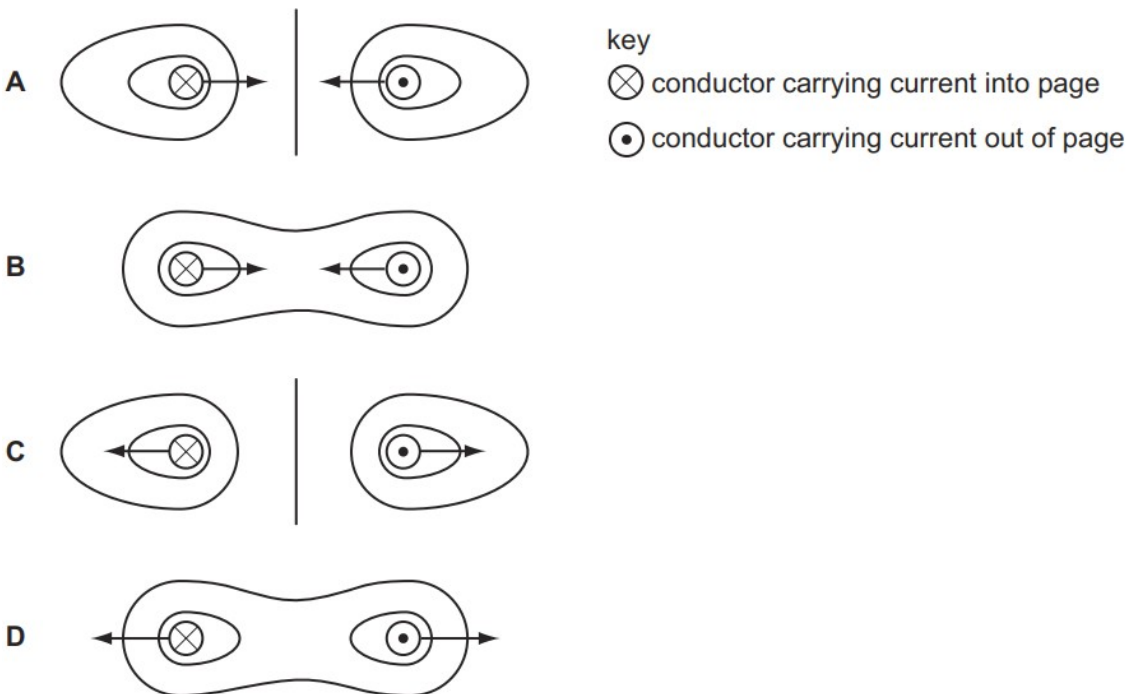
What is inside the box?

- A** a coil alone
- B** a coil connected in series with a cell
- C** a light-dependent resistor (LDR) alone
- D** an LDR in series with a cell

99. O/N/12/11/Q33

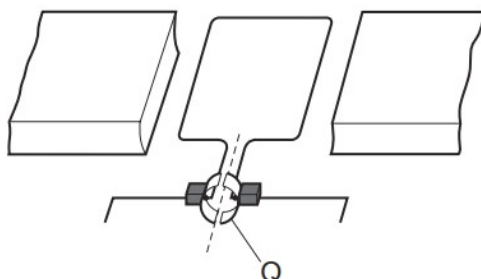
Each of the diagrams shows a cross-section through two parallel, current-carrying conductors.

Which diagram shows the shape of the magnetic field pattern **and** the directions of the forces on the two conductors?



100. O/N/12/22/Q34

The diagram shows a simple d.c. motor.

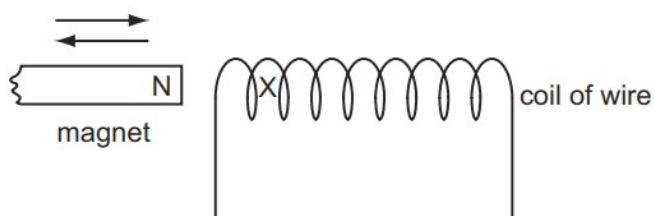


What is the part labelled Q?

- A** a coil
- B** a magnet
- C** a slip ring
- D** a split-ring commutator

101. O/N/12/11/Q35

The diagram shows the N-pole of a magnet moving into, and out of, a coil of wire.



This movement produces a current in the coil of wire. The current produces a magnetic pole at X.

Which pole is produced at X when the magnet is moved in and when it is moved out?

	magnet moved in	magnet moved out
A	N	N
B	N	S
C	S	N
D	S	S

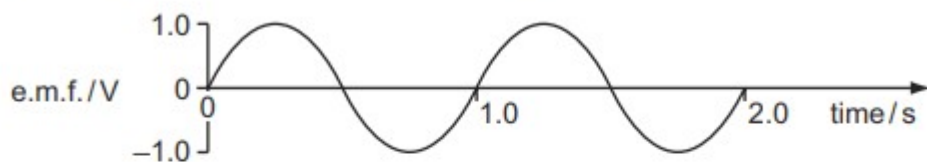
102. O/N/12/11/Q37

What is required to operate a reed relay in a switching circuit?

- A** a capacitor
- B** an electric field
- C** a magnetic field
- D** a transformer

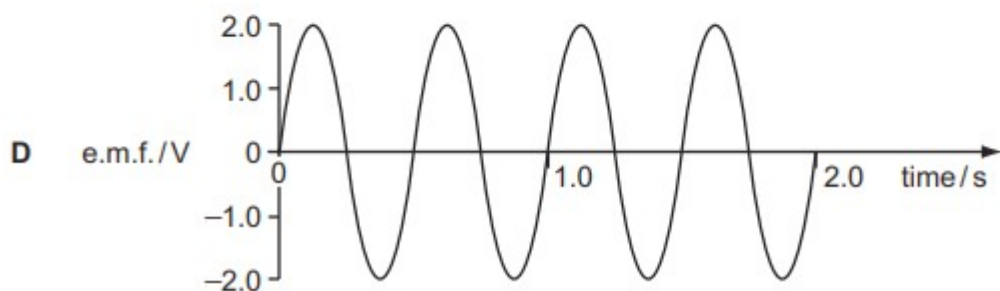
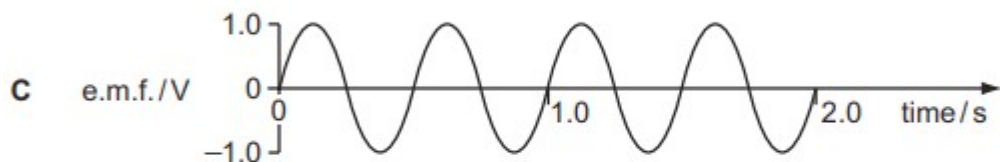
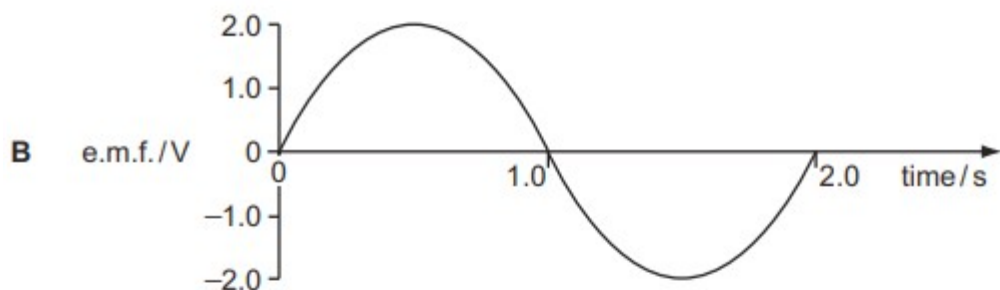
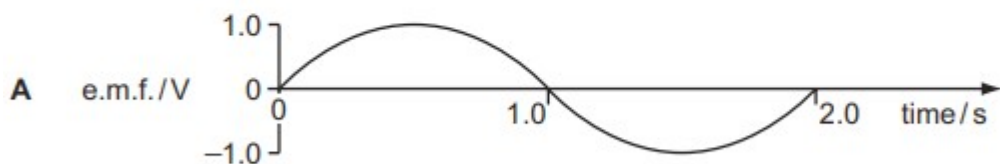
103. O/N/12/11/Q36

A simple a.c. generator produces an alternating e.m.f. as shown.



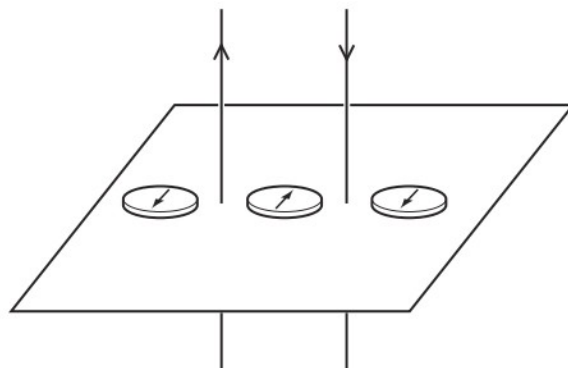
The speed of the generator is doubled.

Which graph best represents the new output?



104. O/N/12/12/Q33

Two parallel wires carry currents in opposite directions. Three plotting compasses are placed in the positions shown.



The currents in **both** wires are reversed. How many compass needles change direction?
(Ignore the effect of the Earth's magnetic field.)

- A** 0 **B** 1 **C** 2 **D** 3

105. O/N/12/12/Q34

Which single-coil motor has the largest turning effect?

	current in coil / A	number of turns in coil	iron core
A	6	100	no
B	10	200	no
C	6	100	yes
D	10	200	yes

106. O/N/12/12/Q35

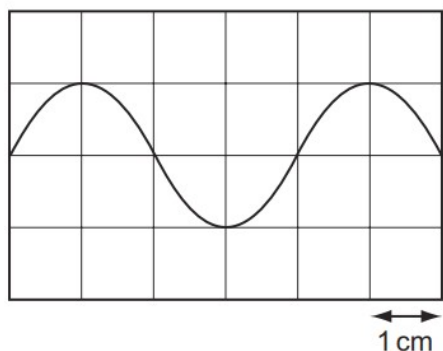
A magnet is moved towards a coil of insulated wire. A voltmeter connected across the coil shows a positive reading.

What produces a higher reading on the voltmeter?

- A** moving the magnet away from the coil at the same speed
B moving the magnet away from the coil at a slower speed
C moving the magnet towards the coil at a faster speed
D moving the magnet towards the coil at a slower speed

107. O/N/12/12/Q36

The diagram shows the output of an a.c. generator as displayed on a cathode-ray oscilloscope. The horizontal scale is 5 ms/cm.

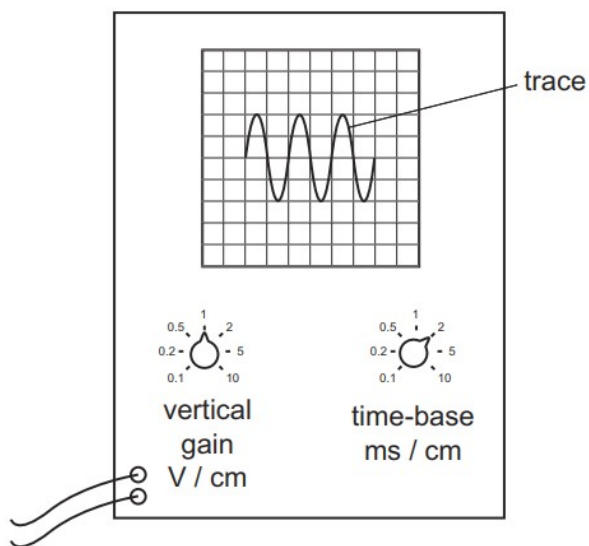


What is the time for one complete rotation of the coil of the generator?

- A** 5 ms **B** 10 ms **C** 20 ms **D** 30 ms

108. O/N/12/12/Q37

The trace of a waveform is seen on the screen of a cathode-ray oscilloscope.



Which statement about the controls is correct?

- A** The amplitude of the trace is changed by adjusting the time-base.
B The amplitude of the trace is changed by adjusting the vertical gain.
C The whole trace is moved to the right by adjusting the time-base.
D The whole trace is moved upwards by adjusting the vertical gain.

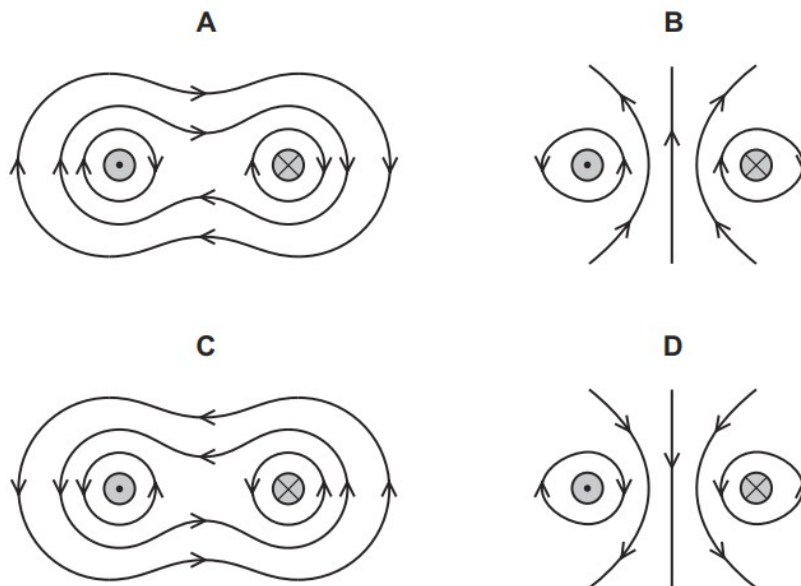
109. M/J/12/12/Q26

Two long, straight wires hang vertically, close to each other.

The wires carry currents in opposite directions.

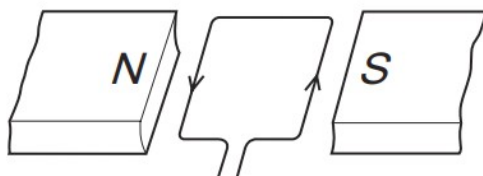


Which diagram shows the magnetic field pattern around the wires?



110. M/J/12/12/Q34

A rectangular coil is placed between the poles of a magnet. A current passes through the coil, as shown.

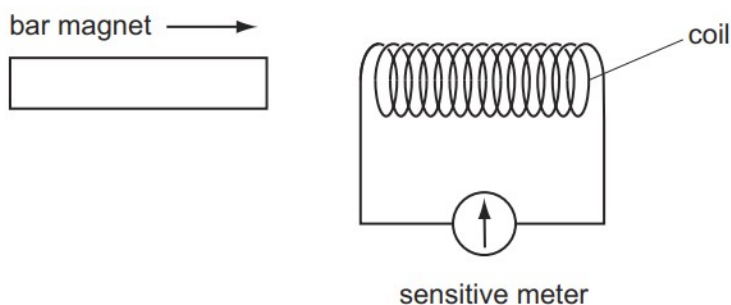


What happens to the coil?

- A** It moves downwards.
- B** It moves upwards.
- C** It rotates anticlockwise.
- D** It rotates clockwise.

111. M/J/12/12/Q35

A bar magnet is pushed into one end of a long coil connected to a sensitive meter.

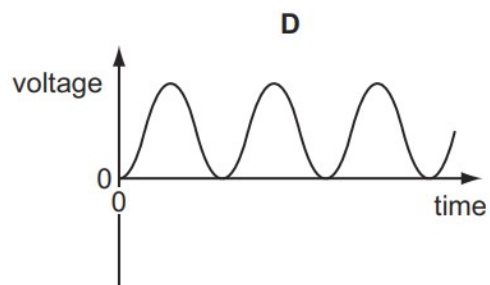
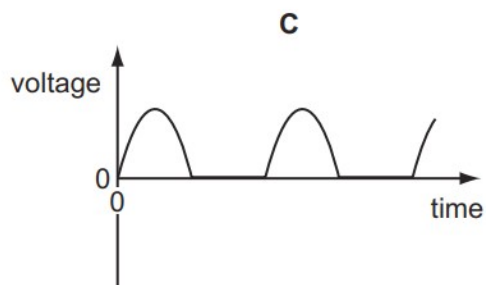
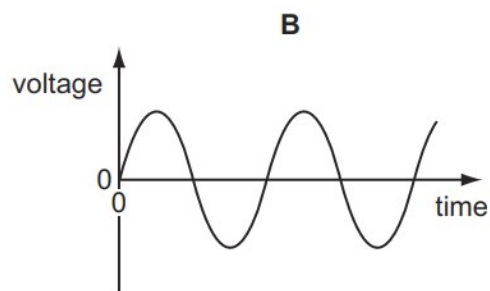
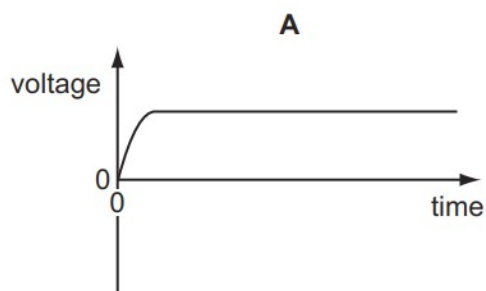


Which of the following affects the **magnitude** of the deflection of the meter?

- A the direction in which the coil is wound
- B the speed with which the magnet enters the coil
- C which end of the coil is used
- D which pole of the magnet enters first

112. M/J/12/12/Q36

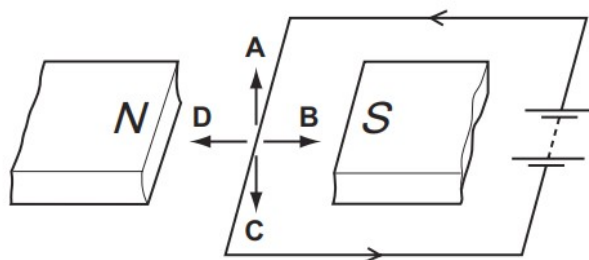
Which graph represents the voltage output of a simple a.c. generator?



113. M/J/12/11/Q33

A current-carrying wire is placed between the poles of a magnet.

What is the direction of the force on the wire due to the current?

**114. M/J/12/11/Q34**

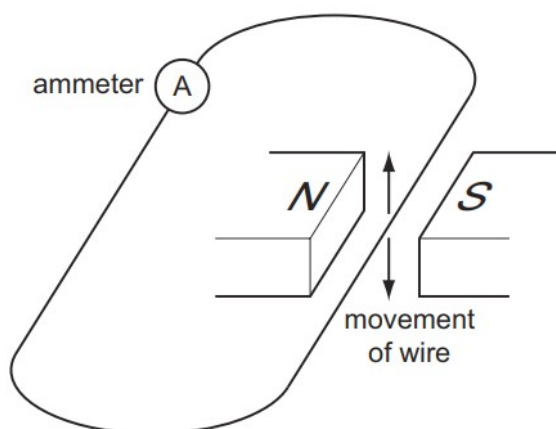
A simple d.c. motor consists of a coil that rotates between the poles of a permanent magnet. The turning effect is increased by winding the coil on a metal cylinder.

Which metals are used to make the magnet and the cylinder?

	magnet	cylinder
A	iron	copper
B	iron	steel
C	steel	copper
D	steel	iron

115. M/J/12/11/Q35

A current is produced when a wire is moved between two magnets as shown.



Which device uses this effect?

- A** a battery
- B** a generator
- C** a motor
- D** an electromagnet

116. M/J/12/11/Q36

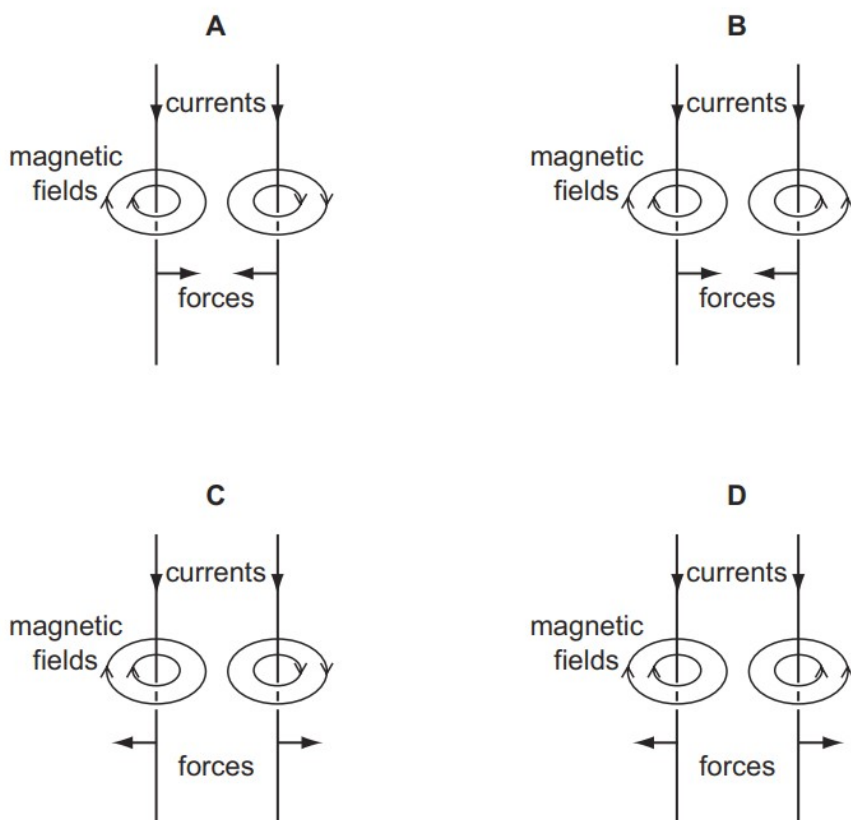
Why is a transformer used to connect a generator in a power station to a long-distance transmission line?

- A** to decrease the voltage and decrease the current
- B** to decrease the voltage and increase the current
- C** to increase the voltage and decrease the current
- D** to increase the voltage and increase the current

117. O/N/11/11/Q34

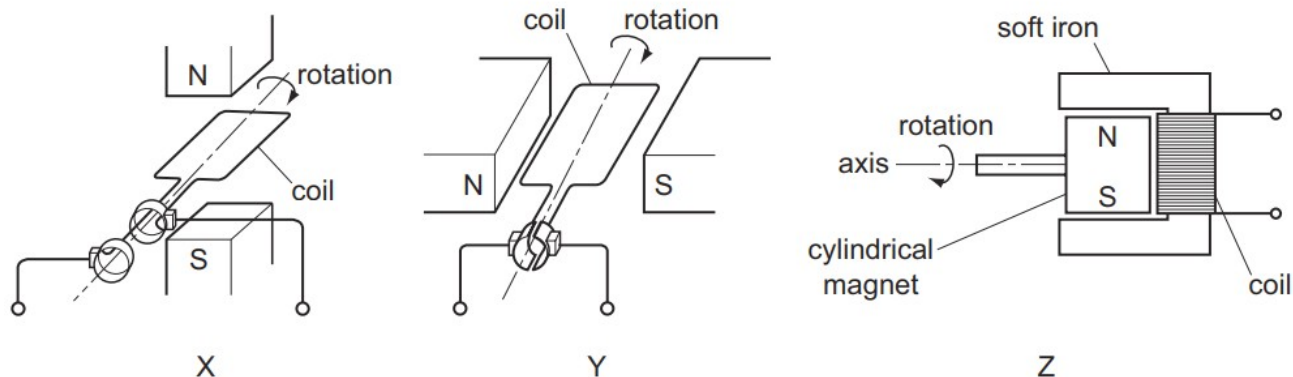
Two parallel wires carry currents in the same direction.

Which diagram shows the magnetic field around each wire and the direction of the force on each wire?



118. O/N/11/11/Q35

The diagrams show three electrical devices, X, Y and Z.



Which devices provide an alternating current (a.c.) output?

- A** X only **B** Y only **C** X and Y **D** X and Z

119. O/N/11/11/Q36

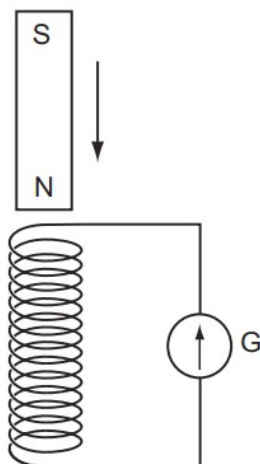
One component of a simple d.c. motor is a split-ring commutator.

Which metal is used to make the commutator, and why is this metal chosen?

	metal	reason
A	copper	it is a good conductor of electricity
B	copper	it is a good conductor of heat
C	iron	it increases the magnetic field strength
D	iron	it is attracted to the brushes

120. O/N/11/11/Q37

A small coil is connected to a galvanometer G, as shown.



When a magnet is allowed to fall towards the coil, the galvanometer pointer gives a momentary deflection to the right of the zero position.

The magnet moves through the coil.

What happens to the galvanometer pointer as the magnet falls away from the coil?

- A** It gives a continuous reading to the left.
- B** It gives a momentary deflection to the left.
- C** It gives a continuous reading to the right.
- D** It gives a momentary deflection to the right.

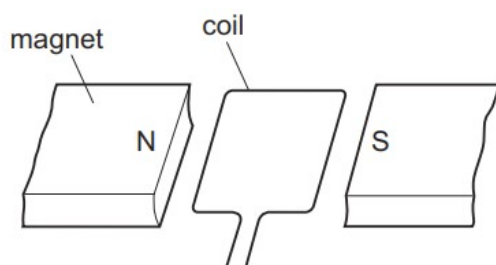
121. M/J/11/12/Q34

The electromotive force (e.m.f.) induced in a conductor moving at right-angles to a magnetic field does **not** depend upon

- A** the length of the conductor.
- B** the resistance of the conductor.
- C** the speed of the conductor.
- D** the strength of the magnetic field.

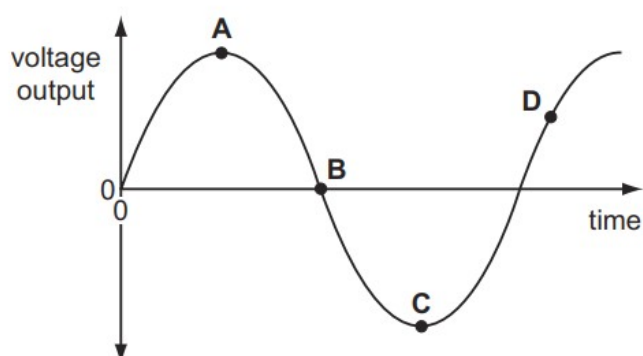
122. M/J/11/12/Q33

The diagram shows part of an a.c. generator when its coil is in a horizontal position.



The graph shows the voltage output plotted against time.

Which point on the graph shows when the coil is in a vertical position?



123. M/J/11/12/Q35

The coil in an electric motor is wound onto a cylinder.

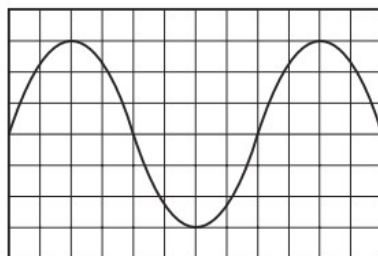
Why is the cylinder made of soft iron?

- A** to deflect the magnetic field away from the coil
- B** to increase the current through the coil
- C** to increase the strength of the magnetic field through the coil
- D** to support the coil and prevent it from collapsing

124. M/J/11/12/Q37

An alternating voltage of frequency 0.5 Hz is applied to the Y-plates of a cathode-ray oscilloscope (c.r.o.).

The diagram shows the screen of the c.r.o.

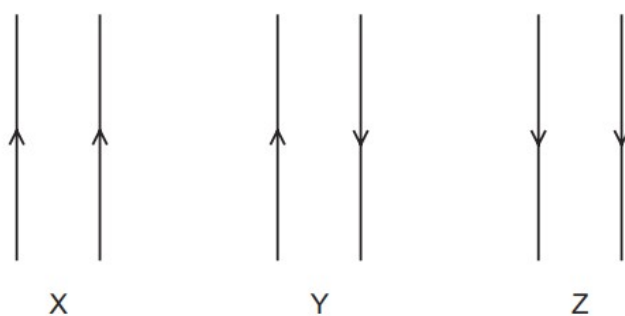


What is the time taken for the spot to cross the screen?

- A** 3 s
- B** 6 s
- C** 15 s
- D** 30 s

125. M/J/11/12/Q36

The diagram shows three pairs of parallel wires with the currents in the directions shown.



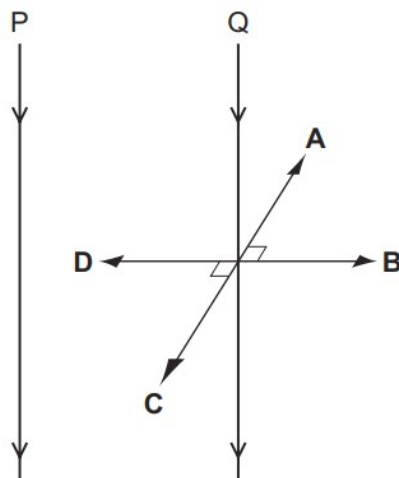
For each pair of wires, what are the forces between the wires?

	X	Y	Z
A	attraction	none	repulsion
B	attraction	repulsion	attraction
C	repulsion	attraction	repulsion
D	repulsion	repulsion	repulsion

126. O/N/10/11/Q32

Two parallel vertical wires P and Q are a small distance apart in air. There is a downwards electric current in both wires. A force acts on Q owing to the current in P. This force is perpendicular to the wire Q.

What is the direction of the force on Q?



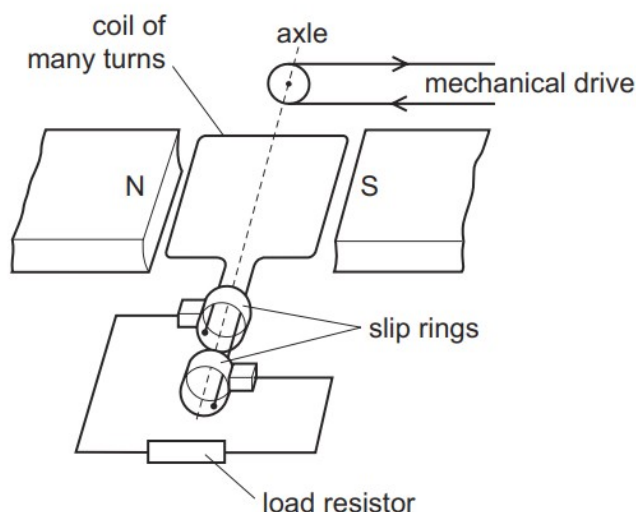
127. O/N/10/11/Q33

What does **not** alter the size of the turning effect on the coil of an electric motor?

- A the direction of the current in the coil
- B the number of turns in the coil
- C the size of the current in the coil
- D the strength of the magnetic field

128. O/N/10/11/Q34

The diagram shows an a.c. generator connected to an electrical circuit (load resistor).



Which statement is correct?

- A The direction of the potential difference across the load resistor is always the same.
- B The size of the induced e.m.f. depends on the number of turns in the coil.
- C The size of the induced e.m.f. does not change as the coil turns.
- D Winding the coil on a soft-iron cylinder makes no difference to the induced e.m.f.

129. O/N/10/11/Q35

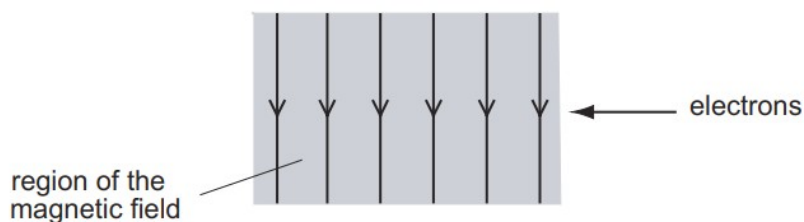
Electric power cables transmit electrical energy over large distances using a high voltage, alternating current.

What are the advantages of using a high voltage and of using an alternating current?

	advantage of using a high voltage	advantage of using an alternating current
A	a higher current is produced in the cable	the resistance of the cable is reduced
B	a higher current is produced in the cable	the voltage can be changed using a transformer
C	less energy is wasted in the cable	the resistance of the cable is reduced
D	less energy is wasted in the cable	the voltage can be changed using a transformer

130. O/N/10/11/Q36

The diagram shows a beam of electrons entering a magnetic field. The direction of the magnetic field is downwards, towards the bottom of the page.

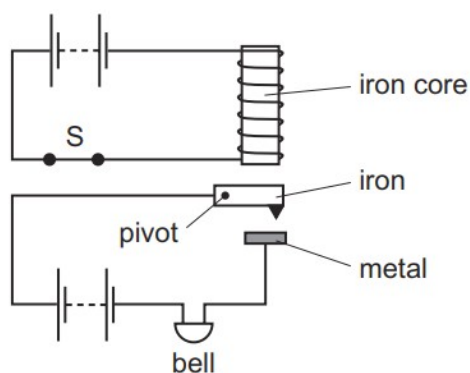


In which direction does the deflection of the electrons occur?

- A** into the page
- B** out of the page
- C** towards the bottom of the page
- D** towards the top of the page

131. O/N/10/11/Q37

The diagram shows an alarm system in which the switch S is shown closed.



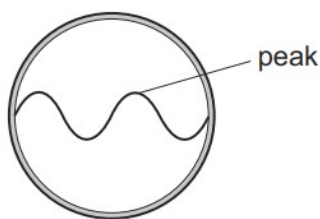
What happens when the switch S is opened?

	iron	bell
A	drops	rings
B	drops	stops ringing
C	moves up	rings
D	moves up	stops ringing

132. M/J/10/12/Q32

A cathode-ray oscilloscope is connected to an a.c. generator.

A wave is seen on the screen of the oscilloscope.



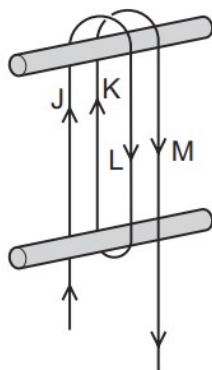
The speed of rotation of the generator is doubled.

What is the effect on the wave?

	number of peaks on the screen	amplitude of wave on the screen
A	doubled	doubled
B	doubled	same
C	same	doubled
D	same	same

133. M/J/10/12/Q33

A long flexible wire is wrapped round two wooden pegs. A large current is passed in the direction shown.



Which two pairs of lengths of wire attract each other?

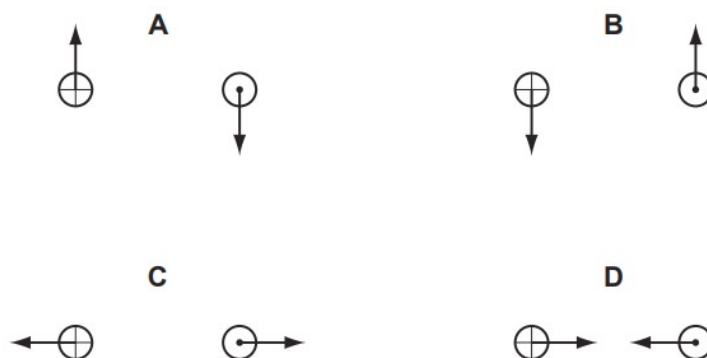
	first pair	second pair
A	J and K	K and M
B	J and K	L and M
C	J and L	K and M
D	J and L	L and M

134. O/N/09/Q31

Each diagram shows a cross-section through two parallel conductors, each carrying an electric current.

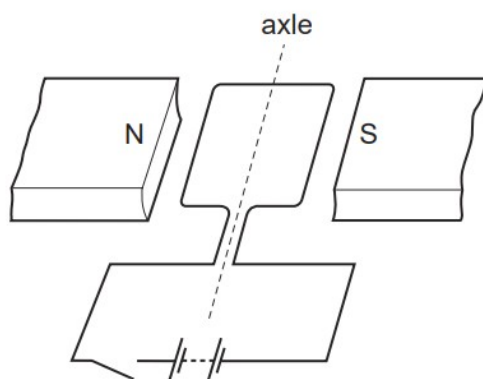
In the conductor on the left, the current is into the page; on the right, it is out of the page.

Which diagram shows the directions of the forces on the two conductors?



135. O/N/09/Q32

A simple model of a d.c. motor is made. By mistake, the split-ring commutator is left out. The coil can turn, but is always connected to the battery in the same way.



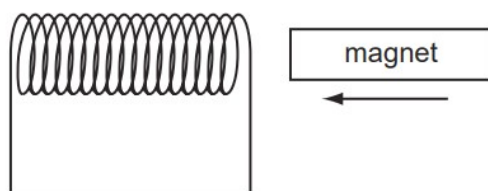
The coil starts in the horizontal position.

What happens to the coil when the circuit is switched on?

- A** It does not move at all.
- B** It moves upwards, out of the magnetic field.
- C** It turns to the vertical position and eventually stops there.
- D** It turns to the vertical position then comes back to the horizontal position.

136. O/N/09/Q33

A magnet is pushed horizontally towards a coil of wire, inducing an e.m.f. in the coil.



In which direction does the induced e.m.f. make the coil move?

- A away from the magnet
- B towards the magnet
- C downwards
- D upwards

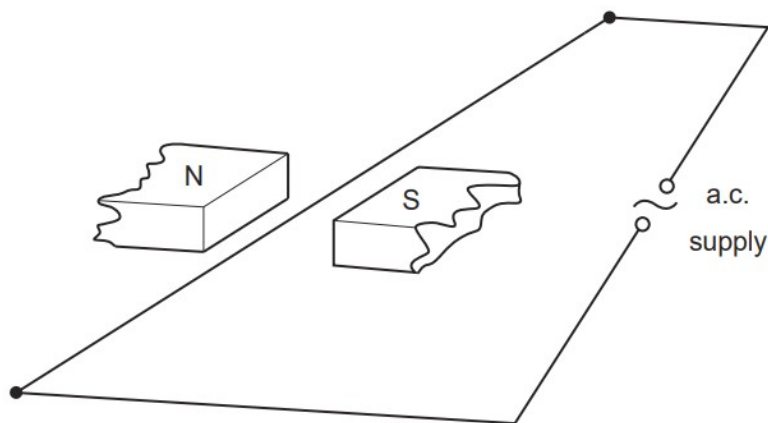
137. O/N/09/Q35

Why is a transformer used to connect a generator in a power station to a long-distance transmission line?

- A to decrease the voltage and decrease the current
- B to decrease the voltage and increase the current
- C to increase the voltage and decrease the current
- D to increase the voltage and increase the current

138. M/J/09/Q31

An a.c. supply is connected to a wire stretched between the poles of a magnet.

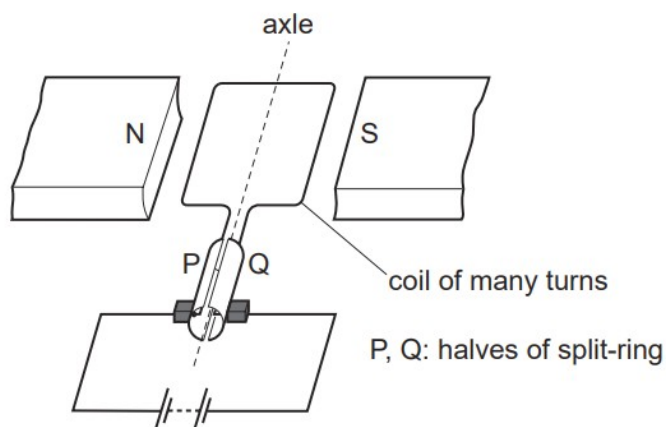


Which way will the wire move?

- A left and right
- B right only
- C up and down
- D up only

139. M/J/09/Q32

A d.c. motor consists of a coil of many turns rotating in a fixed magnetic field. The coil is connected to a d.c. supply through a split-ring commutator.



Some changes are made, one at a time.

- The d.c. supply is reversed.
- The coil is turned before switching on, so that P starts on the right and Q on the left.
- The poles of the magnet are reversed.
- The turns on the coil are increased in number.

How many of these changes make the coil rotate in the opposite direction?

- A** 1 **B** 2 **C** 3 **D** 4

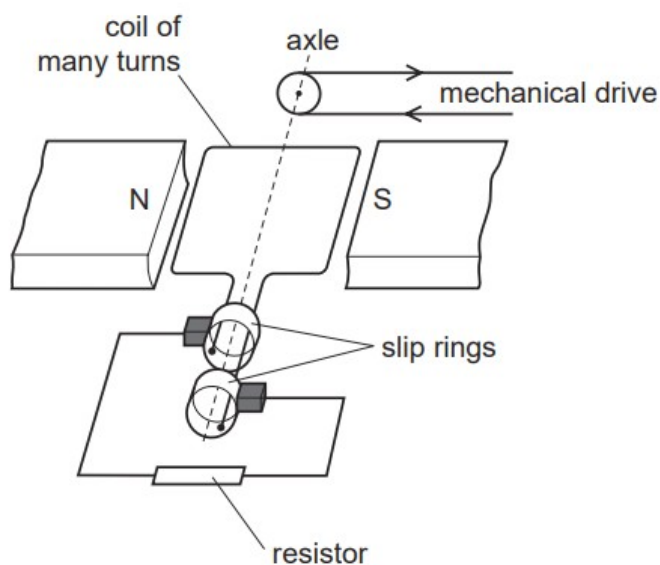
140. M/J/09/Q35

Why is electrical energy usually transmitted at high voltage?

- A** As little energy as possible is wasted in the transmission cables.
- B** The current in the transmission cables is as large as possible.
- C** The resistance of the transmission cables is as small as possible.
- D** The transmission system does not require transformers.

141. M/J/09/Q33

The diagram shows an a.c. generator connected to a resistor.



Some changes are made, one at a time.

- The speed of the drive is changed.
- The strength of the magnets is changed.
- The number of turns in the coil is changed.
- The value of the resistor is changed.

How many of these alter the value of the e.m.f. generated in the coil?

A 1

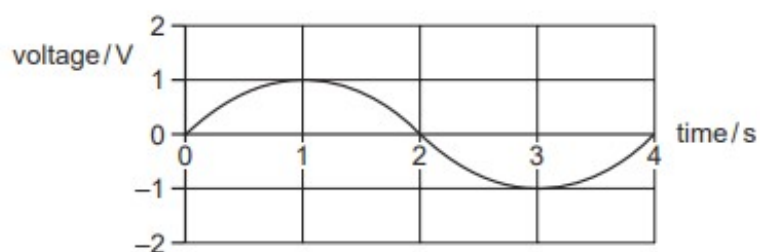
B 2

C 3

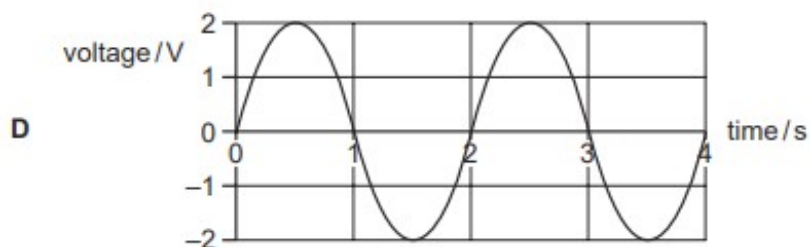
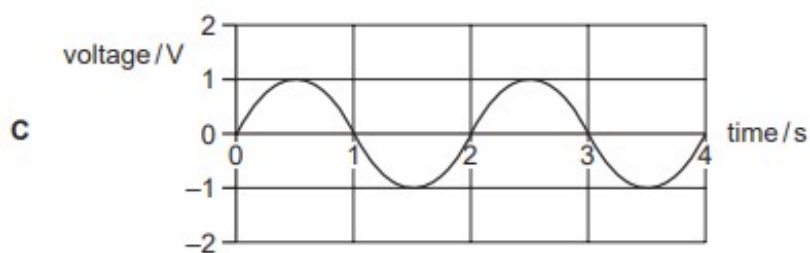
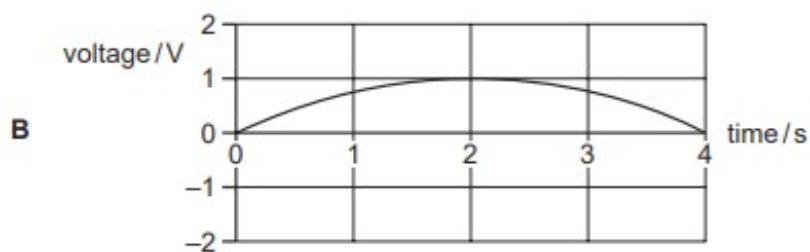
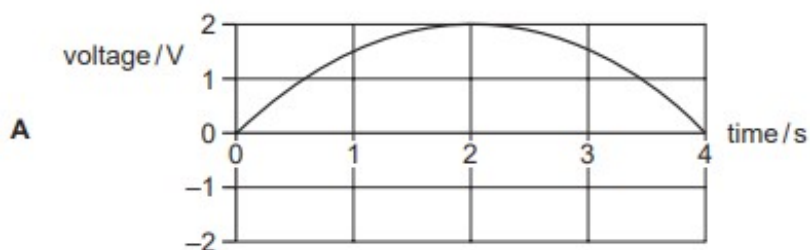
D 4

142. M/J/09/Q34

A simple a.c. generator produces a voltage that varies with time as shown.

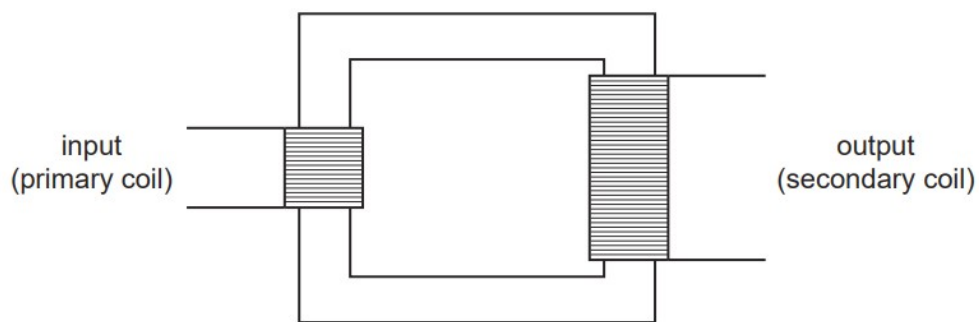


Which graph shows how the voltage varies with time when the generator rotates at twice the original speed?



143. O/N/08/Q36

The diagram shows a working transformer.

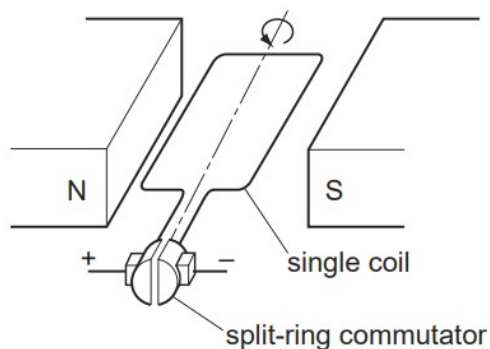


Which statement is correct?

- A** The input voltage is d.c.
- B** The input voltage is greater than the output voltage.
- C** The input voltage is less than the output voltage.
- D** The input voltage is the same as the output voltage.

144. M/J/08/Q33

The diagram shows a simple electric motor.



The split-ring commutator reverses the current in the coil as the coil rotates.

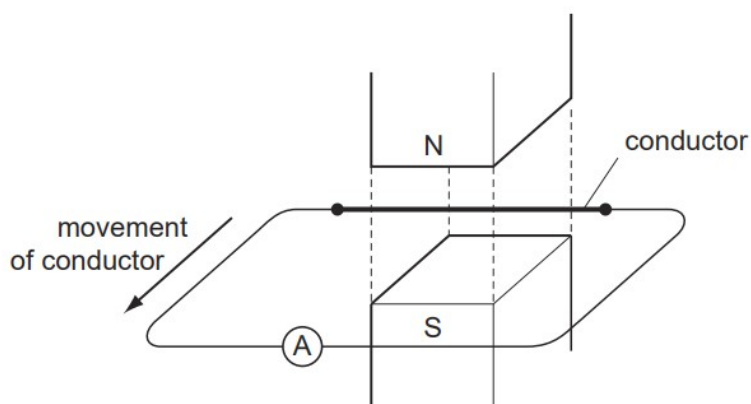
The coil is rotated 360° from the position shown.

How many times is the current in the coil reversed?

- A** 1
- B** 2
- C** 3
- D** 4

145. M/J/08/Q34

A conductor is moving horizontally across a vertical magnetic field.



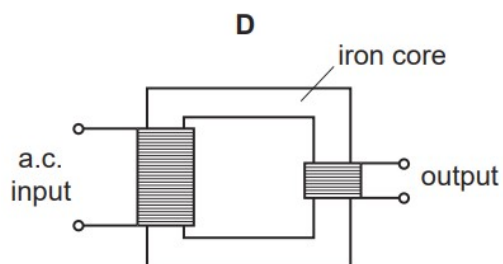
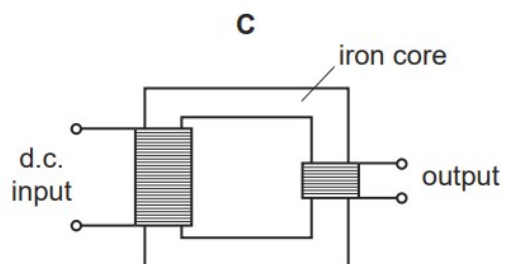
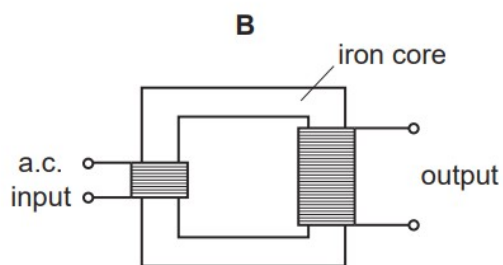
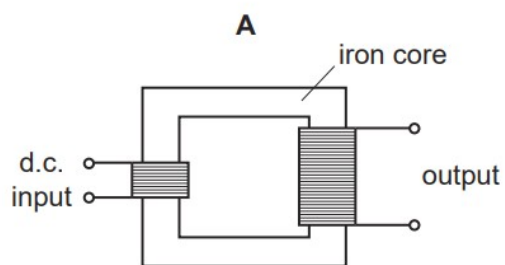
An e.m.f. is induced in the conductor. No deflection is seen on the ammeter.

What is the reason for this?

- A** The ammeter is not between the poles.
- B** The conductor is moving too slowly.
- C** The conductor is not cutting field lines.
- D** The poles are too close together.

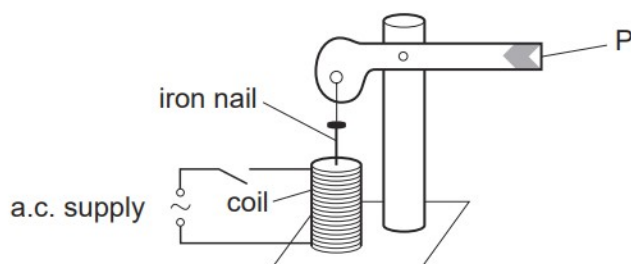
146. M/J/08/Q36

Which transformer arrangement produces an output voltage that is larger than the input voltage?



147. O/N/07/Q24

The diagram shows a model railway signal.



What does the end P do when the switch is closed?

- A It goes down and stays down.
- B It goes up and stays up.
- C It goes down and then returns to its original position.
- D It goes up and then returns to its original position.

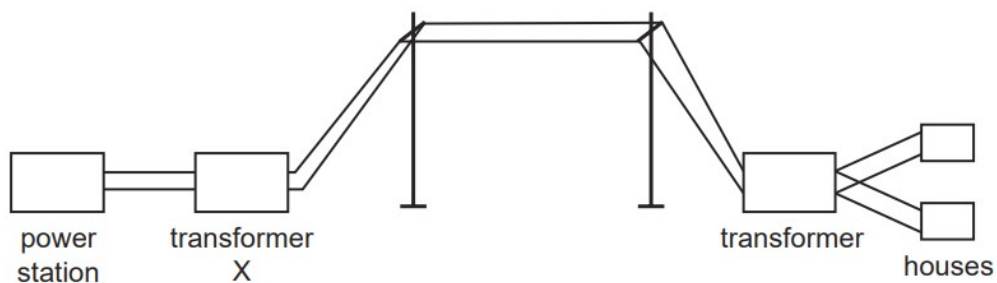
148. O/N/07/Q25

Which part of a video tape recording system does **not** rely on magnetic material for its operation?

- A the drive motor
- B the power lead
- C the transformer
- D the video tape

149. O/N/07/Q36

The diagram shows transformers and cables used to transmit electrical energy over long distances.



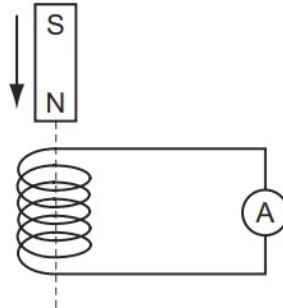
How does transformer X affect the voltage and the current from the power station?

	voltage	current
A	decreases it	decreases it
B	decreases it	increases it
C	increases it	decreases it
D	increases it	increases it

150. O/N/07/Q35

A small coil is connected to a sensitive ammeter. The ammeter needle can move to either side of the zero position.

As the magnet falls towards the coil, the ammeter needle moves quickly to the right of the zero position.



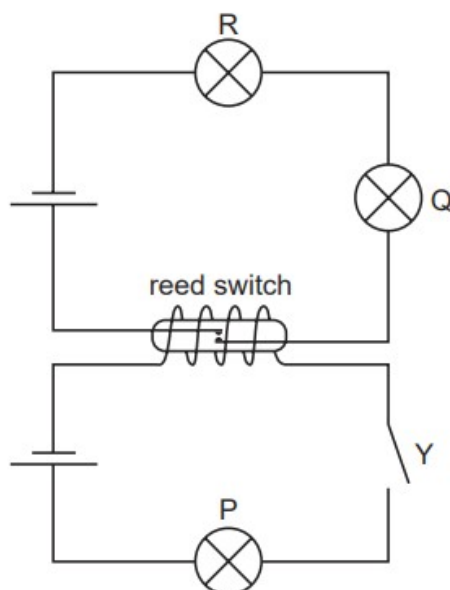
The magnet moves through the coil.

How does the ammeter needle move as the magnet falls away from the coil?

- A** It does not move.
- B** It gives a steady reading to the right.
- C** It moves quickly to the left of the zero position and then returns to zero.
- D** It moves quickly to the right of the zero position and then returns to zero.

151. O/N/07/Q37

In the circuit shown, all lamps are identical and all cells are identical. The resistance of the coil of the reed switch is negligible.



One cell lights one lamp to normal brightness.

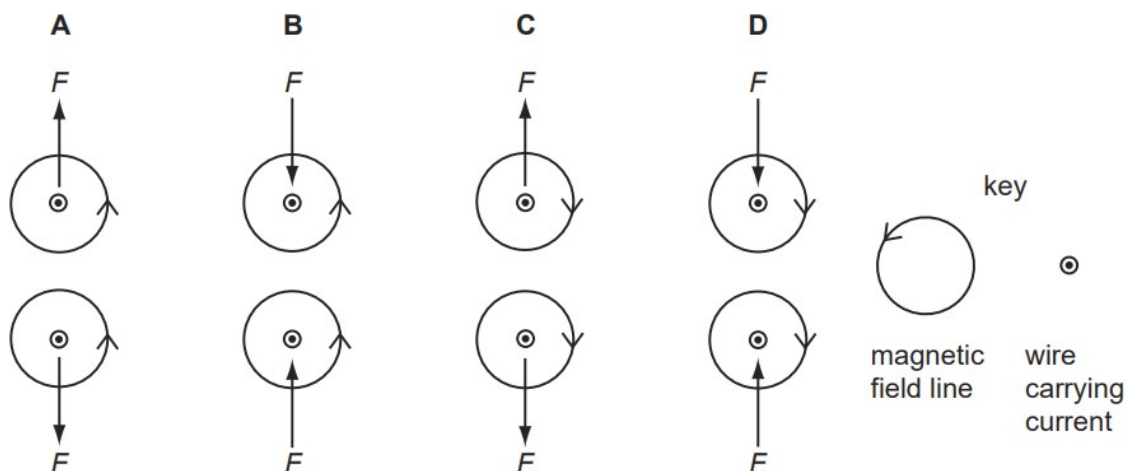
What is the brightness of the lamps when switch Y is closed?

	P	Q	R
A	dim	dim	dim
B	normal	dim	dim
C	normal	off	off
D	off	normal	normal

152. M/J/07/Q33

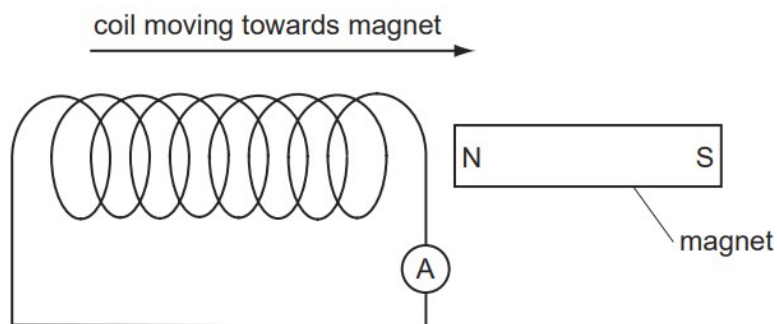
The diagrams show the forces F between two wires carrying currents out of the page. The magnetic fields close to the wires are also shown.

Which diagram is correct?



153. M/J/07/Q35

The diagram shows how a magnet and a coil may be used to induce an electric current.

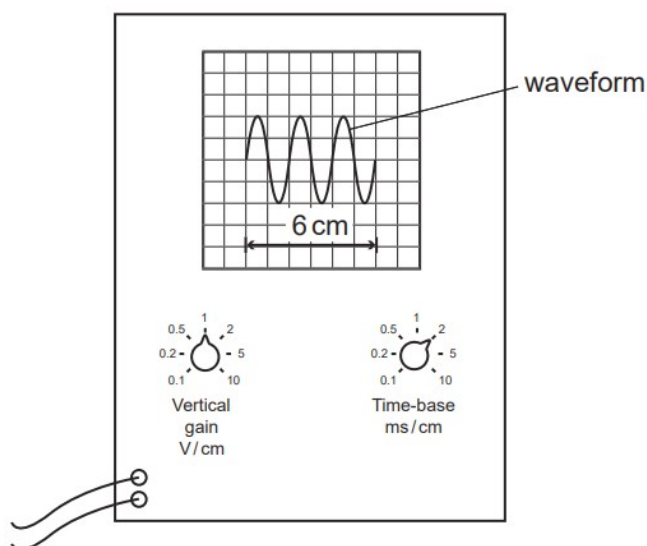


How could the ammeter reading be increased?

- A** Move the coil more slowly.
- B** Put a resistor in series with the ammeter.
- C** Turn the magnet round, then move the coil.
- D** Use a coil with more turns.

154. M/J/07/Q37

A waveform is displayed on a cathode-ray oscilloscope. The length of three cycles of the waveform is 6 cm. The vertical gain and the time-base settings are shown on the diagram.

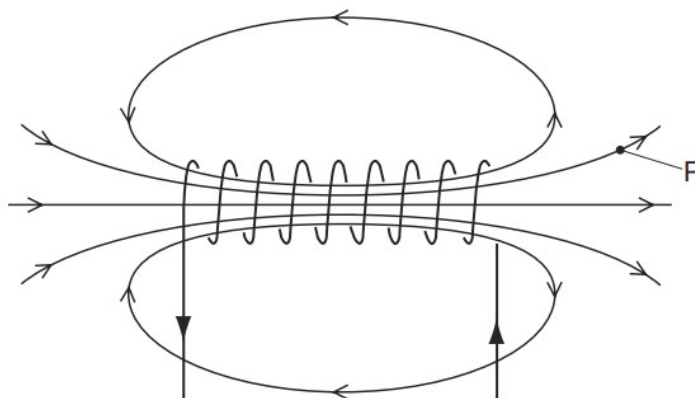


What is the time taken for one cycle of the waveform?

- A** 1 ms
- B** 2 ms
- C** 4 ms
- D** 6 ms

155. O/N/06/Q24

A current in a solenoid creates a magnetic field.

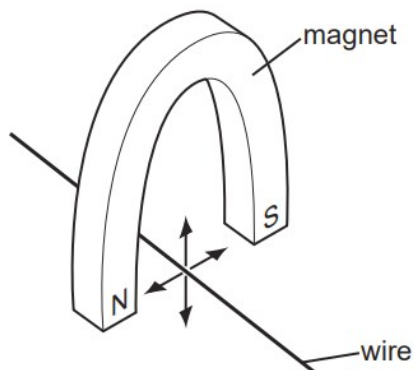


What is the effect on the magnetic field at the point P of using a larger current in the opposite direction?

	field strength	field direction
A	decreases	reverses
B	decreases	unchanged
C	increases	reverses
D	increases	unchanged

156. O/N/06/Q31

A copper wire is held between the poles of a magnet.



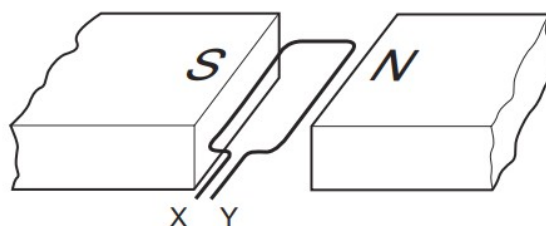
The current in the wire can be reversed. The poles of the magnet can also be changed over.

In how many of the four directions shown can the force act on the wire?

- A** 1 **B** 2 **C** 3 **D** 4

157. O/N/06/Q32

The diagram shows a coil in a magnetic field.

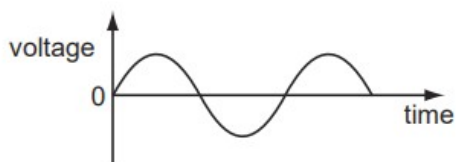


When the coil is part of a d.c. motor, what must be connected directly to X and Y?

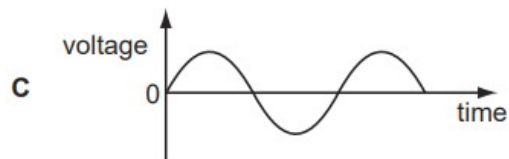
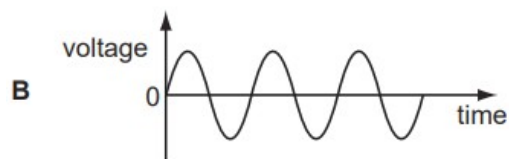
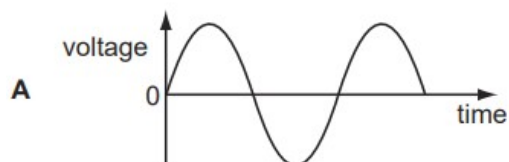
- A d.c. supply
- B slip rings
- C soft-iron core
- D split-ring commutator

158. O/N/06/Q33

A transformer has more turns on the secondary coil than on the primary. The graph shows how the input voltage varies with time.



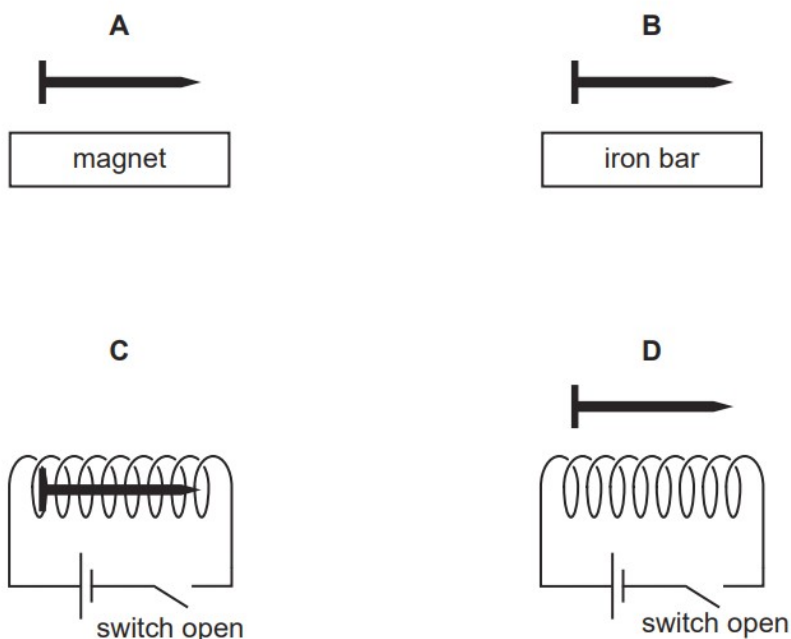
Which graph, drawn to the same scale as the input graph, shows how the output voltage varies with time?



159. M/J/06/Q27

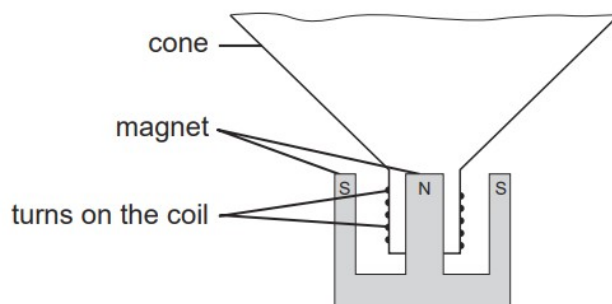
The diagrams show an iron nail in four different situations.

In which diagram will the nail become an induced magnet?



160. M/J/06/Q28

The diagram shows parts of a loudspeaker.



Which type of current is passed through the coil and why?

	current passed through coil	reason why
A	alternating	to keep the magnetic field constant
B	alternating	to make the coil vibrate
C	direct	to keep the magnetic field constant
D	direct	to make the coil vibrate

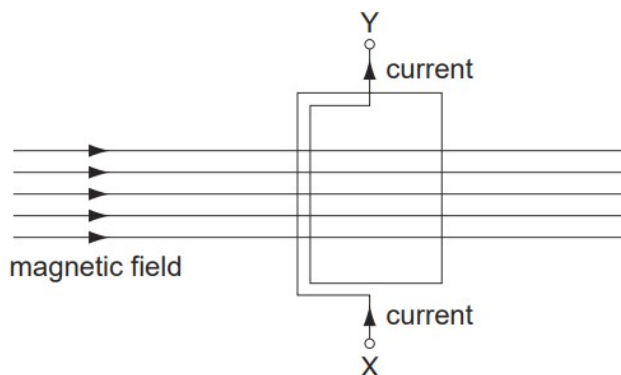
161. M/J/06/Q36

Why is a transformer used to connect a generator in a power station to a long distance transmission line?

- A** to decrease the voltage and decrease the current
- B** to decrease the voltage and increase the current
- C** to increase the voltage and decrease the current
- D** to increase the voltage and increase the current

162. O/N/05/Q32

A coil, carrying a current, is arranged within a magnetic field. The coil experiences forces that can make it move.

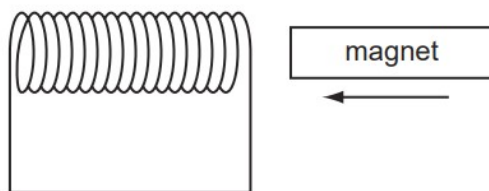


In which direction does the coil move?

- A** along the magnetic field
- B** from X to Y
- C** out of the paper
- D** turns about the axis XY

163. O/N/05/Q33

A magnet is pushed horizontally towards a coil of insulated wire, inducing an e.m.f. in the coil.

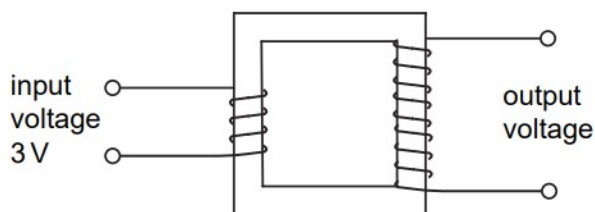


In which direction does the induced e.m.f. make the coil move?

- A** away from the magnet
- B** towards the magnet
- C** downwards
- D** upwards

164. O/N/05/Q34

A step-up transformer with 100 % efficiency has an input voltage of 3 V and an input current of 2 A.



Under these conditions, what output voltage and output current could be obtained?

	output voltage / V	output current / A
A	1	6
B	2	3
C	4	1
D	6	1

165. M/J/05/Q32

X and **Y** are wires carrying electric currents at right angles to the page. **P**, **Q** and **R** are plotting compasses. Any effect of the Earth's magnetic field has been ignored.



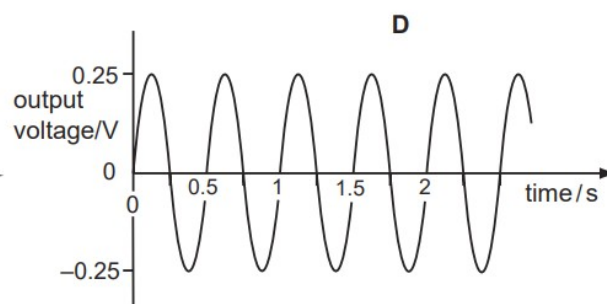
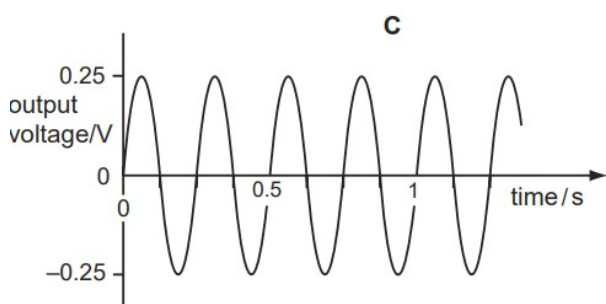
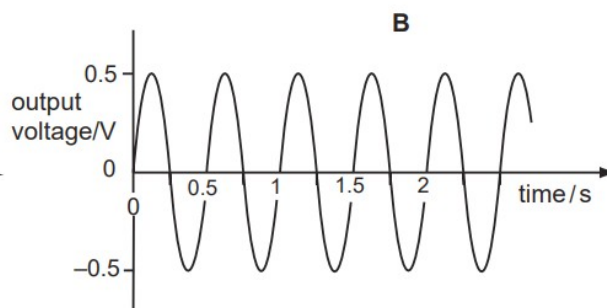
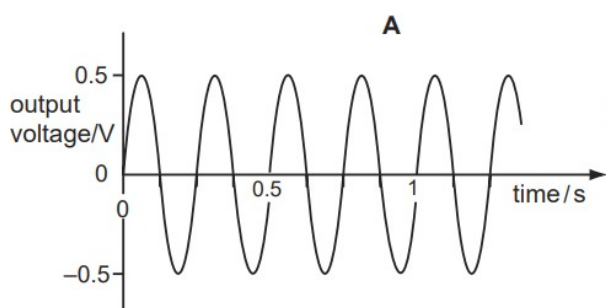
What is true about the direction and size of the currents?

	direction of currents	size of currents
A	same	larger in X than in Y
B	same	smaller in X than in Y
C	different	larger in X than in Y
D	different	smaller in X than in Y

166. M/J/05/Q33

A girl turns the handle of a small a.c. generator four times each second. The generator produces a maximum output voltage of 0.5 V.

Which of the following graphs best shows this?

**167. M/J/05/Q34**

Which statement about the action of a transformer is correct?

- A** An e.m.f. is induced in the secondary coil when an alternating voltage is applied to the primary coil.
- B** An e.m.f. is induced in the secondary coil when there is a steady direct current in the primary coil.
- C** The current in the secondary coil is always larger than the current in the primary coil.
- D** The voltage in the secondary coil is always larger than the voltage in the primary coil.

168. M/J/05/Q35

Electrical energy is transmitted at high alternating voltages.

What is **not** a valid reason for doing this?

- A** At high voltage, a.c. is safer than d.c.
- B** For a given power, there is a lower current with a higher voltage.
- C** There is a smaller power loss at higher voltage and lower current.
- D** The transmission lines can be thinner with a lower current.

ANSWERS

1. A	48. C	97. D	146. B
2. D	49. A	98. A	147. B
3. A	50. B	99. B	148. B
4. C	51. A	100. C	149. C
5. B	52. A	101. D	150. C
6. D	53. A	102. D	151. B
7. A	54. D	103. B	152. B
8. C	55. C	104. D	153. D
9. C	56. C	105. D	154. C
10. C	57. B	106. C	155. C
11. D	58. D	107. C	156. B
12. B	59. A	108. B	157. D
13. B	60. D	109. B	158. A
14. B	61. A	110. D	159. A
15. A	62. D	111. B	160. B
16. D	63. A	112. B	161. C
17. C	64. B	113. A	162. D
18. D	65. B	114. D	163. A
19. D	66. D	115. B	164. D
20. C	67. A	116. C	165. A
21. D	68. B	117. A	166. A
22. C	69. D	118. D	167. A
23. D	70. A	119. A	168. A
24. C	71. D	120. B	
25. A	72. B	121. B	
26. C	73. D	122. B	
27. D	74. B	123. C	
28. C	75. D	124. A	
29. A	76. A	125. B	
30. D	77. D	126. D	
31. D	78. C	127. A	
32. B	79. A	128. B	
33. D	80. B	129. D	
34. D	81. B	130. A	
35. A	82. D	131. A	
36. C	83. C	132. A	
37. B	84. A	133. B	
38. D	85. B	134. C	
39. B	86. D	135. C	
40. C	87. C	136. A	
41. B	88. C	137. C	
42. D	89. B	138. C	
43. B	90. C	139. B	
44. C	91. A	140. A	
45. C	92. C	141. C	
46. C	93. D	142. D	
47. D	94. D	143. C	
	95. A	144. B	
	96. A	145. B	

1. O/N/23/22/Q8

Fig. 8.1 shows a permanent magnet attached to one end of an unstretched spring and held at rest just above a solenoid.

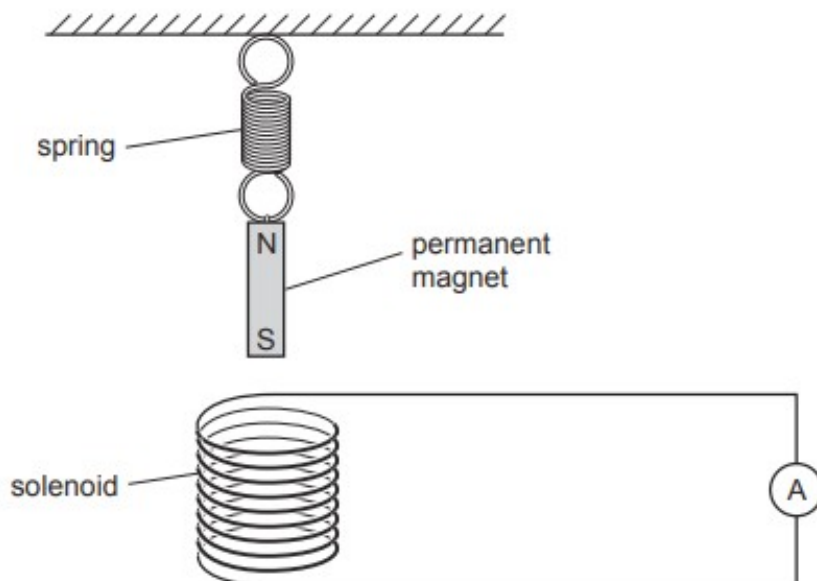


Fig. 8.1

The solenoid is connected to a sensitive ammeter.

- (a) State the name of a substance from which permanent magnets are made.

..... [1]

- (b) The magnet is released and moves downwards towards the solenoid.

The ammeter shows that there is a current in the solenoid when the magnet moves.

- (i) Explain why there is a current in the solenoid.

.....
.....
.....
.....
..... [3]

- (ii) The current causes the top end of the solenoid to become the S pole of an electromagnet.

The bottom end of the permanent magnet is an S pole.

Explain why the top end of the solenoid becomes an S pole as the magnet moves downwards towards the solenoid.

.....

.....

..... [2]

- (c) The magnet oscillates up and down.

Describe what happens to the current in the solenoid.

.....

.....

..... [2]

[Total: 8]

2. O/N/23/21/Q7

An alternating current (a.c.) generator is used in a remote location to supply electricity.

(a) Fig. 7.1 shows the structure of a simple a.c. generator with the coil horizontal.

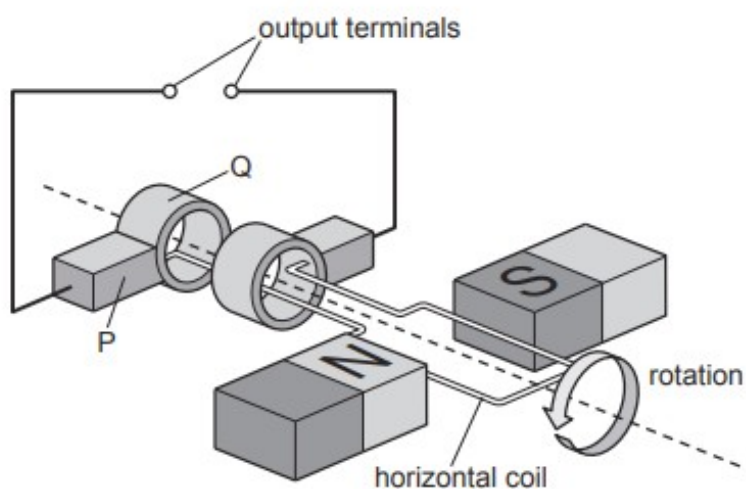


Fig. 7.1

(i) State the name of the components labelled P and Q on Fig. 7.1.

P

Q

[1]

(ii) When the generator is producing an electrical output, P is stationary and Q is rotating.

Explain the purpose of P and Q.

.....

.....

..... [2]

- (b) The output of the generator is connected to an oscilloscope.

On the oscilloscope the Y-gain is set at 5.0 V/cm and the timebase setting is 2.0 ms/cm .

Fig. 7.2 shows the trace on the screen of the oscilloscope.

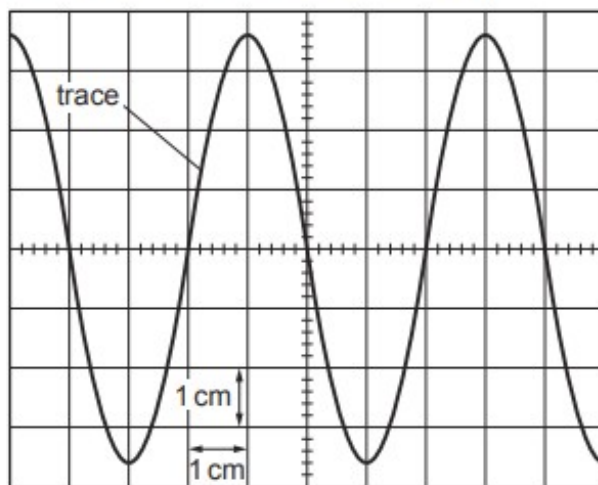


Fig. 7.2

- (i) Write a letter X on the trace shown in Fig. 7.2 to indicate one point when the coil is vertical. [1]
- (ii) Using Fig. 7.2, determine the maximum value of the e.m.f. produced by the generator.

maximum e.m.f. = V [2]

- (iii) Using Fig. 7.2, determine, in milliseconds, the time it takes for one revolution of the coil.

time for one revolution of the coil = ms [1]

- (iv) Determine the frequency of the output of the generator.

frequency = Hz [2]

[Total: 9]

3. M/J/23/22/Q8

Fig. 8.1 shows a step-down transformer used to operate an electric bell.

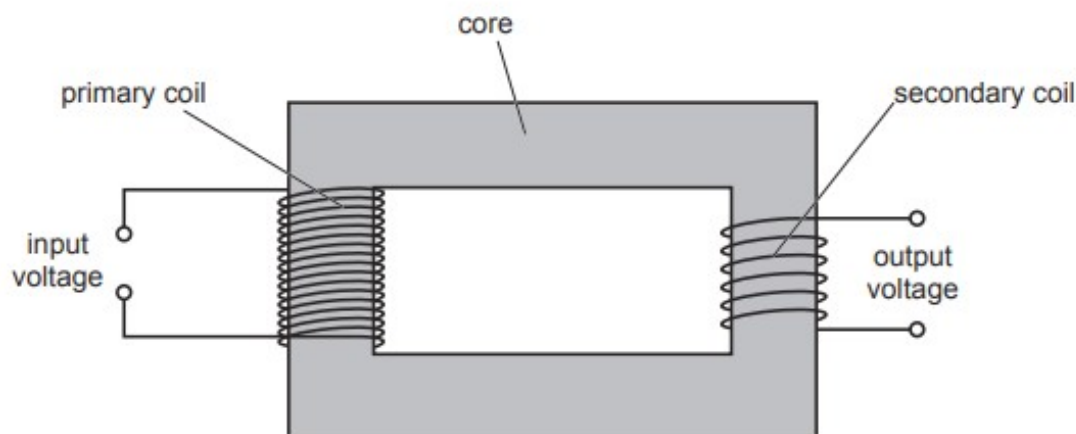


Fig. 8.1

- (a) State the material used for the core of the transformer.

..... [1]

- (b) A current in the primary coil produces a magnetic field in the core.

Explain how an alternating voltage is produced in the secondary coil.

.....

 [3]

- (c) The transformer has 4600 turns on the primary coil which is connected to a mains supply of 230 V.

An output of 5.0 V is used to operate the bell.

Calculate the number of turns needed on the secondary coil.

number of turns = [2]

- (d) State one change that can be made to the transformer shown in Fig. 8.1 so that it can be used as a step-up transformer.

.....
 [1]

4. M/J/23/21/Q9

Fig. 9.1 shows a simple a.c. generator.

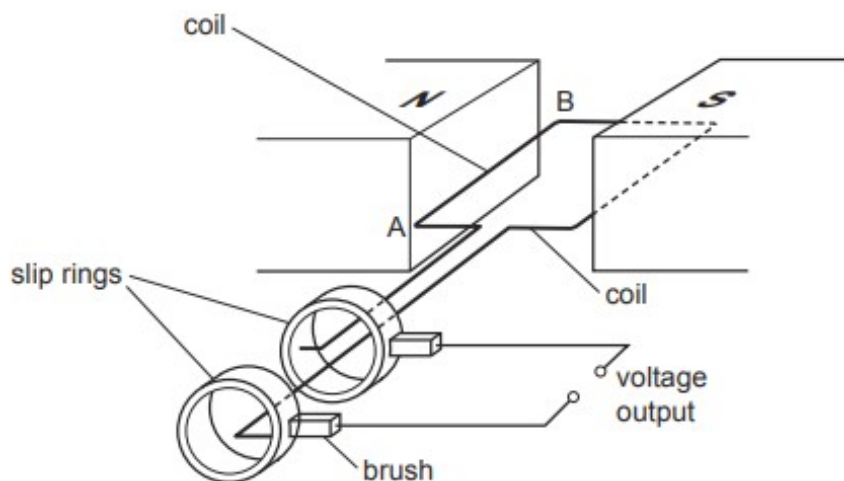


Fig. 9.1

- (a) Explain why there is a voltage induced in the coil when the coil is turned.

.....

.....

.....

..... [2]

- (b) In Fig. 9.1 the coil is horizontal, with side AB on the left. The output voltage is +6.0V.

On Fig. 9.2 draw a line from each of the shaded boxes to one of the circled voltages to show the voltage output when the coil is in different positions.

One line has been drawn for you.

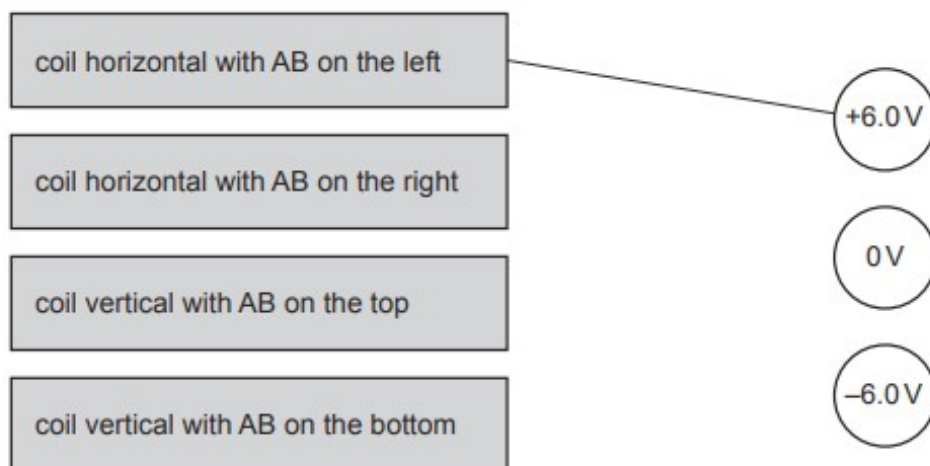


Fig. 9.2

[2]

(c) Both an a.c. generator and a d.c. motor contain a coil and brushes.

(i) Fig. 9.1 shows how the brushes are connected to the coil in an a.c. generator.

Draw a diagram to show how the brushes are connected to the coil in a d.c. motor.

[2]

(ii) State why there are forces on the sides of the coil in a d.c. motor.

.....

..... [1]

[Total: 7]

5. O/N/22/22/Q6

The primary coil of a transformer is connected to the mains supply. The voltage of the a.c. mains supply is 240 V.

Fig. 6.1 is a diagram of the arrangement.

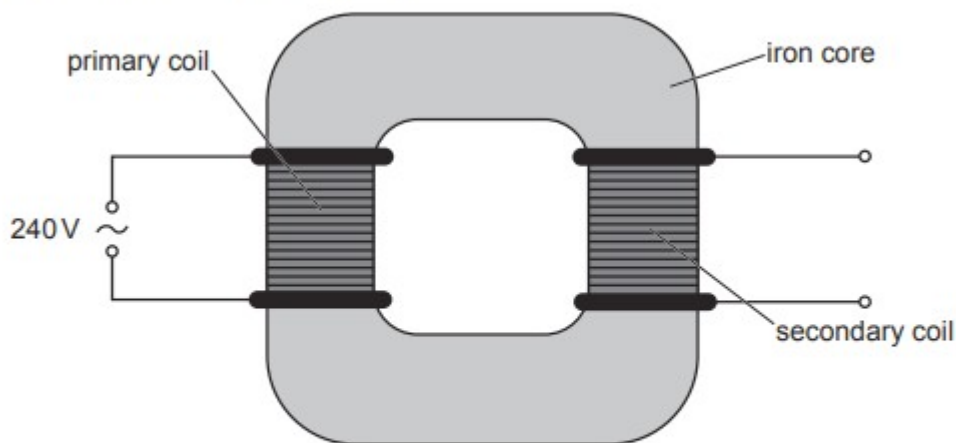


Fig. 6.1

(a) Explain why a voltage is produced in the secondary coil.

.....

.....

.....

..... [3]

(b) There are 5600 turns on the primary coil of the transformer and 350 turns on the secondary coil.

(i) Calculate the output voltage of the transformer.

output voltage = [2]

(ii) The output of the transformer is connected to a 90 W filament lamp which operates at normal brightness.

Calculate the current in the lamp.

current = [2]

6. O/N/22/21/Q6

An electromagnet is used to separate objects that are magnetic from objects that are non-magnetic.

Fig. 6.1 shows the electromagnet suspended from the arm of a crane.

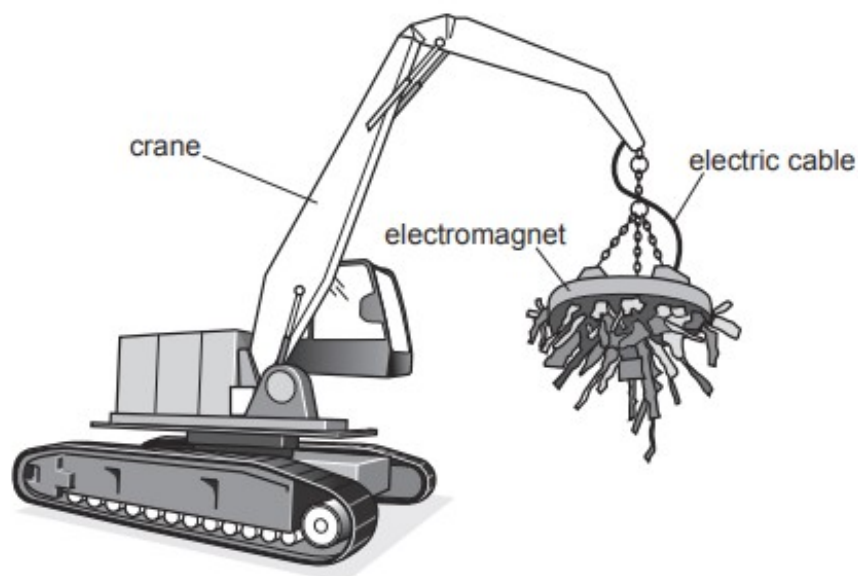


Fig. 6.1

- (a)** State the name of the metal from which the core of the electromagnet is made.

..... [1]

- (b)** The electromagnet is powered by a 220V d.c. (direct current) supply.

The electromagnet is switched on and the current in the circuit is 39A.

- (i)** Calculate the power transferred to the electromagnet.

power = [2]

(ii) The d.c. supply is a set of batteries.

The batteries power the magnet for 4.5 hours before they need to be replaced.

Calculate the charge driven around the complete circuit in this time.

charge = [3]

(c) On a very cold morning, when the electromagnet is first switched on, the current is greater than 39A before decreasing to the usual value.

Explain why the current is initially greater than 39A.

.....

..... [1]

[Total: 7]

7. O/N/22/21/Q9

The power supply in an electric circuit is a battery of electromotive force (e.m.f.) 12 V.

- (a) State **two** ways in which the e.m.f. of a battery differs from that of an alternating current (a.c.) power supply.

1

2

[2]

- (b) The circuit includes three resistors and two open switches, S_1 and S_2 .

Fig. 9.1 shows the circuit.

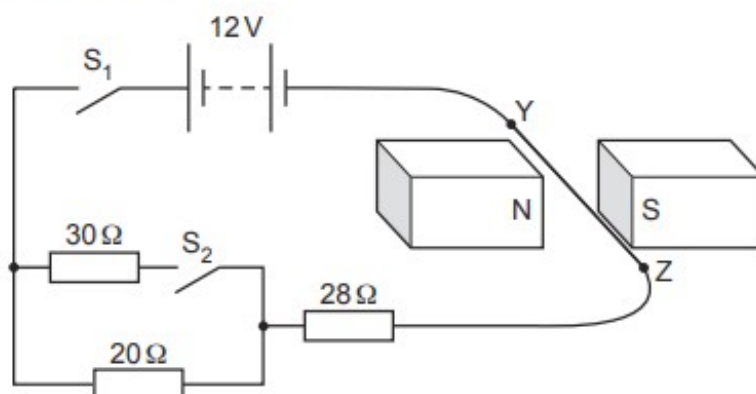


Fig. 9.1

YZ is a straight, horizontal section of connecting wire that lies between two magnets.

S_1 is now closed.

- (i) Calculate the current in YZ.

current = [2]

- (ii) Explain why YZ experiences a force.

.....

.....

..... [2]

(iii) Tick the box which describes the direction of the force on YZ.

towards N ☐

towards Z ☐

towards S ☐

downwards ☐

towards Y ☐

upwards ☐

[1]

(iv) Explain how the direction of the force on YZ is determined.

.....

 [2]

(c) Switch S_2 in the circuit in Fig. 9.1 is now also closed.

(i) Calculate the total resistance of the circuit.

resistance = [3]

(ii) Explain what happens to the force on YZ as switch S_2 is closed.

.....

 [2]

(iii) The current in the $20\ \Omega$ resistor is I_{20} . The current in the $30\ \Omega$ resistor is I_{30} .

State a value for the ratio I_{20}/I_{30} .

ratio $I_{20}/I_{30} =$ [1]

8. M/J/22/22/Q6

- (a) Fig. 6.1 shows part of a toy which contains two ring-shaped, permanent magnets. A plastic rod passes through the centre of both magnets.

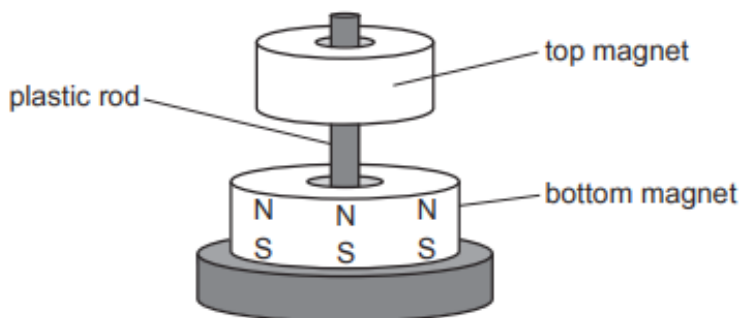


Fig. 6.1

The top magnet can move up and down freely around the plastic rod.

The magnetic poles on the bottom magnet are shown in Fig. 6.1.

- (i) The top magnet floats in the air above the bottom magnet.

On Fig. 6.1, mark the poles on the top magnet and explain why it floats in the air above the bottom magnet.

.....

.....

.....

..... [2]

- (ii) The top magnet is replaced with a ring made of iron.

Explain why the iron ring sticks to the bottom magnet.

.....

.....

.....

..... [2]

(b) A wire carrying a current passes at right angles through a piece of paper.

Fig. 6.2 shows a cross and circle where the current in the wire passes into the plane of the paper.

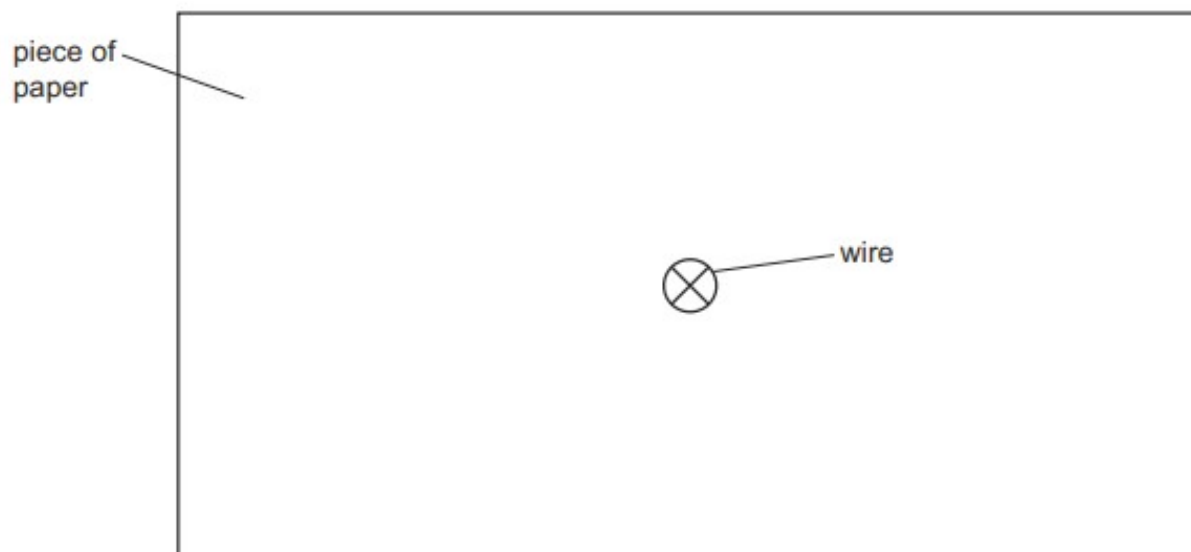


Fig. 6.2

On Fig. 6.2, sketch **three** magnetic field lines to show the magnetic field pattern around the wire.

Show the direction of the field on your sketch.

[2]

[Total: 6]

9. M/J/22/21/Q5

Fig. 5.1 shows a simple d.c. motor used in a toy car.

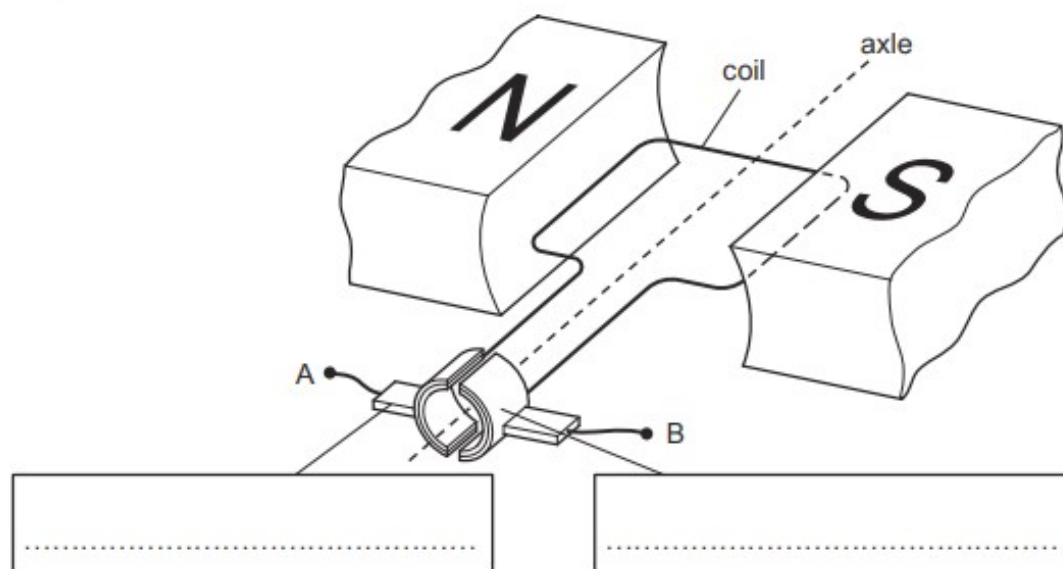


Fig. 5.1

Terminals A and B are connected to a battery and the motor rotates.

(a) On Fig. 5.1, add labels inside the boxes to identify the parts of the motor indicated. [2]

(b) State what happens to the rotation of the coil when:

(i) the number of turns on the coil is increased

..... [1]

(ii) the magnetic field between the poles of the magnet is reversed.

..... [1]

- (c) The power supply to the motor is switched on and off at a steady rate.

Fig. 5.2 shows how the speed of the toy car varies with time as a result of the power supply being switched on and off.

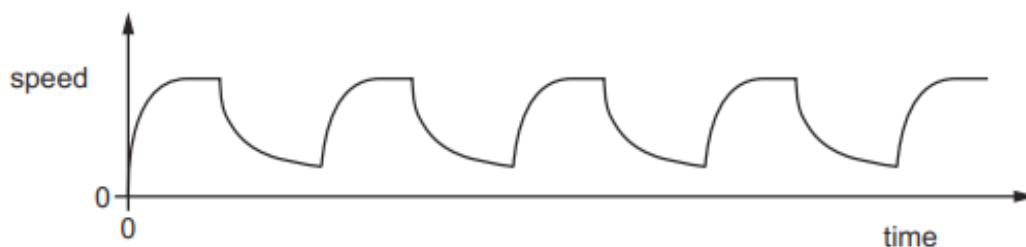


Fig. 5.2

- (i) Describe the motion of the toy car.

.....

.....

.....

..... [2]

- (ii) The voltage supplied to the motor is switched on for a longer period of time and off for a longer period of time, at a steady rate.

Suggest what happens to the motion of the toy car.

.....

.....

..... [1]

[Total: 7]

10. O/N/21/21/Q7

Fig. 7.1 shows a solenoid connected to an ammeter.

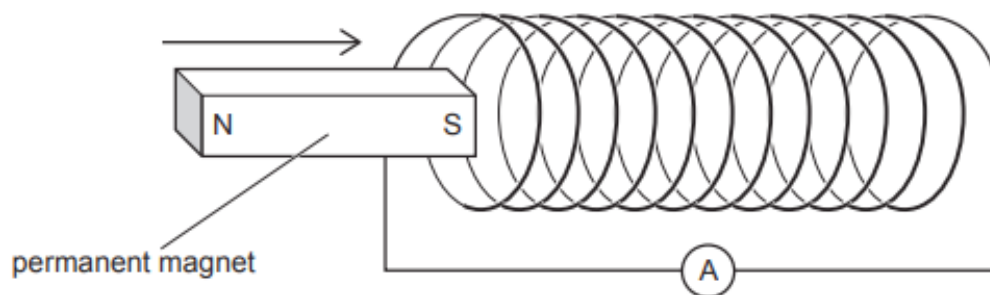


Fig. 7.1

The S-pole of a permanent magnet moves into the left-hand end of the solenoid.

The ammeter reading shows that there is a small positive current in the circuit.

(a) Explain why there is a current in the circuit when the magnet moves.

.....

.....

.....

..... [3]

(b) When the magnet is inside the solenoid, it stops moving. It is then pulled back out of the solenoid.

Explain what happens to the ammeter reading as the magnet moves out of the left-hand end of the solenoid.

.....

.....

..... [2]

[Total: 5]

11. M/J/21/22/Q7

Fig. 7.1 shows part of a simple d.c. electric motor.

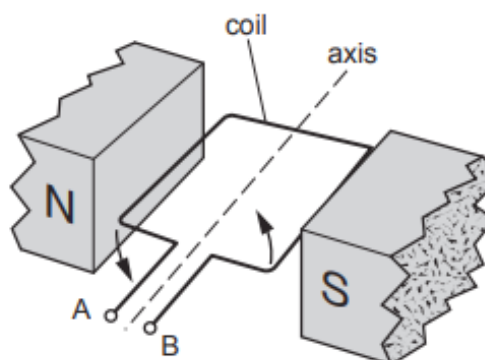


Fig. 7.1

The motor is connected to a battery with the positive terminal of the battery connected to terminal A.

(a) Explain why the coil turns in the direction shown.

.....

.....

.....

.....

..... [3]

(b) The turning effect is increased when the coil is wound around a soft-iron cylinder.

(i) Explain why this happens.

.....

.....

..... [2]

(ii) Suggest **one** other way to increase the turning effect of the motor.

..... [1]

[Total: 6]

12. M/J/21/21/Q6

A student moves a metal bar upwards between the poles of a magnet, as shown in Fig. 6.1.

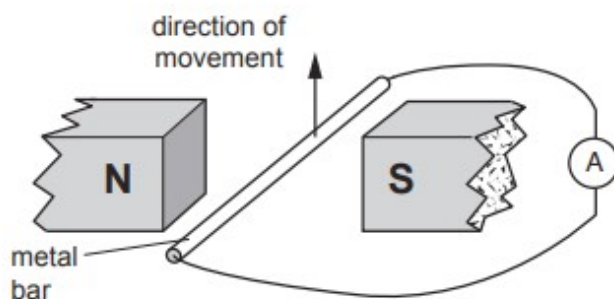


Fig. 6.1

The ammeter connected to the metal bar shows a small positive reading as the bar moves.

(a) Explain why there is a reading on the ammeter.

.....
 [2]

(b) Complete Table 6.1 by stating what the ammeter shows when the metal bar moves in the direction shown by the arrow in each diagram.

Table 6.1

	reading shown on the ammeter
	
	
	

[3]

- (c) The student finds that using a stronger magnet increases the reading on the ammeter.

State **one** other way in which the student can produce a larger reading on the ammeter, using the same rod.

.....

..... [1]

- (d) Describe how Lenz's law applies when the bar is moved upwards.

.....

.....

..... [1]

[Total: 7]

13. O/N/20/22/Q9

Fig. 9.1 shows a permanent magnet lying on a piece of paper.

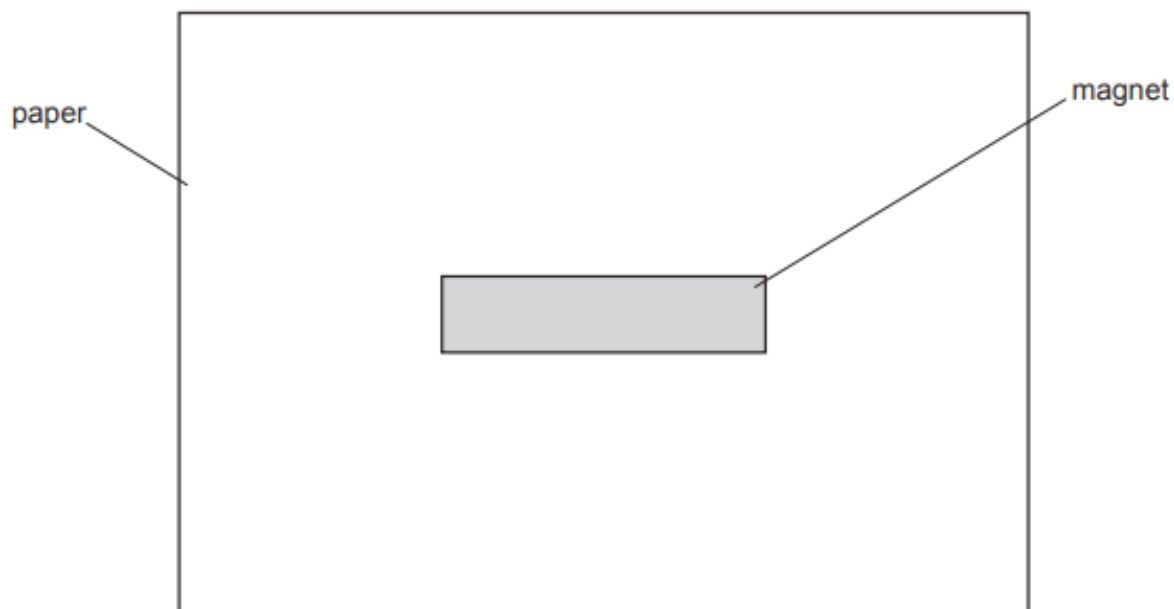


Fig. 9.1

- (a) Underline the material in the list from which it is possible to make a strong, permanent magnet.

aluminium copper glass iron mercury plastic steel

[1]

- (b) Describe an experiment to plot the pattern and the direction of the magnetic field surrounding the magnet. You may draw on Fig. 9.1, if you wish.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (c) Fig. 9.2 shows the N-pole of a magnet placed in front of the S-pole of a second magnet.

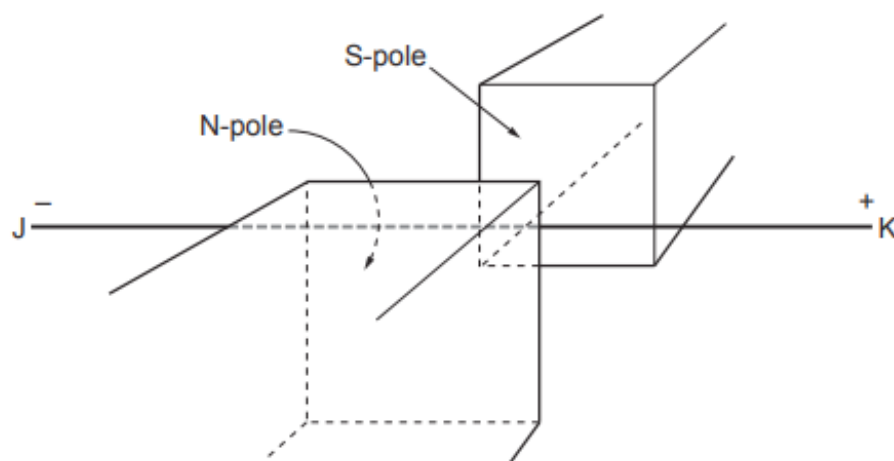


Fig. 9.2

A section of a horizontal, metal wire JK lies in the magnetic field between the two magnetic poles. End K of the metal wire is connected to the positive terminal of a battery and end J is connected to the negative terminal.

- (i) Explain in terms of electrons, why there is a current in the wire and state the direction of the conventional current.

.....

.....

..... [2]

- (ii) The part of JK that is in the magnetic field experiences a force F .

State the direction of F and describe how this direction is deduced.

.....

.....

.....

.....

..... [4]

- (iii) The equipment in Fig. 9.2 is used in a similar experiment.

The part of JK that lies between the poles of the magnets, now passes through a long iron tube that is fixed in position.

The tube is shown in Fig. 9.3.

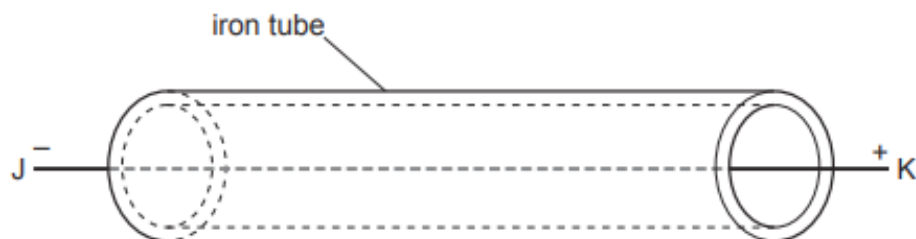


Fig. 9.3

JK is connected to the battery in the same way as before.

State what happens to force F on the wire.

Explain your answer.

.....

.....

..... [2]

(d) The iron tube and the wire JK are removed.

A square, vertical coil is placed between the poles so that the plane of the coil lies in the magnetic field as shown in Fig. 9.4.

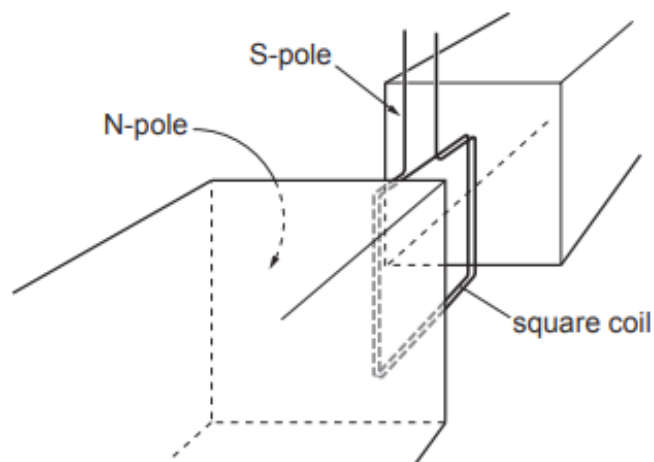


Fig. 9.4

Explain why the coil tries to rotate when there is a current in the coil.

.....

.....

.....

..... [2]

[Total: 15]

14. O/N/20/21/Q6

When electricity is transmitted over large distances, a transformer is used to increase the voltage before transmission. A second transformer is used at the destination to decrease the voltage to the usual mains value.

- (a) Sketch a labelled diagram to show the structure of a transformer that is used to **increase** voltage.

[2]

- (b) Describe the principle of operation of a transformer.

.....

.....

.....

.....

..... [3]

- (c) Explain **one** advantage of transmitting electricity at a high voltage.

.....

.....

..... [2]

[Total: 7]

15. M/J/20/22/Q9

(a) Fig. 9.1 shows a simple relay used to switch a mains electric motor on and off.

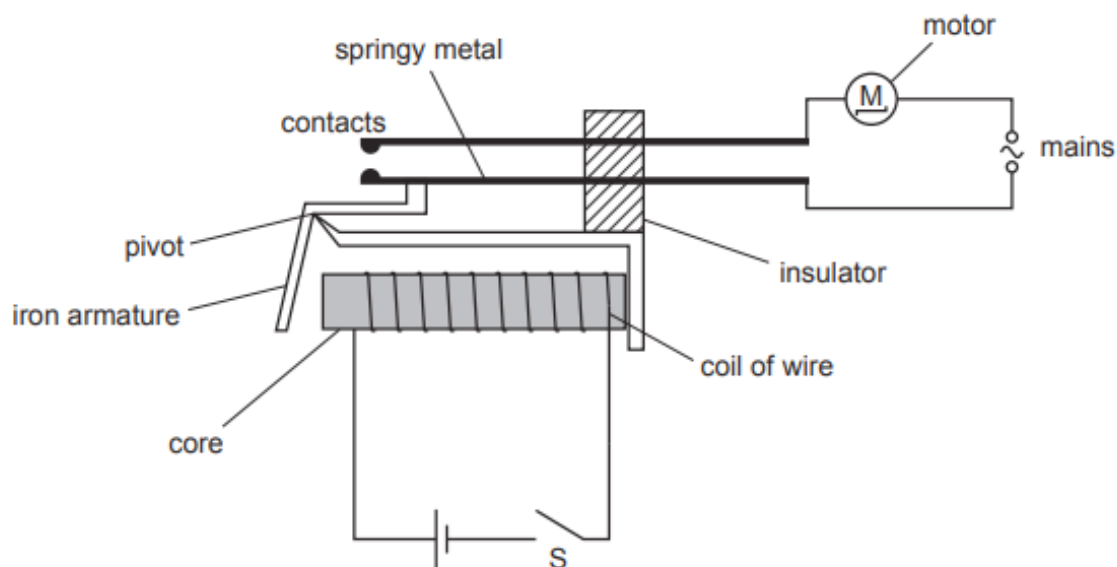


Fig. 9.1

(i) Explain why the motor switches on when switch S is closed.

.....

.....

.....

..... [3]

(ii) Explain why the core is made of iron rather than steel.

.....

.....

..... [2]

(iii) A student suggests that the motor can be turned on and off without a relay.

He suggests connecting the mains to a simple switch in series with the motor.

Suggest **one** reason why, in some situations, using the relay is better.

.....

..... [1]

(b) Fig. 9.2 shows the coil of wire wrapped around a cardboard tube. There is no core.

There is an electric current in the wire in the direction shown by the arrows.

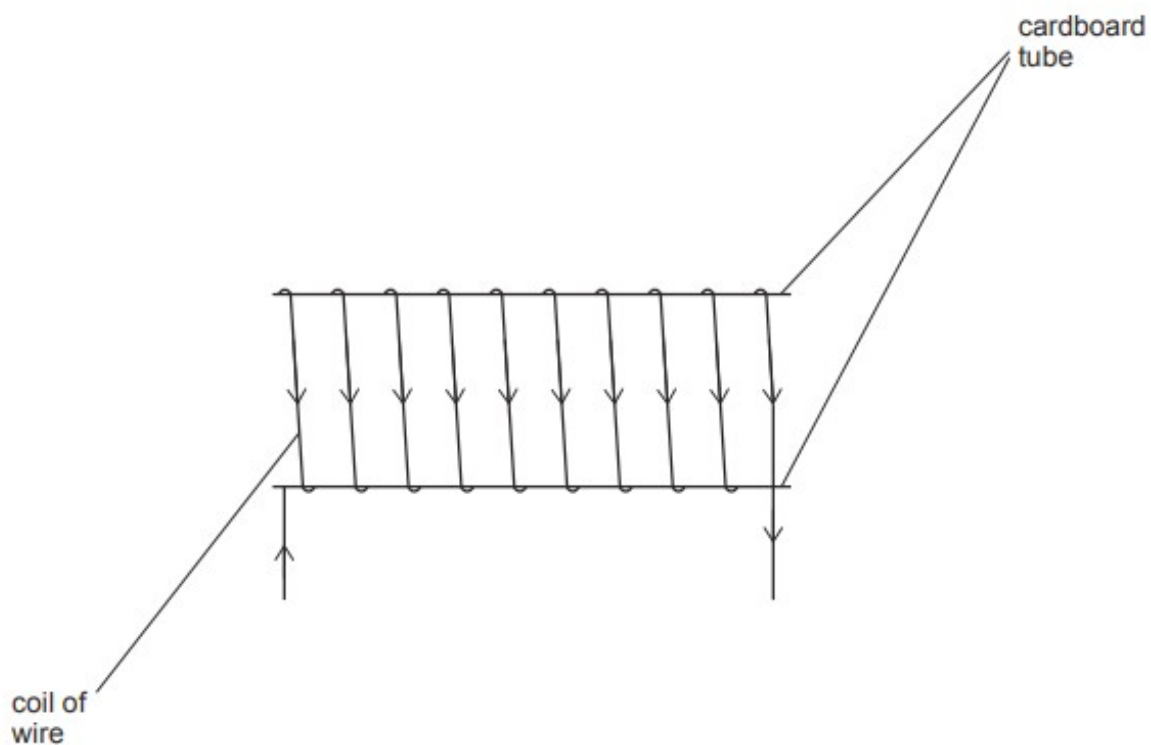


Fig. 9.2

- (i)** On Fig. 9.2, draw the pattern of the magnetic field inside and around the coil. Mark the direction of this magnetic field. [4]
- (ii)** On Fig. 9.2, mark the N-pole of the coil. [1]

- (c) The supply of current to the coil is removed.

The ends of the coil are connected to a sensitive ammeter, as shown in Fig. 9.3.

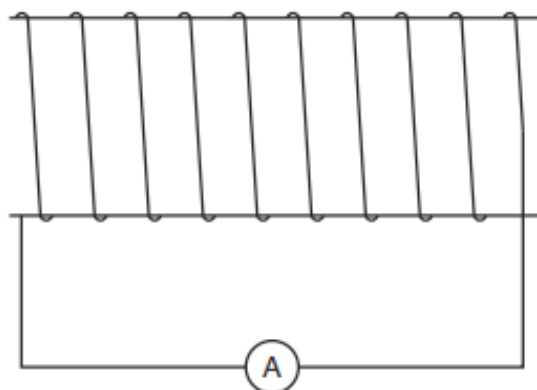


Fig. 9.3

- (i) Describe **how** a permanent magnet is used to produce a **large** reading on the ammeter.

.....

.....

.....

..... [2]

- (ii) Explain **why** a current is produced in (i).

.....

.....

..... [2]

[Total: 15]

16. M/J/20/21/Q10

Fig. 10.1 shows a motor lifting a mass. Fig. 10.2 shows part of the circuit diagram of the connections to the motor.

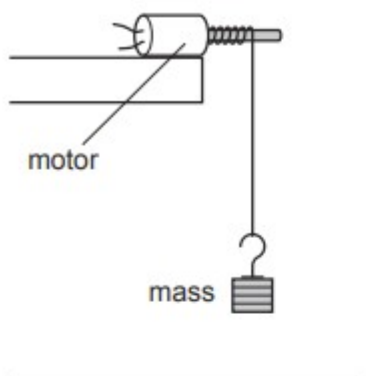


Fig. 10.1

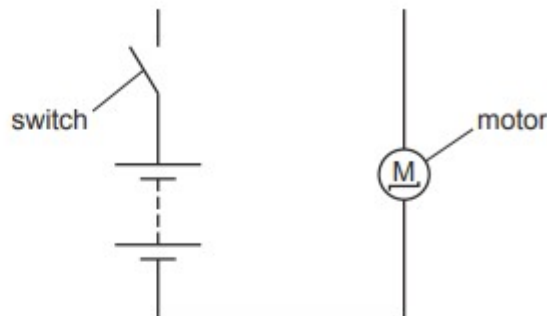


Fig. 10.2

(a) The current in the motor is 1.5 A and the voltage supplied by the battery is 8.0 V .

(i) Complete the circuit diagram in Fig. 10.2 to show an ammeter and a voltmeter in the correct positions to take these measurements while the motor is working. [2]

(ii) The motor takes 4.0 s to lift the mass.

Calculate the electrical energy transferred to the motor in this time.

energy = [2]

(iii) The motor lifts the 150 g mass through a height of 80 cm in the 4.0 s .

Calculate the gravitational potential energy gained by the mass.

The gravitational field strength $g = 10\text{ N/kg}$.

gravitational potential energy = [3]

(iv) State **two** reasons why the gravitational potential energy gained by the mass is less than the electrical energy supplied to the motor.

1.

2.

[2]

(b) Fig. 10.3 shows the structure of the motor.

When the mass reaches the top of its motion, the switch is opened. This disconnects the battery and causes the mass to fall. The coil turns as the mass falls.

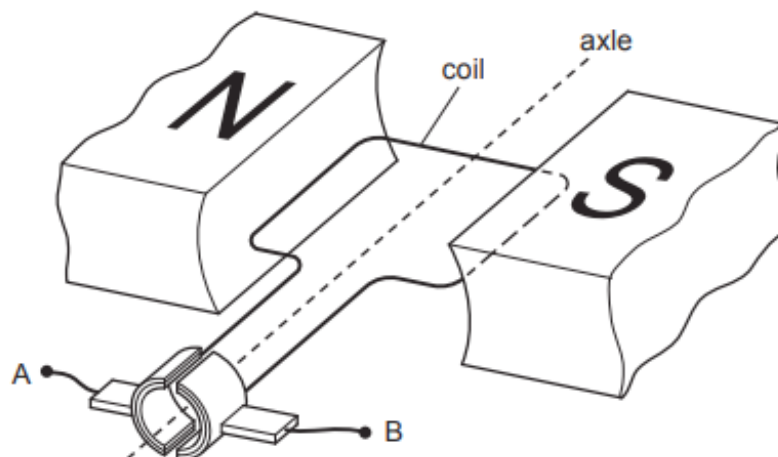


Fig. 10.3

As the coil turns, a small voltage is produced.

(i) Explain why a voltage is produced as the coil turns.

.....

.....

..... [3]

(ii) As the mass falls, a student connects a wire between the points A and B shown in Fig. 10.3.

He notices that the mass takes a longer time to fall when the wire is connected.

The student suggests that this is an example of Lenz's law.

State Lenz's law and suggest why the mass takes longer to fall.

.....

.....

.....

.....

.....

..... [3]

17. O/N/19/22//Q6

An electric circuit contains a $700\ \Omega$ resistor and a light-dependent resistor (LDR). Fig. 6.1 is the circuit diagram.

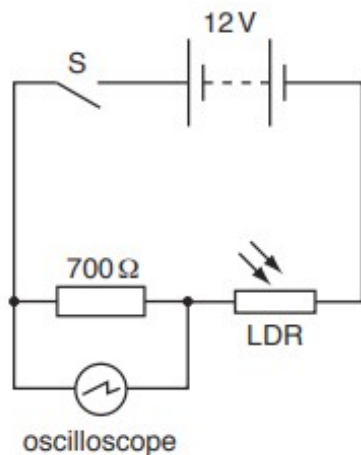


Fig. 6.1

The electromotive force (e.m.f.) of the battery is 12V.

An oscilloscope is connected across the fixed resistor. Fig. 6.2 shows the oscilloscope, including the settings of the timebase and the Y-gain controls. Line Q shows the position of the trace on the oscilloscope when the switch S is open.

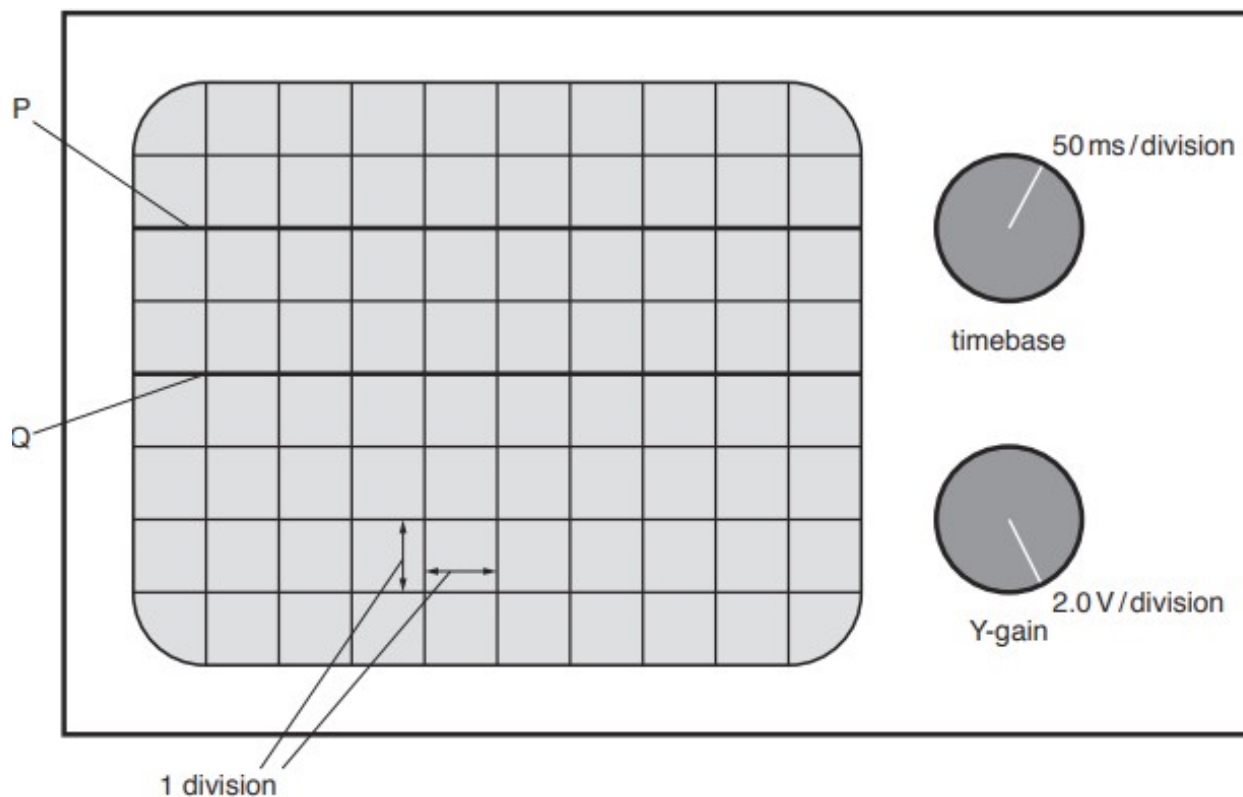


Fig. 6.2

The switch S is closed and the trace on the oscilloscope moves to the position shown by line P in Fig. 6.2.

- (a) (i) Determine the potential difference (p.d.) across the $700\ \Omega$ resistor.

p.d. = [1]

- (ii) Determine the resistance of the LDR.

resistance = [3]

- (b) The intensity of the light incident on the LDR gradually increases.

State and explain how the trace on the oscilloscope screen moves.

.....
.....
.....
..... [3]

[Total: 7]

18. O/N/19/22/Q7

Fig. 7.1 shows a horizontal, rectangular coil ABCD placed between a magnetic N-pole and an S-pole.

There is a current in the coil.

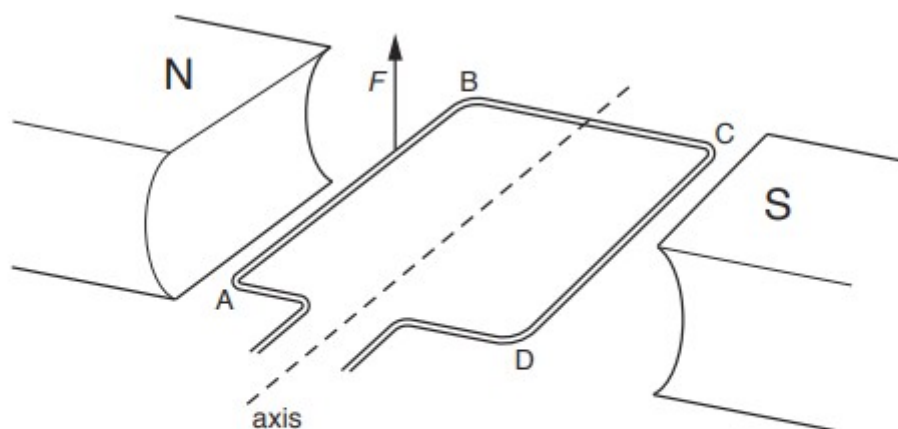


Fig. 7.1

An upward force F of size $9.6 \times 10^{-3} \text{ N}$ acts on side AB of the coil.

- (a) (i) Explain why there is a force on side AB.

.....
 [1]

- (ii) Determine the direction of the current in AB and state how the direction is deduced.

.....

 [2]

- (b) Both side AB and side CD of the coil are 2.5 cm from the axis.

Determine the total moment acting on the coil.

moment = [2]

- (c) The coil in Fig. 7.1 is part of a direct current (d.c.) motor.

- (i) State the name of the device that connects the coil of a d.c. motor to the electricity supply.

..... [1]

- (ii) State **one** change to the arrangement in Fig. 7.1 that produces a greater turning effect in coil ABCD.

..... [1]

19. O/N/19/21/Q8

- (a)** A small electric component in a circuit needs to be protected from a strong, external magnetic field.

Describe how this is done and state any material used.

.....

.....

..... [2]

- (b)** Electromagnets have many domestic and industrial applications.

- (i)** In the space, draw a diagram of a simple electromagnet.

[3]

- (ii)** State and describe in detail one use of an electromagnet.

.....

.....

.....

.....

..... [3]

- (c) Fig. 8.2 shows a solenoid connected to a sensitive ammeter.

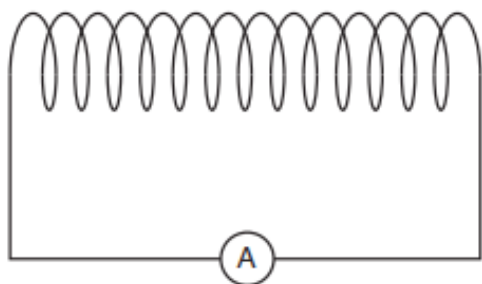


Fig. 8.2

Explain what is observed on the ammeter when a student:

- (i) inserts an N-pole of a bar magnet slowly into the right-hand end of the solenoid

.....
.....
.....
..... [3]

- (ii) holds the bar magnet stationary in the solenoid

.....
.....
..... [1]

- (iii) withdraws the N-pole of the magnet **quickly** from the right-hand end of the solenoid.

.....
.....
.....
..... [3]

[Total: 15]

20. M/J/19/22/Q7

Fig. 7.1 shows a copper wire placed on two copper rods in the magnetic field between the poles of a magnet.

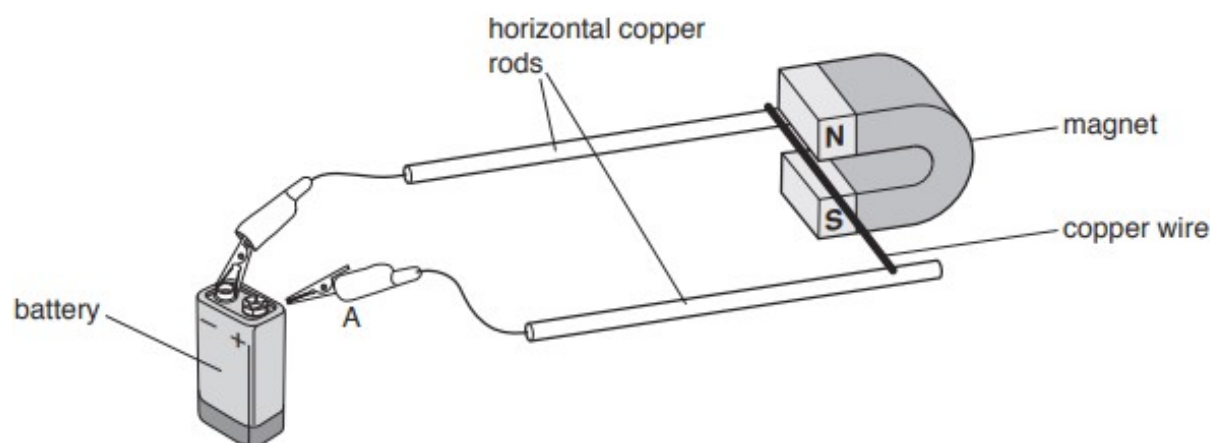


Fig. 7.1

The crocodile clip A touches the positive terminal of the battery. This causes the copper wire to move.

(a) On Fig. 7.1, mark with an arrow the direction of the current in the copper wire. [1]

(b) Explain why the copper wire moves along the copper rods.

.....

.....

.....

..... [2]

(c) Name **two** different devices that use this effect.

1.

2. [2]

[Total: 5]

21. M/J/19/21/Q7

Fig. 7.1 shows part of a torch. The torch does not contain a battery.

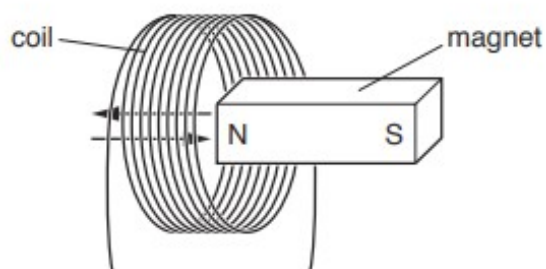


Fig. 7.1

The torch is shaken and this causes the magnet to move backwards and forwards through the coil.

- (a) (i) Explain why an electromotive force (e.m.f.) is induced in the coil when the magnet moves.

.....
 [2]

- (ii) State one way to increase the e.m.f. induced.

.....
 [1]

- (b) The current in the coil is used to produce light from a light-emitting diode (LED).

- (i) In the space below, draw the circuit symbol for an LED.

[1]

- (ii) An LED is a *more efficient* source of light than a filament lamp.

Explain what this statement means.

.....
 [1]

[Total: 5]

22. O/N/18/22/Q7

Fig. 7.1 shows the structure of a transformer.

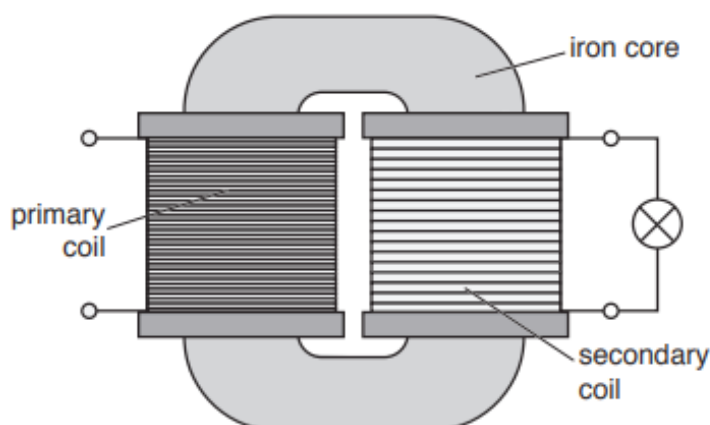


Fig. 7.1

A lamp is connected to the secondary coil.

(a) Explain why the core of the transformer is made from iron.

.....

 [2]

(b) When there is an alternating current (a.c.) in the primary coil, the lamp is lit.

When there is a direct current (d.c.) in the primary coil, the lamp is **not** lit.

(i) State two ways in which an alternating current differs from a direct current.

1.

 2.
 [2]

(ii) Explain why the lamp is **not** lit when there is a direct current in the primary coil.

.....

 [2]

23. M/J/18/22/Q7

A computer hard disk contains a layer of a magnetic material.

- (a) Describe how a magnet is used to find out if a sample of material is magnetic or non-magnetic.

.....

[1]

- (b) Data is stored on the disk as a series of N-poles and S-poles.

Fig. 7.1 shows part of the hard disk. The thin layer of magnetic material contains small regions. Each region has an N-pole and an S-pole. Some magnetic field lines are shown on Fig. 7.1.

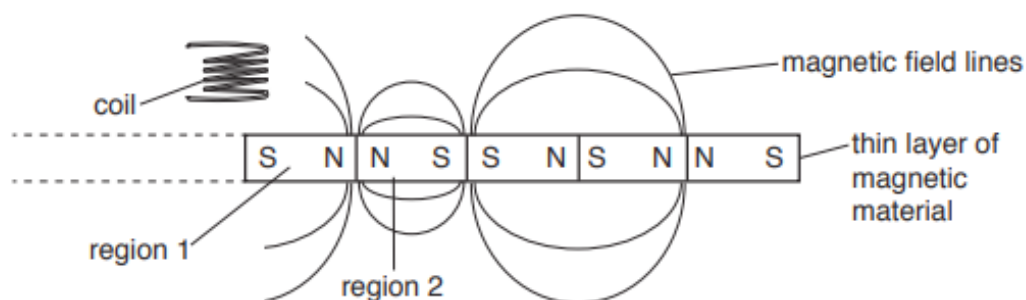


Fig. 7.1 (not to scale)

- (i) Region 2 causes a magnetic force on region 1.

State the direction of the magnetic force on region 1 and explain why it acts.

.....

[2]

- (ii) On Fig. 7.1, draw arrows on the field lines to show the direction of the magnetic field near the boundary between region 1 and region 2. [1]

- (iii) The coil shown in Fig. 7.1 is fixed in position. The layer of magnetic material passes quickly under the coil.

A voltage is induced in the coil as some of the boundaries between the regions pass under the coil.

1. Explain why a voltage is induced in the coil.

.....
.....
.....[1]

2. Suggest why the coil must be close to the layer.

.....
.....
.....[1]

24. O/N/17/22/Q11

The metals in the list below have many different uses.

aluminium

copper

iron

silver

steel

(a) State which metal from the list is used for

(i) a compass needle,

.....

(ii) magnetic screening,

.....

(iii) the core of a transformer.

.....

[2]

(b) (i) Describe one use for a transformer.

.....

.....

.....

[2]

(ii) An a.c. generator supplies an input voltage of 220 V to a transformer.

1. In the space below, sketch a graph of the output voltage against time for the a.c. generator. [2]

2. The transformer has 1700 turns on the primary coil and 85 turns on the secondary coil.

Calculate the output voltage of the transformer.

output voltage = [2]

(c) Fig. 11.1 shows a magnet next to one end of a solenoid.

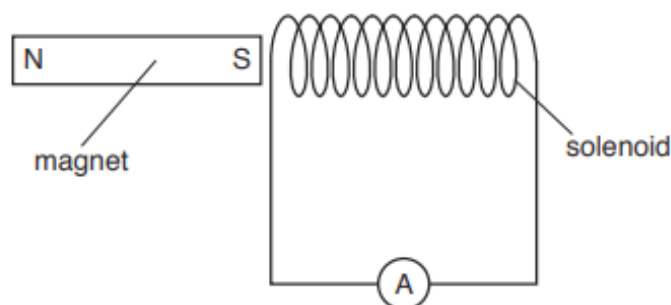


Fig. 11.1

The terminals of the solenoid are connected to a very sensitive ammeter.

(i) The magnet is moved to the right at a constant speed and a reading is observed on the ammeter.

1. Explain why there is a current in the ammeter.

.....

.....

.....

.....

.....[3]

2. Explain how the current in the ammeter opposes the change producing it.

.....

.....

.....

.....[2]

(ii) The magnet stops when the S-pole reaches the middle of the solenoid.

The reading on the ammeter is observed when the magnet is moved to the left at a constant speed that is less than its speed in (c)(i).

State two ways in which the reading on the ammeter differs from the reading observed in (c)(i) as the magnet moves to the left.

1.

2.

[2]

25. O/N/17/21/Q7

An electric circuit consists of a battery, a variable resistor and connecting wires. Sections PQ and QR of the connecting wire are in a region of space where there is a uniform magnetic field.

In Fig. 7.1, the shaded area represents the magnetic field.

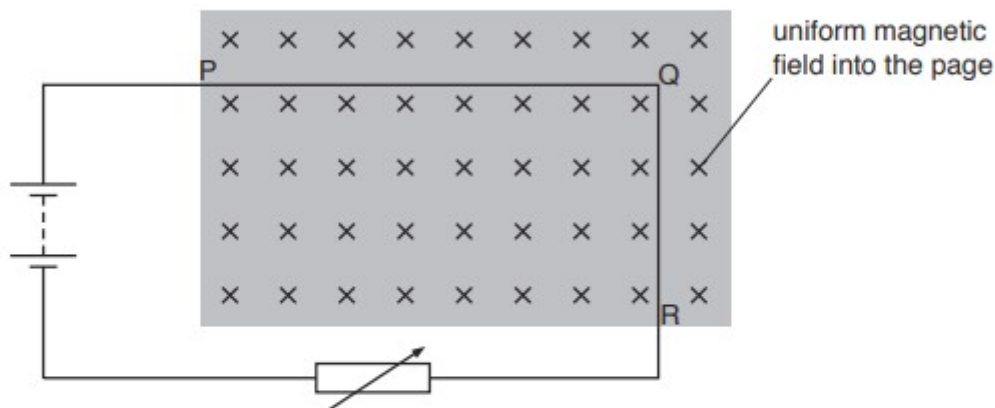


Fig. 7.1

The direction of the magnetic field is into the page.

A current-carrying wire in a magnetic field can experience a force.

- (a) In the table, mark **one** tick (✓) in each column to indicate the force on PQ and the force on QR.

	section PQ	section QR
no force on this section		
a force towards the top of the page		
a force towards the bottom of the page		
a force out of the page		
a force into the page		
a force towards the left of the page		
a force towards the right of the page		

[2]

- (b) The battery in the circuit in Fig. 7.1 is reversed.

State and explain the effect of this on PQ and QR.

.....

.....

.....

..... [2]

(c) The variable resistor is adjusted and the current in the circuit decreases.

Suggest one effect of decreasing the current.

.....
..... [1]

26. M/J/17/22/Q7

Fig. 7.1 shows part of a d.c. electric motor.

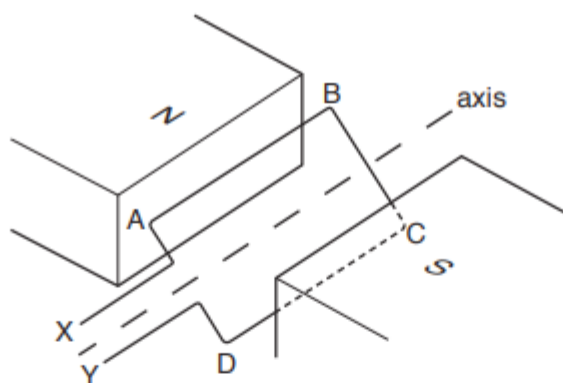


Fig. 7.1

- (a) A coil of wire ABCD is placed between the poles of a magnet. Ends X and Y are connected to a commutator and a battery.

(i) State why there is a force on side AB of the coil.

.....
[1]

(ii) Describe how the commutator keeps the coil rotating in one direction.

.....

[2]

- (b) The current in the motor is 2.0A and the battery has an electromotive force (e.m.f.) of 12 V. In a time of 8.0s, the motor does 140J of work.

Calculate

(i) the energy supplied to the motor in 8.0s,

energy supplied =[2]

(ii) the efficiency of the motor.

efficiency =[1]

27. M/J/17/21/Q7

Fig. 7.1 shows a metal rod placed between the poles of a magnet.

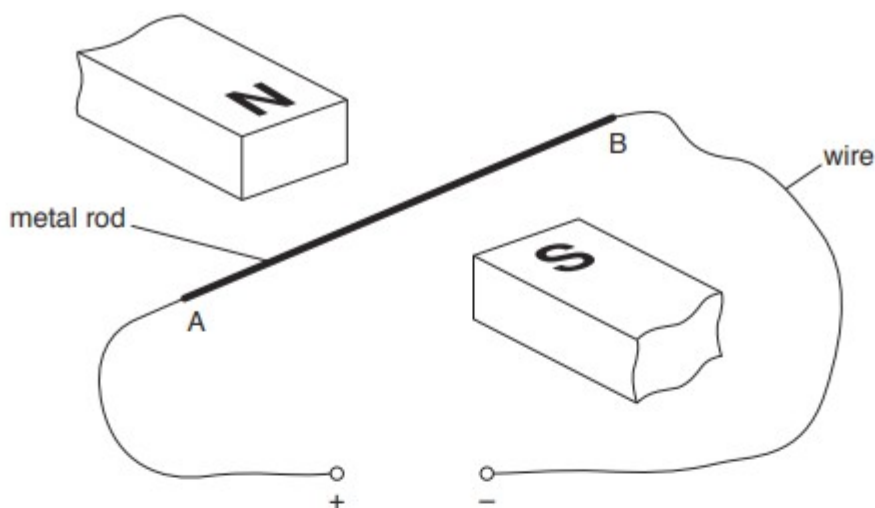


Fig. 7.1

- (a) End A of the rod is connected to the positive terminal of a d.c. power supply and end B is connected to the negative terminal. The current in the rod and the magnetic field produce a force on the rod.

On Fig. 7.1 draw arrows to show

- (i) the direction of the current in the metal rod; label this arrow C,
- (ii) the direction of the magnetic field that acts on the metal rod; label this arrow M,
- (iii) the direction of the force on the rod; label this arrow F.

[3]

- (b) The power supply is removed and a sensitive voltmeter is connected between the ends A and B.

When the rod is moved upwards or downwards there is a reading on the voltmeter.

- (i) Explain why there is a reading on the voltmeter.

.....

[2]

- (ii) The reading on the voltmeter is increased by using a stronger magnetic field.

State one other way of producing a larger reading on the voltmeter.

.....
[1]

28. O/N/16/21/Q6

Two flexible iron strips, WX and YZ, are placed close to each other inside a solenoid (long coil). The end W of WX and the end Z of YZ are held firmly in position.

Fig. 6.1 shows that the solenoid is connected to a d.c. power supply and a switch.

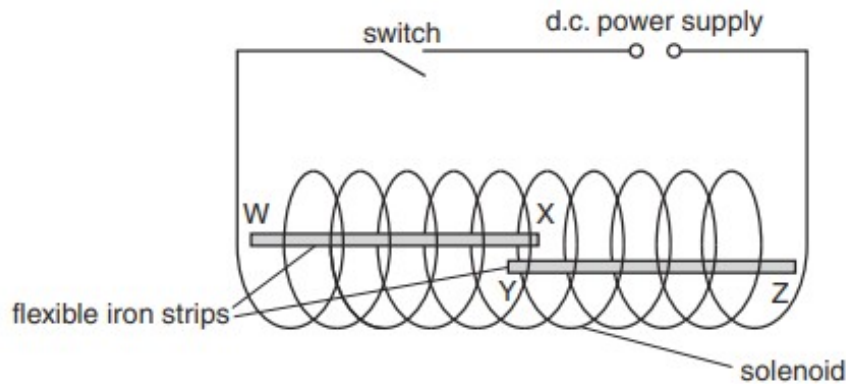


Fig. 6.1

The switch is closed and there is an electric current in the solenoid.

End W of flexible iron strip WX becomes a magnetic S-pole.

(a) (i) State the type of magnetic pole produced at X, Y and Z.

1. X-pole
2. Y-pole
3. Z-pole

[2]

(ii) State and explain what happens to X and Y because the flexible iron strips are magnetised.

.....
[1]

(b) A relay operates a switch in one circuit using the current in a different circuit.

State one example of a relay used in this way and explain why a relay is used.

.....

[2]

29. M/J/16/22/Q8

A transformer and a diode are used to charge a battery.

Fig. 8.1 shows the transformer, which contains a soft-iron core and two coils.

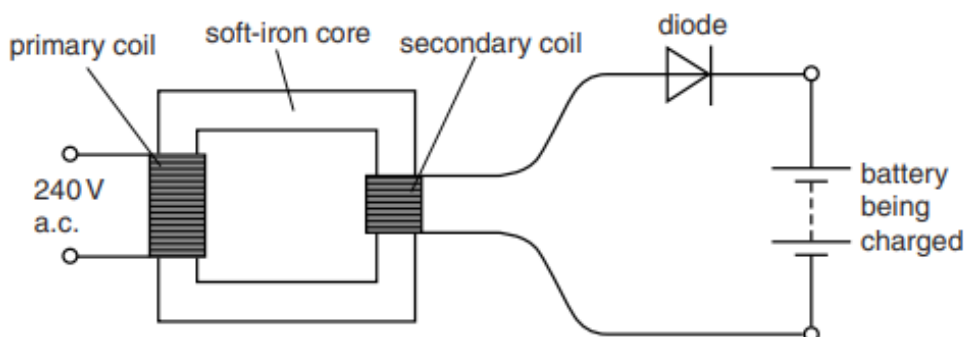


Fig. 8.1

The primary coil is connected to the 240V a.c. mains supply. The secondary coil is connected in series with the diode and the battery.

(a) Explain why an electromotive force (e.m.f.) is induced in the secondary coil.

.....

.....

.....

.....

.....[2]

(b) The e.m.f. induced in the secondary coil is less than 240V.

Suggest why.

.....

.....[1]

(c) Suggest why steel is not used as the core of a transformer.

.....

.....[1]

(d) Describe the action of the diode.

.....

.....[1]

30. M/J/16/21/Q7

Fig. 7.1 shows part of a loudspeaker. A coil is attached to a paper cone and is suspended freely within a cylindrical magnet.

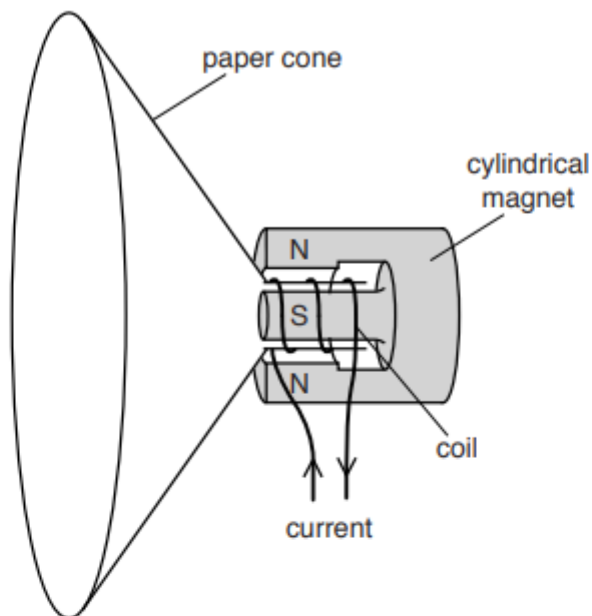


Fig. 7.1

- (a) Explain why a current in the coil causes the paper cone to move.

.....

[2]

- (b) Describe and explain the movement of the cone when there is an alternating current in the coil.

.....

[2]

- (c) The loudspeaker produces sound of frequency 500Hz. The speed of sound is 320m/s. Calculate the wavelength of the sound produced.

wavelength =[2]

31. M/J/16/21/Q11

(a) Induced currents can be formed in a coil of wire by electromagnetic induction.

(i) Describe how to demonstrate the formation of an induced current in a coil of wire.

Sketch and label a diagram of the arrangement of the apparatus.

.....
.....
.....[3]

(ii) State two changes to the apparatus that increase the induced current.

1.
2.
[2]

(iii) The direction of the induced current is determined by Lenz's Law.

1. State Lenz's Law.

.....
.....
.....[2]

2. Describe how Lenz's Law applies in the experiment you described in **(a)(i)**.

.....
.....
.....
.....
.....[2]

(b) An electrical generator supplies power to a distant factory.

(i) The factory receives 500 kW of electrical power at a voltage of 33 kV.

Calculate

1. the current in the wires supplying the power,

current =[2]

2. the electrical energy supplied to the factory in one hour.

energy =[2]

(ii) The 33 kV voltage supplied to the factory is very high.

Transmitting electrical energy at high voltage reduces energy losses in the transmission lines.

Explain why.

.....
.....
.....
.....
.....[2]

32. O/N/15/22/Q10

Thin wire, covered in plastic insulation, is used to make a solenoid (long coil). The solenoid is connected to a sensitive ammeter. Fig. 10.1 shows the N-pole of a steel magnet placed next to the solenoid.

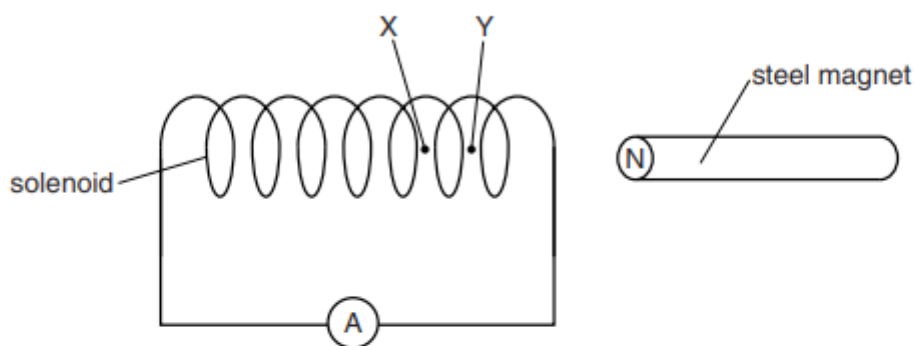


Fig. 10.1

Point X and point Y are on the axis of the solenoid.

(a) (i) Explain why plastic is an electrical insulator.

.....

 [1]

(ii) Explain why the magnet is **not** made from

1. aluminium,

..... [1]

2. iron.

..... [1]

(b) In one experiment, the magnet in Fig. 10.1 is moved to the left and passes into the solenoid. The N-pole of the magnet travels from Y to X at a constant speed. As it moves, the ammeter shows a small current.

(i) Explain why there is a current in the solenoid when the magnet is moving.

.....

 [3]

- (ii) The N-pole travels from Y to X in 0.14 s. As it moves, the current shown on the ammeter is 0.045 mA. The resistance of the solenoid is $1.2\ \Omega$.

Calculate

1. the potential difference (p.d.) across the solenoid,

potential difference = [2]

2. the charge that passes through the solenoid as the N-pole moves from Y to X.

charge = [2]

- (c) In a second experiment, the speed of the N-pole is greater than its speed in the first experiment. It now takes only 0.070 s to travel from Y to X.

A current in the same direction is shown on the ammeter.

- (i) State and explain how the size of this current compares with the size of the current in the first experiment.

.....
.....
..... [2]

- (ii) The same quantity of charge passes through the coil in both the first and second experiments.

Explain why this is the case.

.....
.....
..... [1]

- (d) State two ways in which the equipment shown in Fig. 10.1 can be used to produce a current in the solenoid that is in the opposite direction.

1.
2.

[2]

33. O/N/15/21/Q10

- (a) Fig. 10.1 shows a solenoid (long coil) X connected in series with a battery, a switch S and a variable resistor (rheostat).

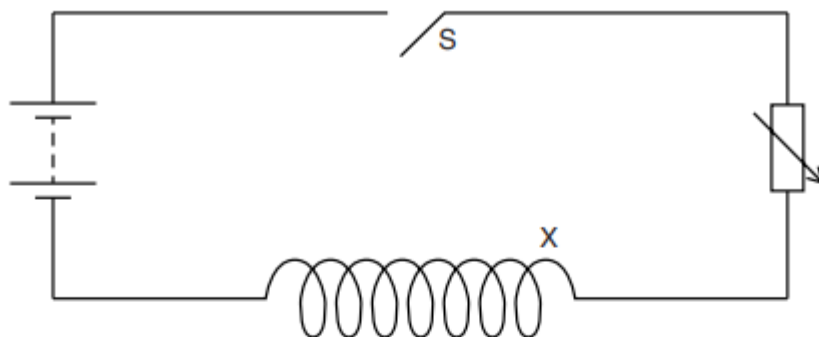


Fig. 10.1

The switch S is closed and there is a magnetic field due to the current in the solenoid.

- (i) On Fig. 10.1, draw the pattern of the magnetic field in, above and below the solenoid. [3]
- (ii) A second solenoid Y is connected to a sensitive centre-zero ammeter. Solenoid Y is placed in the magnetic field of X.
1. The resistance of the variable resistor is gradually decreased and the pointer of the sensitive ammeter deflects slightly to one side.

Explain why this happens.

.....

.....

.....

.....

.....

.....

..... [3]

2. The switch S is now opened.

Describe and explain what happens to the deflection on the ammeter as the switch is opened.

.....

.....

.....

.....

.....

.....

..... [3]

- (b) Fig. 10.2 shows a transformer that consists of two coils wound on an iron core. The transformer is connected to an electricity transmission cable.

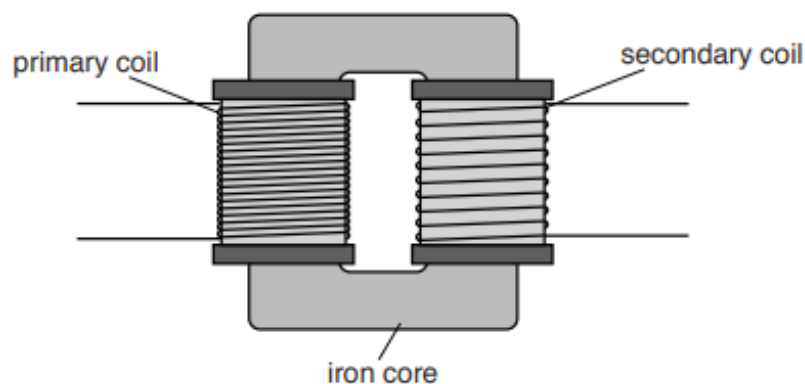


Fig. 10.2

- (i) Explain the purpose of the iron core in the transformer.

.....

.....

.....

.....

.....

..... [2]

- (ii) The transformer supplies electrical power to a factory at 33 000 V. The current supplied is 85 A.

Calculate

1. the electrical power that the factory is receiving,

power = [2]

2. the electrical energy the factory receives in 1.0 hour.

energy = [2]

34. M/J/15/22/Q6

Fig. 6.1 shows a simple a.c. generator.

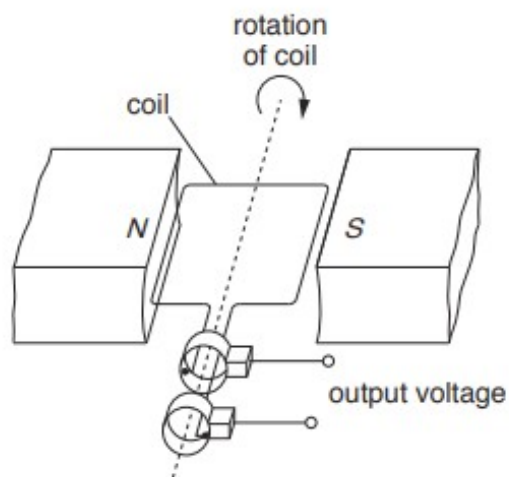


Fig. 6.1

- (a) The coil rotates and an alternating electromotive force (e.m.f.) is induced in the coil.

Fig. 6.2 shows how the alternating e.m.f. varies with time as the coil rotates.

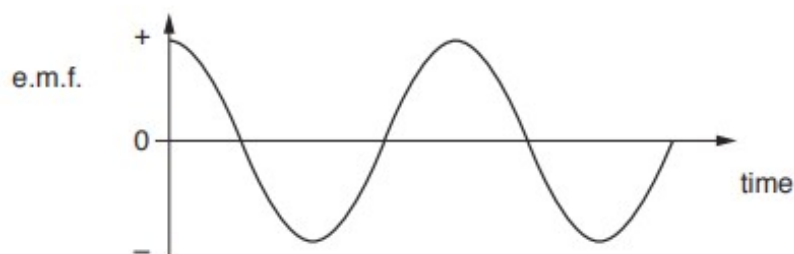


Fig. 6.2

Explain

- (i) why an e.m.f. is induced,

.....

 [2]

- (ii) why the e.m.f. is sometimes positive and sometimes negative.

.....

 [1]

(b) Changes are made to the a.c. generator, one at a time:

- stronger magnets are used
- more turns are wound on the coil
- the coil is turned faster.

Complete the table in Fig. 6.3 to show what happens to the maximum value of the e.m.f. and to the frequency of the alternating e.m.f.

changes made	what happens to the maximum value of the e.m.f.	what happens to the frequency of the e.m.f.
stronger magnets		
more turns on the coil		
the coil is turned faster		

Fig. 6.3

[3]

35. M/J/15/22/Q10

Fig. 10.1 shows a relay connected to a cell and a switch.

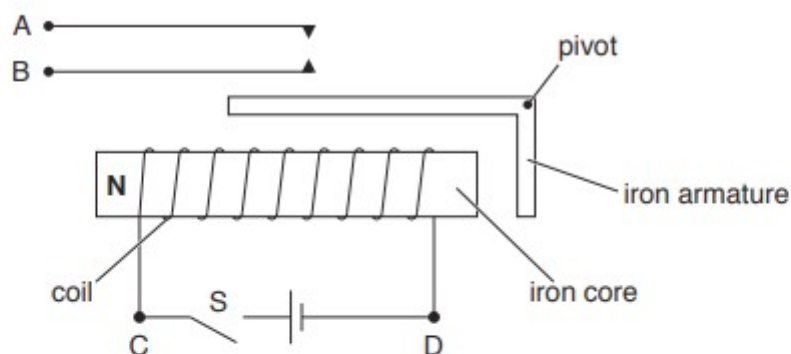


Fig. 10.1

- (a) When switch S is closed, the iron core is magnetised. The left side of the core is an N-pole, as shown in Fig. 10.1. The iron armature is attracted to the core.

(i) On Fig. 10.1, mark

1. the S-pole of the iron core
2. the N-pole and the S-pole of the iron armature.

[2]

(ii) The cell is reversed.

State what happens to the poles and to the armature.

.....
 [2]

(iii) Explain why the core is made of iron and not steel.

.....

 [2]

- (b) Fig. 10.2 shows the relay connected in a circuit to a 12V battery. The bell is not ringing.

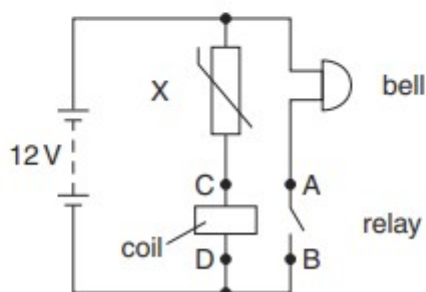


Fig. 10.2

- (i) State the name of component X.

..... [1]

- (ii) Explain why the bell rings when the temperature of X rises.

.....

.....

..... [2]

- (iii) When the resistance of X is 2000Ω , the current in the coil is 1.5mA . This causes the contacts in the relay to close. The resistance of the bell is 200Ω .

Calculate

1. the potential difference (p.d.) across X,

p.d. = [2]

2. the p.d. across the coil,

p.d. = [1]

3. the current in the battery.

current = [2]

- (iv) Component X is removed from the circuit and replaced by a different component Y. The bell now rings when bright light shines on Y.

State the name of component Y.

..... [1]

36. M/J/15/21/Q7

A simple apparatus used to demonstrate electromagnetic induction is shown in Fig. 7.1.

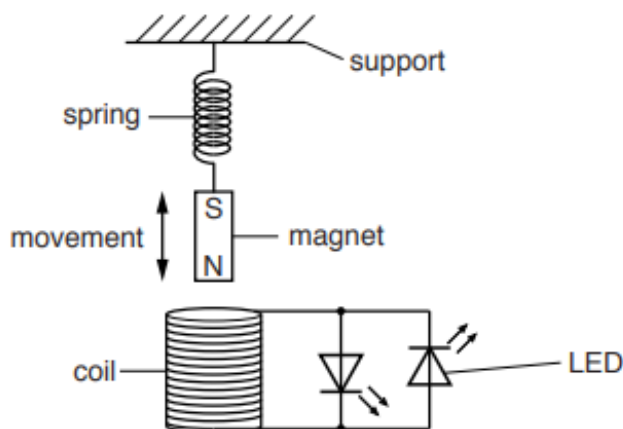


Fig. 7.1

The coil is connected to two light-emitting diodes (LEDs). The magnet moves into and out of the coil.

- (a)** Explain why there is an induced e.m.f. in the coil when the magnet moves.

.....

[2]

- (b)** Explain why one LED lights up when the magnet moves into the coil and the other LED lights up when the magnet moves out of the coil.

.....

[2]

- (c)** The LEDs are brighter when the magnet moves faster.
 Explain why.

.....

[1]

37. O/N/14/22/Q8

Fig. 8.1 shows the structure of a simple alternating current (a.c.) generator.

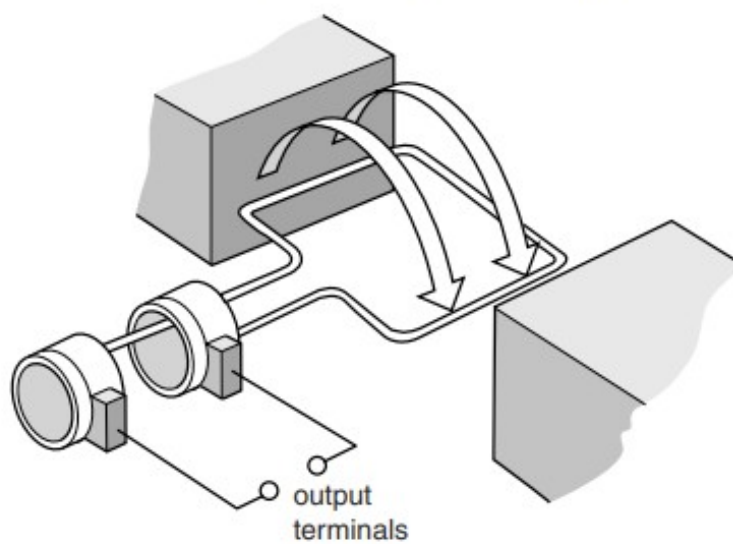


Fig. 8.1

(a) On Fig. 8.1, label

- the coil of the generator with the letter C,
- a slip ring with the letter S,
- a carbon brush with the letter B.

[2]

(b) The a.c. generator is operating and the arrows on Fig. 8.1 show the direction of rotation.

Explain why there is an electromotive force (e.m.f.) between the two output terminals.

.....

.....

.....

.....[3]

- (c) The output terminals of the a.c. generator are connected to a cathode-ray oscilloscope (c.r.o.). Fig. 8.2 shows the trace on the screen of the c.r.o.

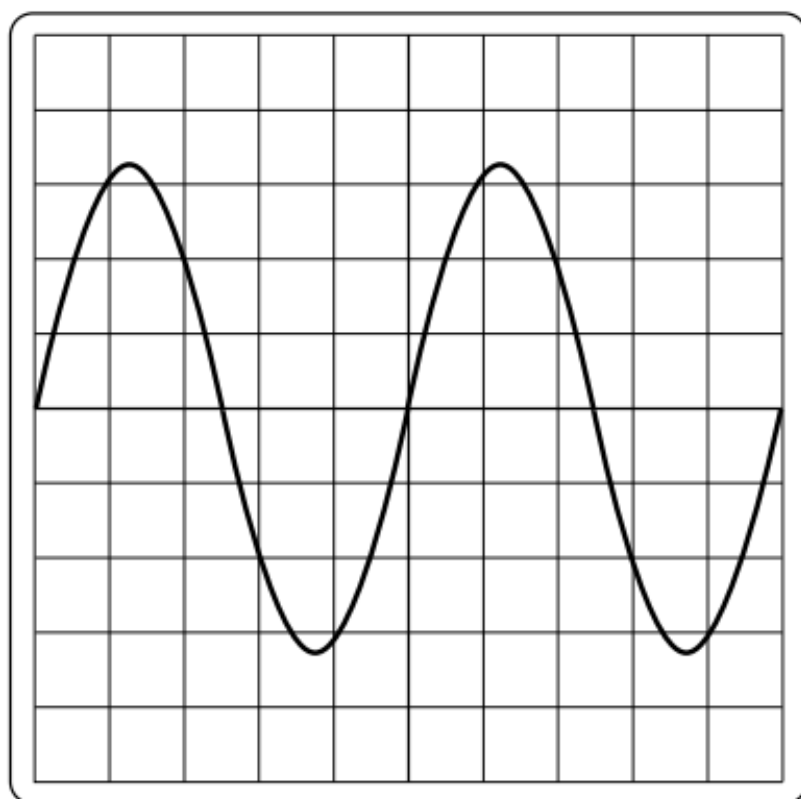


Fig. 8.2

Describe how the trace and a setting on the c.r.o. are used to find the time for one revolution of the coil of the a.c. generator. You may draw on Fig. 8.2 if you wish.

.....
.....
.....[2]

38. O/N/14/21/Q7

A straight length of copper wire lies horizontally between the poles of a U-shaped magnet. Fig. 7.1 shows the two ends of the wire connected to a very sensitive, centre-zero ammeter.

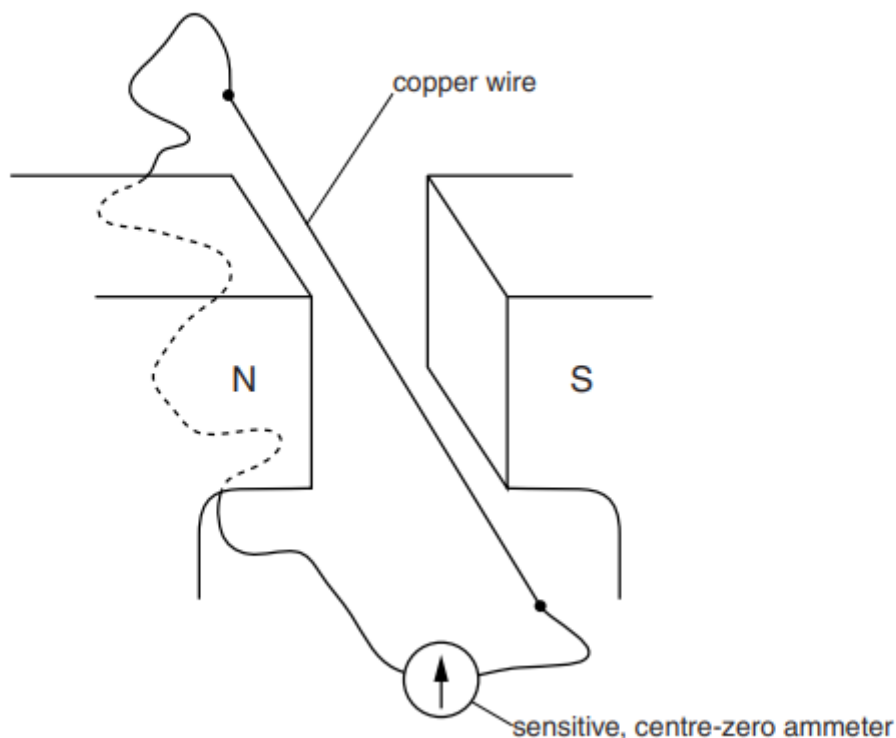


Fig. 7.1

The copper wire is moved upwards slowly between the two magnetic poles. The needle on the ammeter deflects to the right.

- (a) Explain why the needle on the ammeter deflects.

.....

[2]

- (b) The wire is moved downwards very quickly between the two magnetic poles.

State what happens to the needle on the ammeter.

.....
[1]

- (c) State what happens to the needle on the ammeter when the copper wire is moved horizontally between the two poles.

.....
[1]

39. M/J/14/21/Q6

Fig. 6.1 shows a coil of wire connected by flexible leads to a switch and a battery.

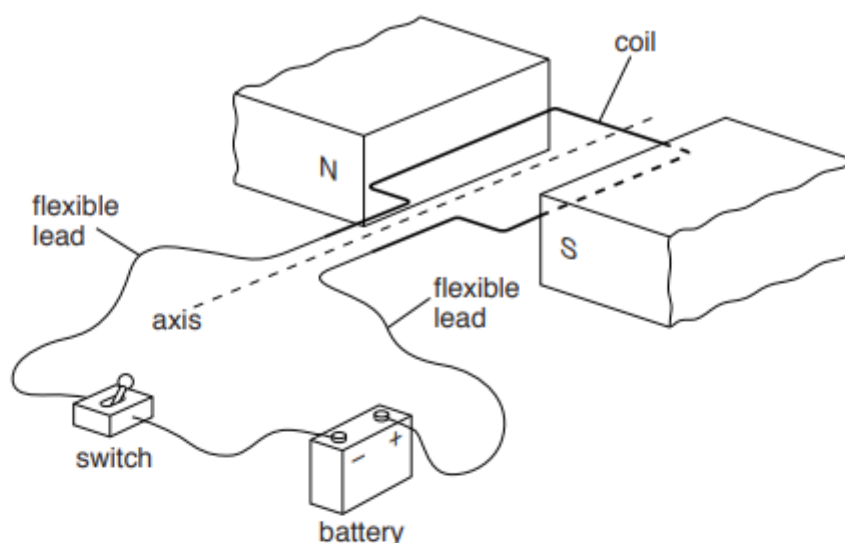


Fig. 6.1

The coil is placed between the poles of a permanent magnet and is free to turn about the axis.

When the switch is closed, forces due to the current act on the sides of the coil. The coil starts to turn.

(a) On Fig. 6.1, draw arrows to show the directions of the forces. [2]

(b) The coil stops when it is vertical. Explain why the turning effect of the forces is zero at this position.

..... [1]

(c) In order for the coil to rotate continuously, a split-ring commutator is connected between the battery and the coil.

Explain how the split-ring commutator enables the coil to rotate continuously. Include a diagram in your answer.

..... [4]

40. M/J/13/22/Q7

Fig. 7.1 shows a view, from above, of two wires X and Y. These wires carry equal currents vertically downwards through a piece of card.

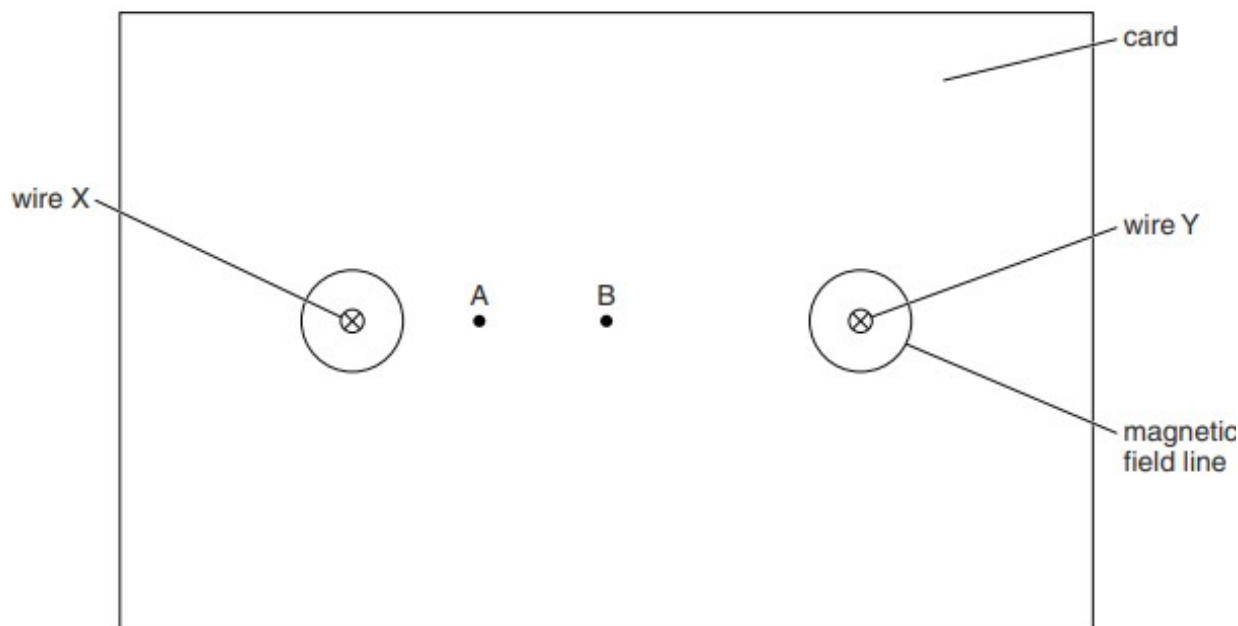


Fig. 7.1

One complete magnetic field line is drawn around each wire.

In this question, ignore the effects of the Earth's magnetic field.

- (a) On Fig. 7.1, draw the complete magnetic field line due to the current in wire X that passes through point A.

Mark the direction of this field line. [2]

- (b) Point B is midway between the two wires. Explain why the magnetic field at B is zero.

.....
 [1]

- (c) There is a force on wire Y due to the current in wire X.

- (i) State the direction of the force on wire Y.

..... [1]

- (ii) Explain why there is a force on wire Y.

.....
 [1]

41. M/J/13/22/Q8

Fig. 8.1 shows a simple a.c. generator. The coil is turning and an e.m.f. is induced in the coil.

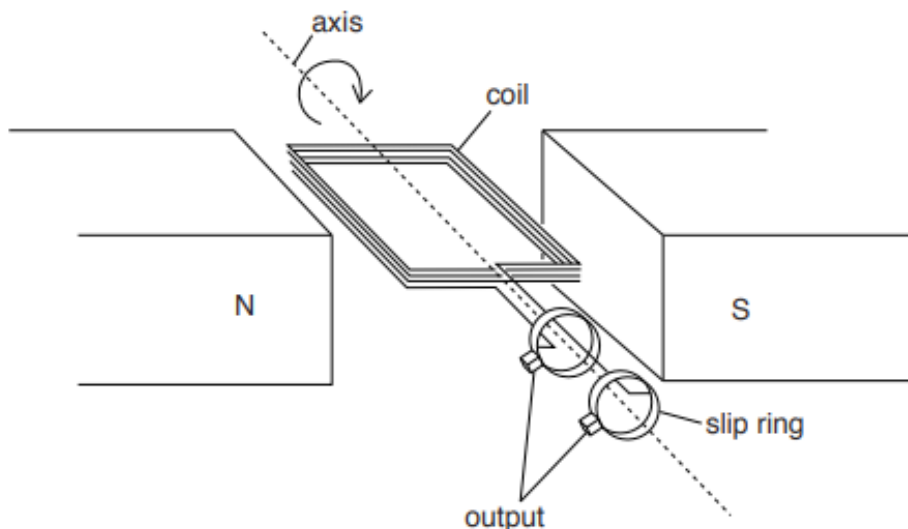


Fig. 8.1

- (a) The generator contains a permanent magnet. State the name of a metal used in a permanent magnet.

..... [1]

- (b) At the instant shown in Fig. 8.1, the induced e.m.f. is a maximum.

- (i) Explain why the induced e.m.f. is a maximum.

.....

 [2]

- (ii) State the position of the coil where there is no induced e.m.f.

.....
 [1]

42. M/J/13/21/Q8

Fig. 8.1 shows a simple transformer.

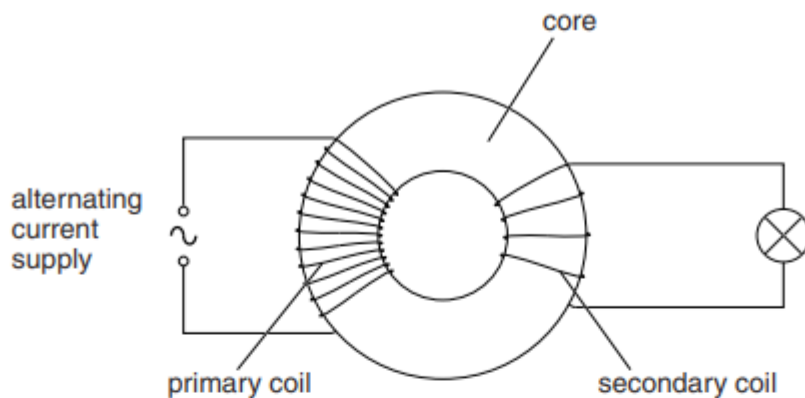


Fig. 8.1

- (a) State the metal used for the core of a transformer.

..... [1]

- (b) Explain how an alternating current in the primary coil causes the lamp to light.

.....

 [3]

- (c) Transformers are used to produce high voltages for the transmission of electrical power over long distances.

State one advantage of high voltage transmission.

.....

 [1]

43. O/N/12/22/Q11

(a) A wire is wound around a soft-iron core forming a solenoid, as shown in Fig. 11.1.

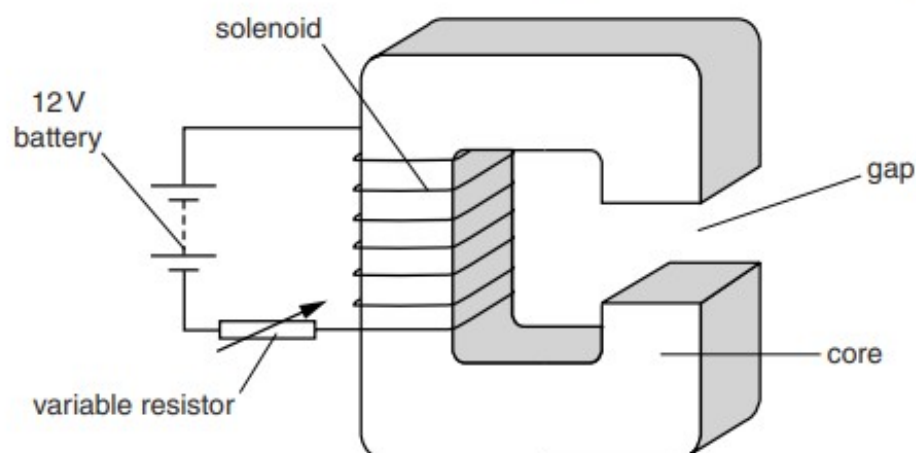


Fig. 11.1

There is a gap in the core. The solenoid is connected in series with a 12V battery and a variable resistor (rheostat). The resistance of the solenoid is $0.30\ \Omega$ and the variable resistor is set so that it has a resistance of $4.5\ \Omega$.

(i) Calculate the current in the solenoid.

current = [3]

(ii) The current in the solenoid magnetises the soft-iron core.

Explain how the electric circuit is used to increase the strength of the magnetic field.

.....

 [2]

(iii) Fig. 11.2 shows a horizontal, current-carrying wire PQ in the gap.

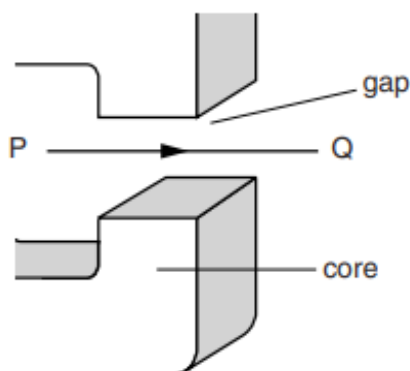


Fig. 11.2

1. The magnetic field in the gap is uniform and vertically upwards. The current in PQ is from left to right. Describe the effect of the magnetic field on PQ.

.....
 [2]

2. State the effect on PQ of increasing the strength of the magnetic field in the gap.

.....
 [1]

- (b) The starter motor in a car is powered by a 12V battery that is positioned next to the motor. The current in the motor is 75 A.

- (i) Calculate the power supplied by the battery.

power = [2]

- (ii) Suggest and explain why the wires that connect the motor to the battery are very thick.

.....

 [2]

- (c) A relay is used to switch on a starter motor in a car. Describe how the relay works. You may include a diagram in your answer.

.....

.....

.....

..... [3]

44. O/N/12/21/Q7

(a) Fig. 7.1 shows a solenoid made from wire wound around a plastic cylinder.

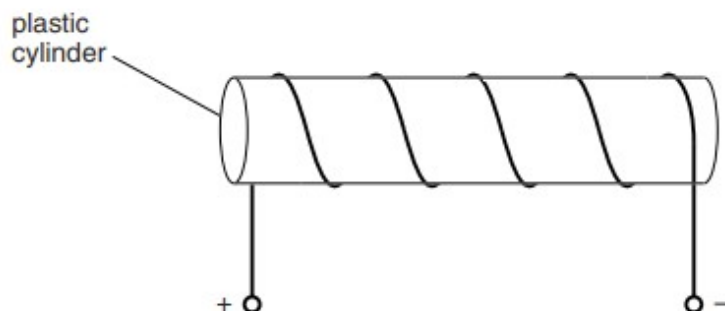


Fig. 7.1

A current in the solenoid produces a magnetic field.

On Fig. 7.1, draw the pattern of the magnetic field lines inside and outside the cylinder. [2]

(b) Fig. 7.2 shows a beam of beta-particles, in a vacuum, passing into a magnetic field.

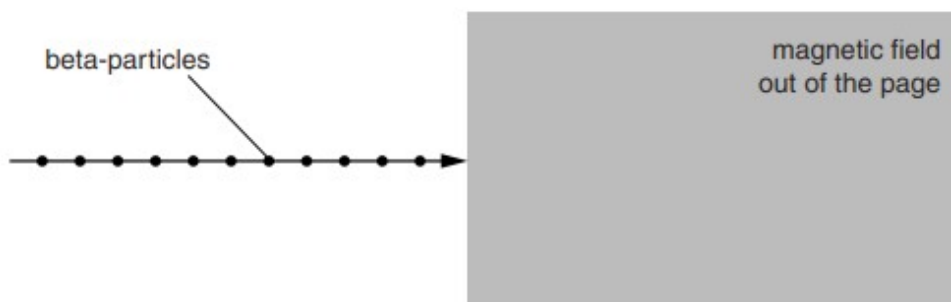


Fig. 7.2

The movement of the beta-particles from left to right is an electric current.

(i) On Fig. 7.2, draw an arrow to show the direction of the conventional current. [1]

(ii) A solenoid is used to produce the magnetic field that lies within the shaded region of Fig. 7.2. The direction of the field is out of the page.

1. On Fig. 7.2, draw the path followed by one of the beta-particles in the shaded region. [2]

2. The direction of the current in the solenoid is reversed. State what happens to the path of the beta-particle.

.....
 [1]

45. M/J/12/22/Q9

Fig. 9.1 shows two coils of insulated wire wound on an iron ring. Coil A is connected to a battery and a switch. The switch is open. Coil B is connected to a sensitive centre-zero voltmeter.

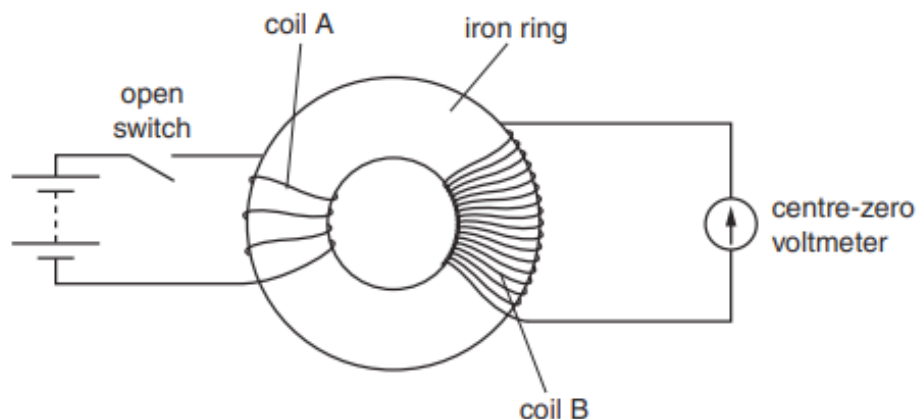


Fig. 9.1

The switch is closed. There is a current in coil A.

(a) On Fig. 9.1,

(i) mark the direction of the current in coil A, [1]

(ii) draw the magnetic field lines produced in the iron ring. [3]

(b) As the switch is closed, the voltmeter deflects to the right and then returns to zero.

(i) Explain why there is a deflection on the voltmeter.

.....

[2]

(ii) The switch is opened. State and explain what happens to the deflection on the voltmeter.

.....

[2]

(iii) Without changing coil A, state two changes to the apparatus that cause a greater deflection of the voltmeter.

1.

2.

[2]

- (c) The battery in Fig. 9.1 is replaced by an alternating current (a.c.) supply. The output from coil B is used to power a lamp that is a long distance away. Each lead from coil B to the lamp has a resistance of 2.5Ω . These leads are represented by the two resistors shown in Fig. 9.2.

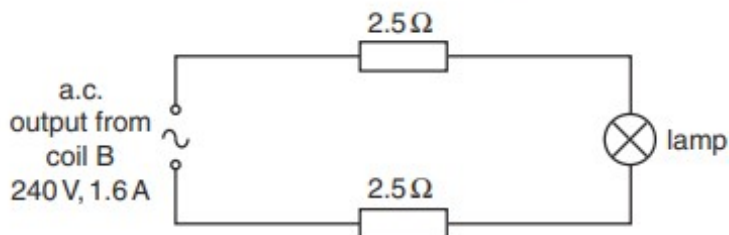


Fig. 9.2

The output voltage of coil B is 240V and the current in the circuit is 1.6 A.

- (i) Calculate the electrical power produced by coil B.

power =[2]

- (ii) Calculate the total power loss in the leads to the lamp.

power loss =[3]

46. M/J/12/21/Q8

- (a) A magnet is placed on a bench, as shown in Fig. 8.1a. End P of a rod is held above each end of the magnet in turn, as shown in Fig. 8.1b and in Fig. 8.1c. One end of the magnet is lifted off the bench in both cases.

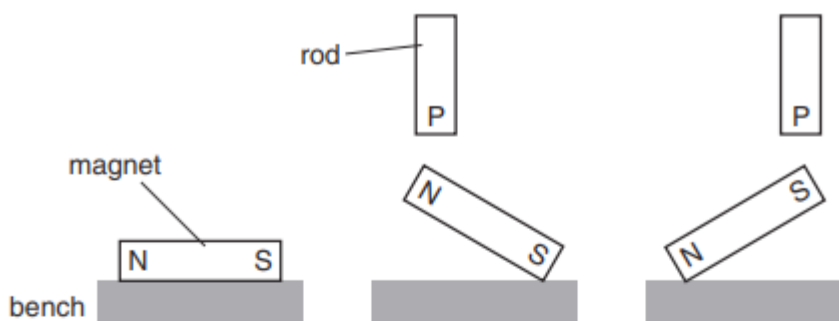


Fig. 8.1a

Fig. 8.1b

Fig. 8.1c

- (i) Suggest what material the rod is made from.

.....[1]

- (ii) Explain how the rod lifts each end of the magnet off the bench.

.....

[2]

- (b) Fig. 8.2 and Fig. 8.3 show views of a wire carrying a current downwards through a horizontal board.

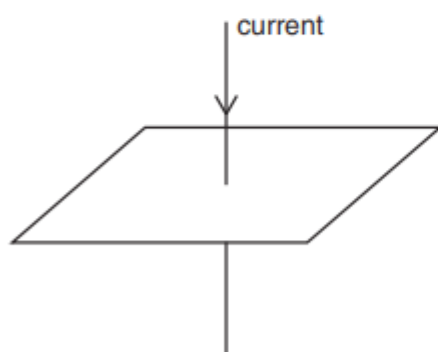


Fig. 8.2

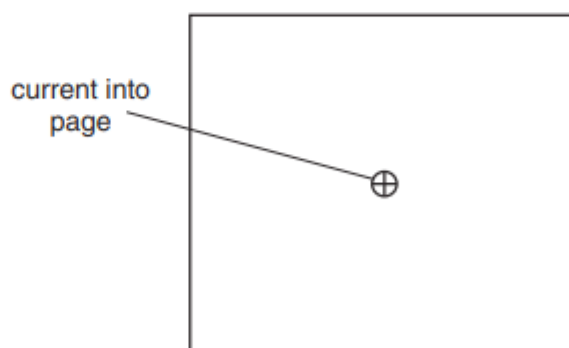


Fig. 8.3 (viewed from above)

- (i) On Fig. 8.3, draw the magnetic field due to the current in the wire. [2]

- (ii) The magnetic field is stronger closer to the wire. State how the magnetic field lines indicate that the field is stronger.

.....
[1]

47. O/N/11/22/Q5

Fig. 5.1 shows part of an electric bell.

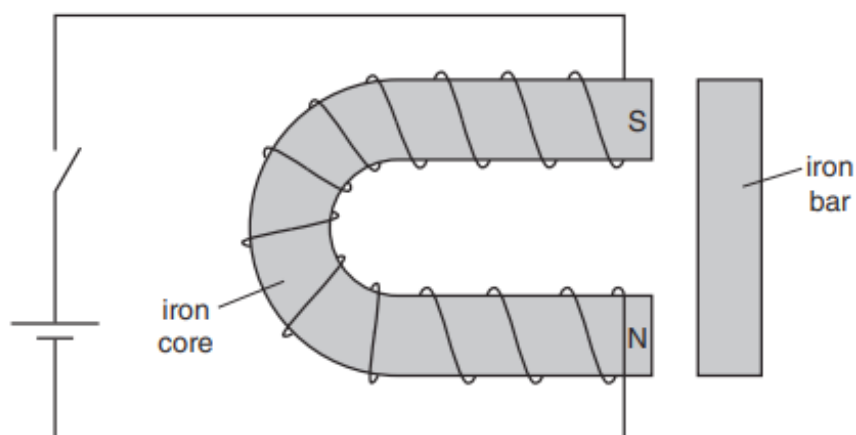


Fig. 5.1

A switch and a cell are in series with a length of wire coiled around an iron core.

The switch is closed and the current in the wire produces a south pole S and a north pole N at the ends of the core, as shown in Fig. 5.1. Magnetic poles are also produced in a small iron bar, placed near to the ends of the core.

- (a) (i) On Fig. 5.1, mark with an N the position of the north pole produced in the iron bar and mark with an S the position of the south pole produced in the iron bar. [1]
- (ii) State and explain what happens to the iron bar once it is magnetised.

.....
[2]

- (b) The switch is opened and there is no current in the wire. State what happens to the magnetic poles in the iron bar.

.....
[1]

48. O/N/11/21/Q6

When a car is moving, its electrical equipment is powered by an a.c. generator.

(a) The coil of the a.c. generator is rotated by the car engine.

(i) On the axes in Fig. 6.1, sketch a graph of the output voltage of the coil against time for two rotations of the coil of the generator.



[1]

Fig. 6.1

(ii) The speed of the car increases and so does the speed of rotation of the coil.

State two changes that this causes to the output voltage.

change 1

.....

change 2

.....

[2]

(b) When the car engine is off, the current in a lamp from a 12V battery is 0.50 A. Calculate the resistance of the lamp.

resistance =[2]

49. M/J/11/21/Q7

Fig. 7.1 shows the structure of a circuit-breaker that uses an electromagnet. The circuit-breaker operates when the current is 10 A.

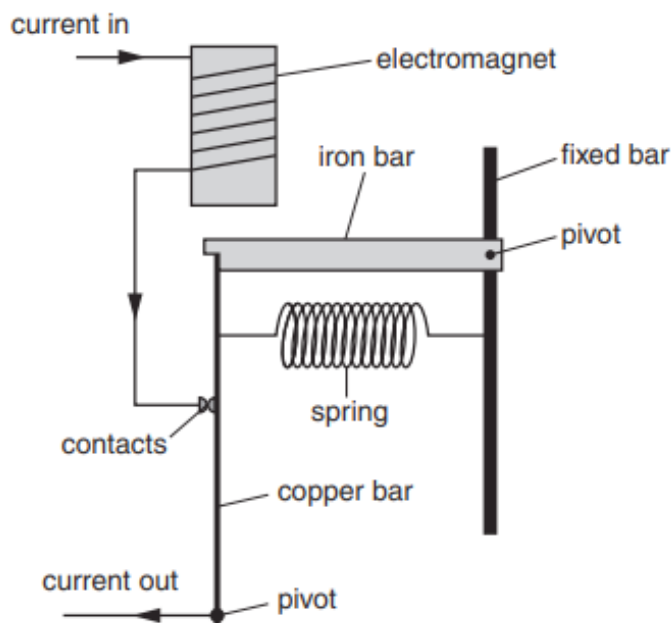


Fig. 7.1

(a) On Fig. 7.1, mark with an arrow the force on the iron bar caused by the electromagnet. [1]

(b) Suggest one reason why the iron bar does not move when the current is less than 10 A.

..... [1]

(c) When the current is greater than 10 A, the circuit-breaker stops the current. Explain what happens in the circuit-breaker when this occurs.

..... [3]

(d) State and explain how the electromagnet can be altered so that the circuit-breaker stops the current at less than 10 A.

..... [2]

50. M/J/11/21/Q8

- (a) A wire carrying a current in a magnetic field experiences a force due to the current. On Fig. 8.1, insert the words **current**, **field** and **force** in the boxes to show the relative directions of the current, the magnetic field and the force.

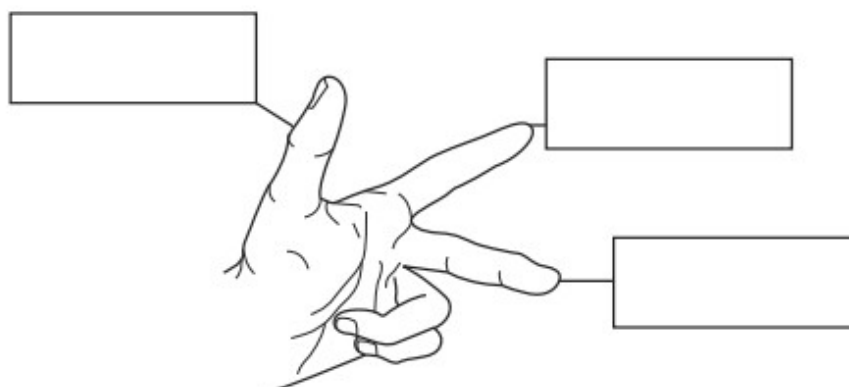


Fig. 8.1

[1]

- (b) Fig. 8.2 shows a current-carrying coil ABCD in a magnetic field.

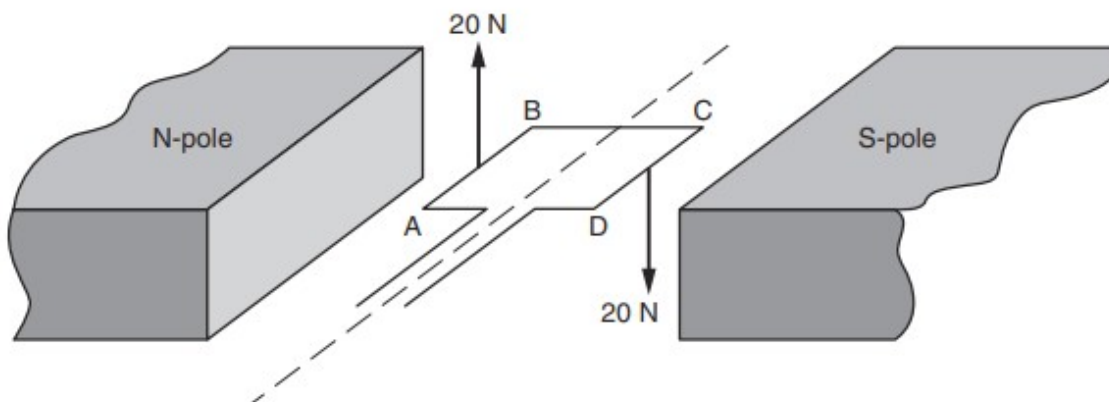


Fig. 8.2

Each side of the coil is 4.0 cm in length. The force on AB is 20 N and the force on CD is 20 N.

- (i) Calculate the total moment caused by these forces.

moment = [2]

- (ii) The moment is increased by using a stronger magnetic field.

State two other ways to increase the moment.

1.

2.

[2]

51. M/J/11/21/Q9

Fig. 9.1 shows a rotating magnet in an alternating current generator that is used to power a lamp.

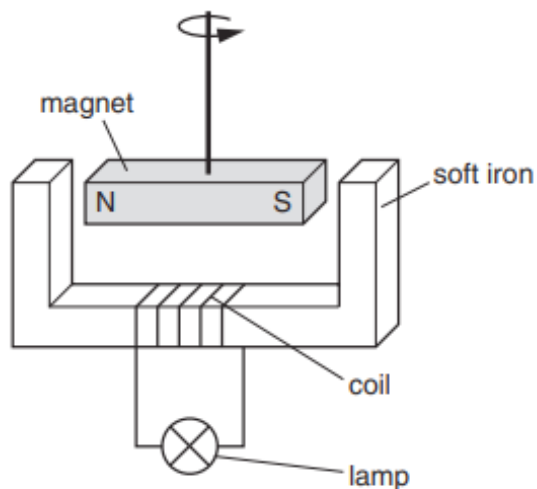


Fig. 9.1

- (a) (i) State how an *alternating* current differs from a *direct* current.

.....

 [1]

- (ii) Explain, in detail, how **alternating** current is produced by the apparatus shown in Fig. 9.1.

.....

 [4]

- (iii) State two ways in which the current in the lamp may be increased.

1.
 2.

[2]

- (b)** The generators at a power station produce a voltage of 25 000V. This voltage is stepped up to 400 000V by a transformer for long-distance transmission on overhead power lines. The voltage is later stepped down to 240V.

(i) State and explain why the voltage is stepped up for long-distance transmission.

.....
.....
..... [2]

(ii) Calculate the ratio of the number of turns in the primary coil of the step-up transformer to the number of turns in its secondary coil.

ratio = [1]

(iii) State one advantage and one disadvantage of using thicker wire in the overhead power lines.

advantage:

.....

disadvantage:

..... [2]

(iv) An electric drill of power 1000W is used in a country where the mains voltage is 240V. State and explain the most appropriate fuse to use with this drill.
You should select a fuse from the following values: 1 A, 3 A, 4 A, 13 A.

.....

.....

.....

..... [3]

52. O/N/10/22/Q6

Fig. 6.1 shows a metal bar placed inside a vertical solenoid.

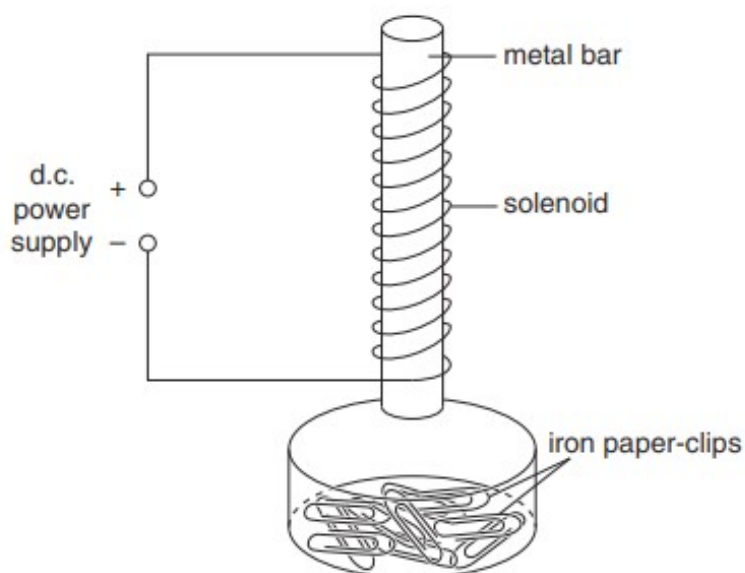


Fig. 6.1

The solenoid is a coil of several turns of insulated wire. A d.c. power supply is connected to the solenoid so that there is a current in it when the supply is switched on. The metal bar is a short distance above a small pile of iron paper-clips in a glass dish.

The power supply is

- switched on,
- left on for several seconds,
- then switched off.

Describe the behaviour of the paper-clips when this procedure is carried out using a metal bar of

(a) aluminium,

.....
.....[1]

(b) iron,

.....
.....[2]

(c) steel.

.....
.....[2]

53. O/N/10/21/Q6

Fig. 6.1 shows the coil of a loudspeaker attached to a cardboard cone. One pole of a stationary cylindrical magnet lies near to the coil.

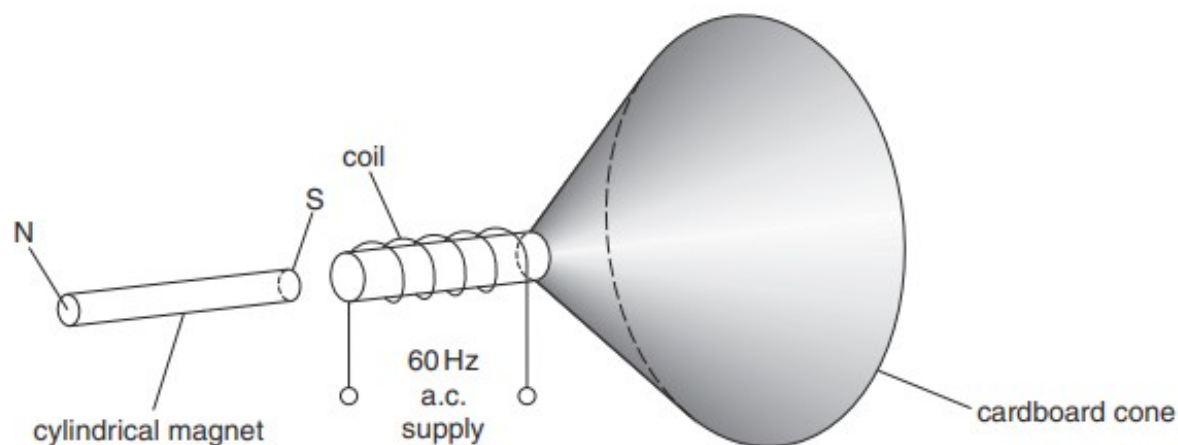


Fig. 6.1

There is an alternating current in the coil of the loudspeaker. A student hears the note produced.

- (a) (i) Explain why the cone of the loudspeaker vibrates.

.....

.....

.....

..... [3]

- (ii) Explain how the vibrating cone produces sound waves in the air.

.....

.....

..... [2]

- (b) A stronger cylindrical magnet is now used. State the difference in the note heard.

.....

..... [1]

54. M/J/10/22/Q7

Fig. 7.1 shows a compass.

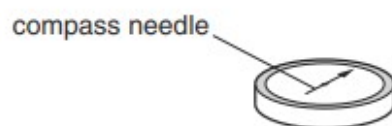


Fig. 7.1

The compass needle is a small magnet free to rotate. The head of the arrow on the compass needle is an N-pole.

(a) A bar magnet is placed between two compasses, as shown in Fig. 7.2.

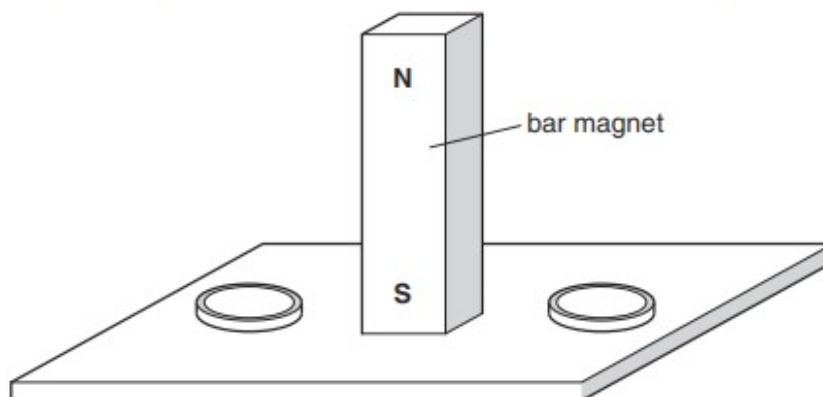


Fig. 7.2

On Fig. 7.2, draw the needles inside the two compasses and mark the N-pole of both compass needles. [2]

(b) Fig. 7.3 shows the structure of a relay.

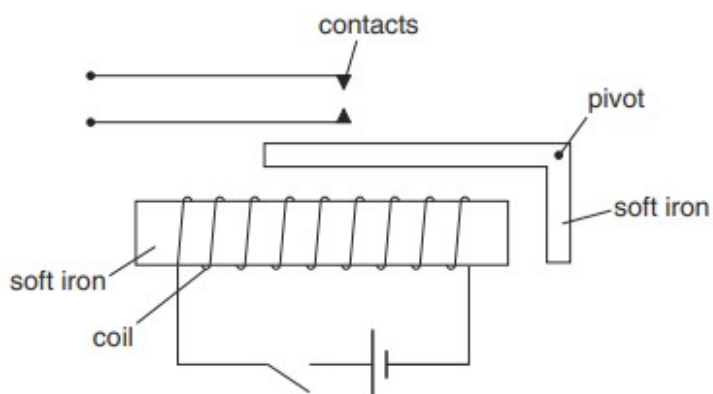


Fig. 7.3

Explain how closing the switch causes the contacts to close.

.....

.....

.....[2]

(c) Fig. 7.4 is a circuit that includes a relay.

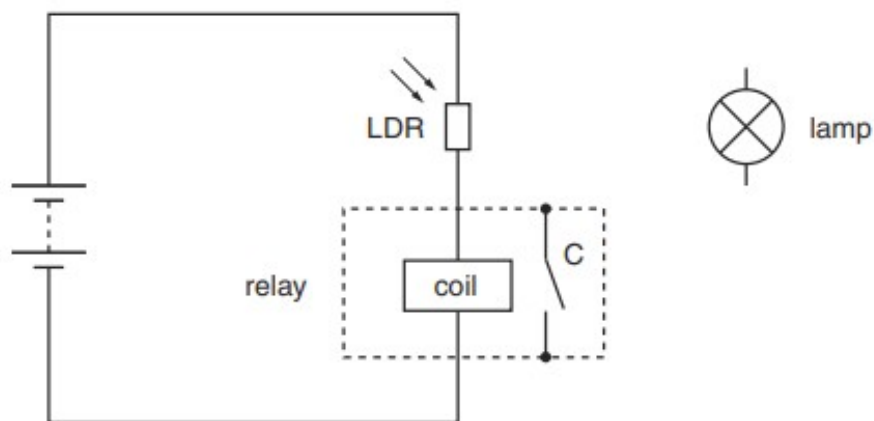


Fig. 7.4

When light shines on the light-dependent resistor (LDR), the relay contacts C close.

(i) State what happens to the resistance of the LDR when light falls on it.

.....[1]

(ii) The circuit for the lamp, as shown in Fig. 7.4, is not complete.

On Fig. 7.4, draw the connections to the lamp, the contacts C and the battery that cause the lamp to switch on when light shines on the LDR. [2]

55. M/J/10/21/Q8

Fig. 8.1 shows two coils of wire wound on an iron ring. One coil is connected in series to a switch and a d.c. supply. The other coil is connected to a very sensitive centre-zero ammeter.

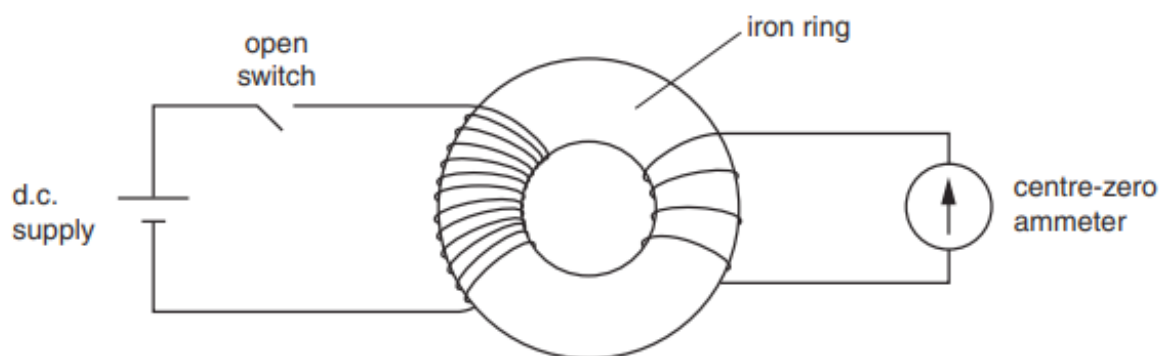


Fig. 8.1

At first the switch is open, as shown in Fig. 8.1.

The following actions are taken in turn.

Describe and explain what happens to the reading on the ammeter in each case.

(a) The switch is closed.

.....

.....

.....

..... [3]

(b) The switch is left closed for a long time.

.....

.....

..... [1]

(c) The switch is opened.

.....

.....

..... [2]

56. M/J/09/Q7

Fig. 7.1 shows two pieces of soft iron in the magnetic field of a strong permanent magnet.

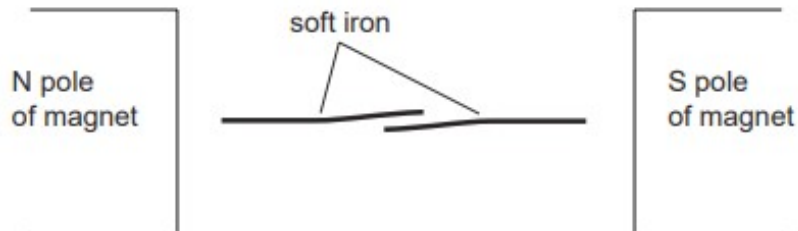


Fig. 7.1

The pieces of soft iron become magnetised.

(a) On Fig. 7.1, mark the magnetic poles produced at each end of both pieces of soft iron. [1]

(b) Fig. 7.2 shows a reed switch.

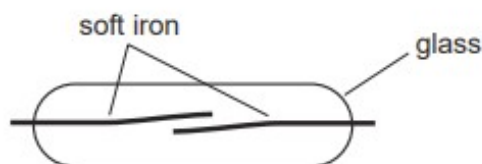


Fig. 7.2

The reed switch is placed between the poles of the strong permanent magnet. State and explain what happens.

.....

 [2]

(c) Fig. 7.3 shows two separate electrical circuits.

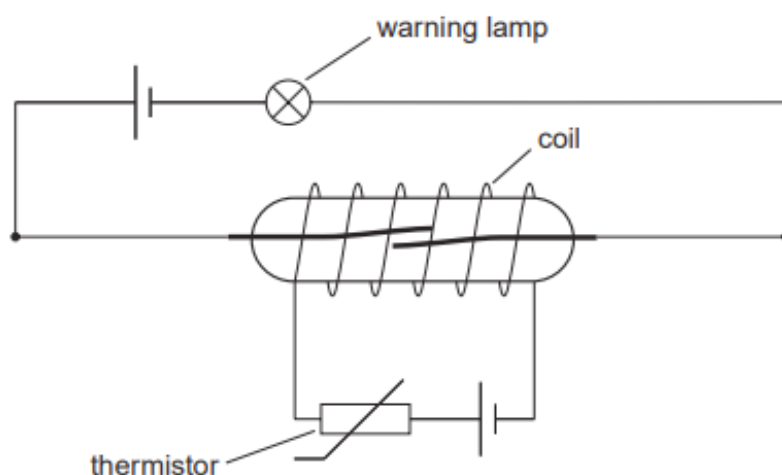


Fig. 7.3

One circuit consists of a reed switch, a cell and a warning lamp. The other circuit consists of a thermistor, another cell, and a coil wound round the reed switch. The thermistor is at the same temperature as the air around it.

(i) State what happens to the thermistor when the temperature of the air rises.

..... [1]

(ii) Explain why the warning lamp lights up when the air temperature rises.

.....
.....
.....
..... [2]

57. M/J/08/Q7

- (a) Fig. 7.1 shows a straight wire between the poles of a magnet. The wire carries a current into the page.

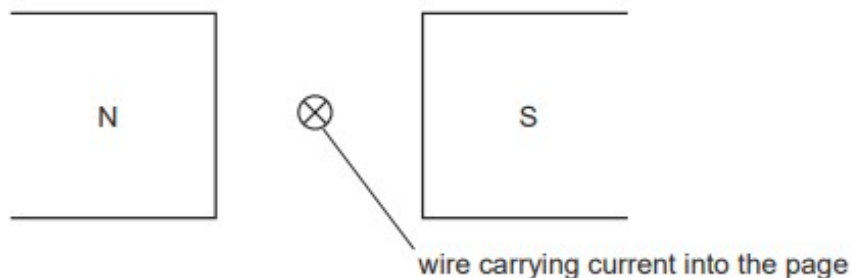


Fig. 7.1

- (i) State the direction of the magnetic field between the poles of the magnet.
 [1]
- (ii) On Fig. 7.1, draw an arrow to show the direction of the force acting on the wire. [1]
- (b) Fig. 7.2 shows two wires.
 Each wire carries a current into the page.



Fig. 7.2

- (i) On Fig. 7.2, draw the magnetic field due to the currents in the wires. [3]
- (ii) There is a force on each wire due to the current in the other wire.
 On Fig. 7.2, draw an arrow on each wire to show these forces. [1]

58. O/N/07/Q7

Fig. 7.1 shows apparatus that can be used to make an electromagnet or a permanent magnet.

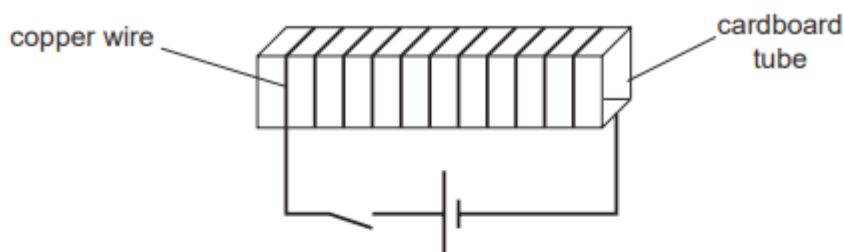


Fig. 7.1

Four rods are available. They are made of aluminium, soft iron, steel and wood.

- (a) (i) State which rod is used to make a permanent magnet.

..... [1]

- (ii) Describe how the apparatus is used to make a permanent magnet.

.....
 [1]

- (b) A computer component is screened from external magnetic fields by placing it in a box, as shown in Fig. 7.2.

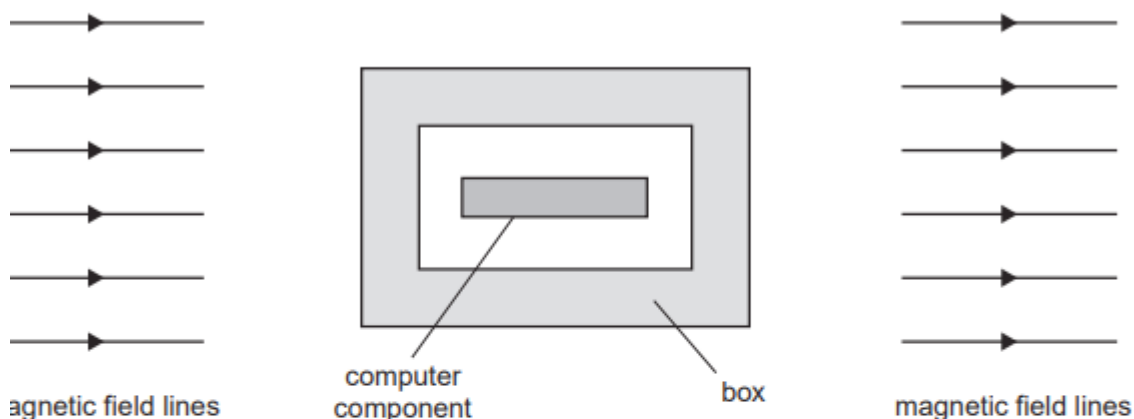


Fig. 7.2

There is a strong magnetic field outside the box. The magnetic field lines have not been drawn near the box.

- (i) State the best choice for the material of the box.

..... [1]

- (ii) On Fig. 7.2, join the magnetic field lines on the left of the box to those on the right, showing the pattern of the magnetic field. [2]

59. M/J/07/Q6

Fig. 6.1 shows a coil of wire wound on a cardboard tube.

There is a d.c. current in the coil. The direction of the current is shown in the key.

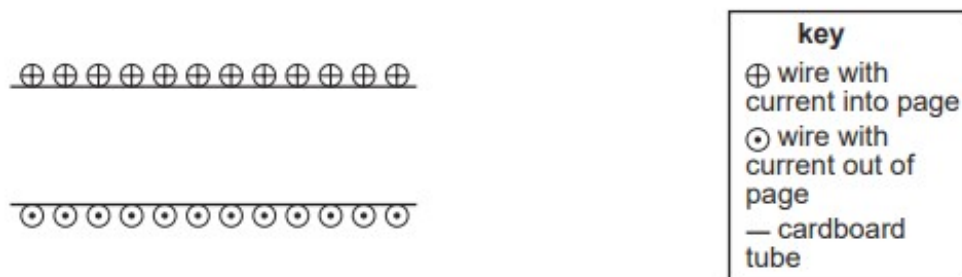


Fig. 6.1

- (a) On Fig. 6.1, draw the magnetic field produced by the coil. [3]
- (b) Fig. 6.2 shows a simple loudspeaker that uses the coil shown in Fig. 6.1 attached to a paper cone.

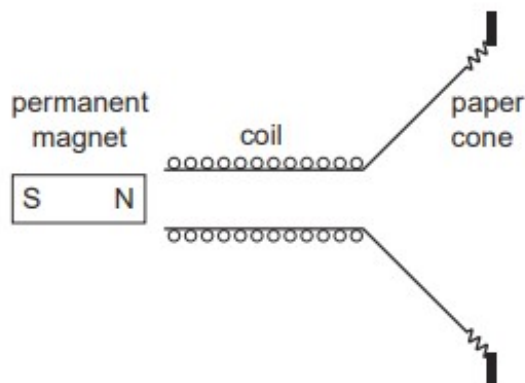


Fig. 6.2

The coil is connected to a signal generator.

There is an alternating current of frequency 100Hz in the coil.

- (i) State what is meant by a *frequency of 100Hz*.

.....

..... [1]

(ii) Describe and explain the movement of the coil.

.....

.....

.....

.....

.....

..... [3]

60. O/N/06/Q7

Fig. 7.1 shows one way to demonstrate an electrical effect.

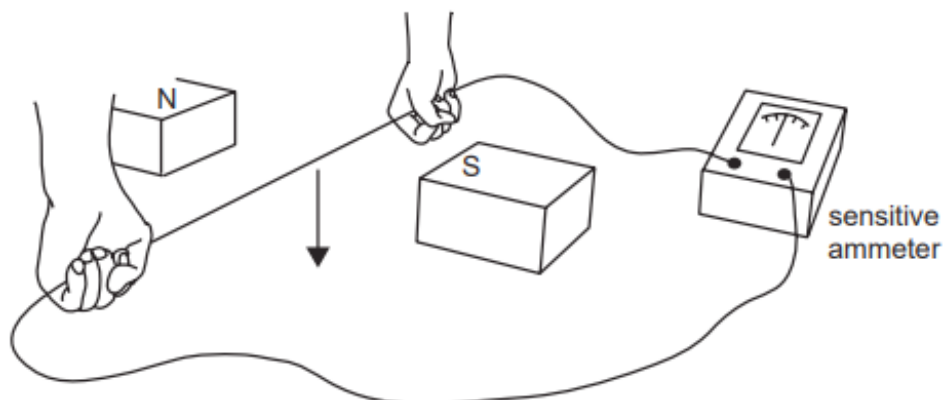


Fig. 7.1

As the wire is moved downwards between the magnetic poles, the needle of the ammeter deflects to the right.

(a) State the name of this electrical effect.

..... [1]

(b) State what happens to the needle of the ammeter when the wire is moved upwards between the magnetic poles.

.....
..... [1]

(c) State and explain what happens when the wire is held stationary between the magnetic poles.

.....
.....
.....
.....
..... [2]

61. M/J/06/Q3

Fig. 3.1 shows the construction of a simple a.c. generator. When the coil is rotated an e.m.f. is induced in the coil.

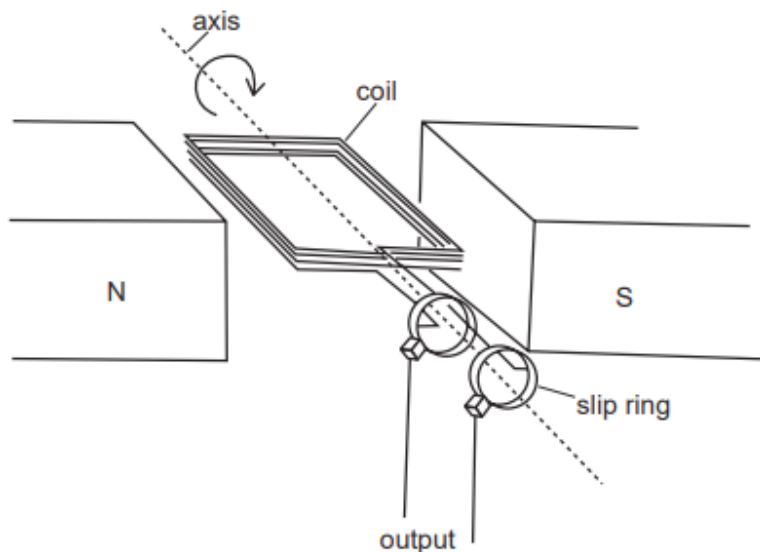


Fig. 3.1

- (a) Explain why an e.m.f. is induced.

.....

.....

.....

..... [2]

- (b) State the purpose of the slip rings.

.....

..... [1]

- (c) The direction of the current in the coil can be found from Lenz's law.

State Lenz's law.

.....

.....

.....

..... [1]

- (d) The induced e.m.f. can be increased by rotating the coil faster. State one other way in which the e.m.f. can be increased.

.....

..... [1]

62. M/J/06/Q7

Fig. 7.1 shows an electrical circuit and a cathode-ray oscilloscope (C.R.O.).

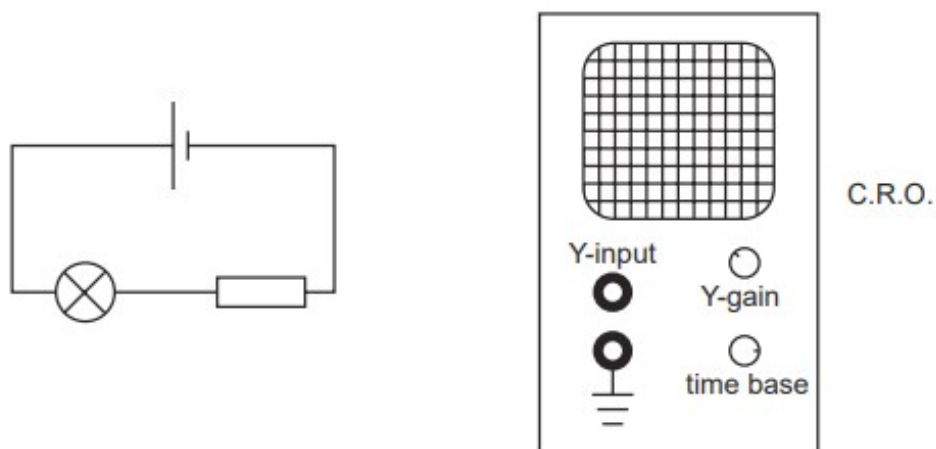


Fig. 7.1

(a) On Fig. 7.1, draw the connections you would make to enable the C.R.O. to measure the potential difference (p.d.) across the resistor. [1]

(b) Fig. 7.2 shows the trace on the screen before and after the connections are made.

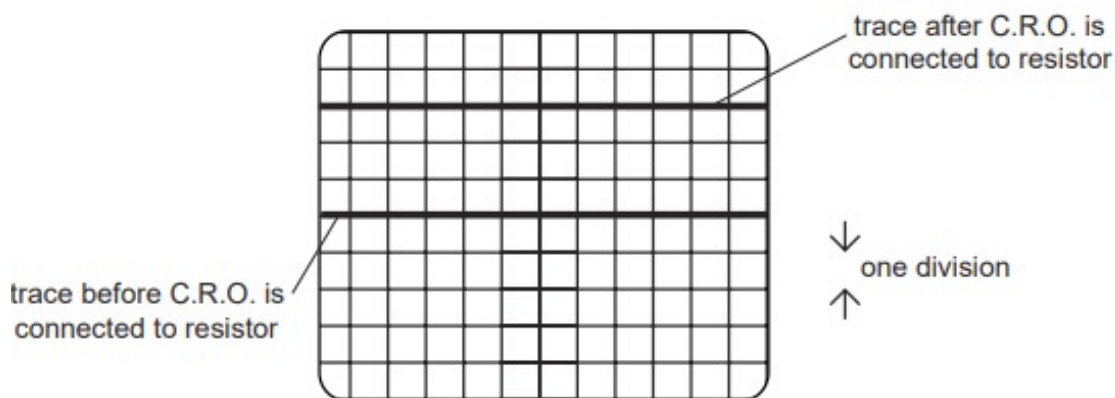


Fig. 7.2

The Y-gain control is set at 2 V for each division on the screen.

(i) State the value of the p.d. across the resistor.

p.d. = [2]

(ii) The Y-gain control is altered to 4 V for each division.

On Fig. 7.2, draw the new trace seen on the screen.

[1]

63. O/N/05/Q7

Fig. 7.1 shows a coil of wire wound around a rectangular tube.

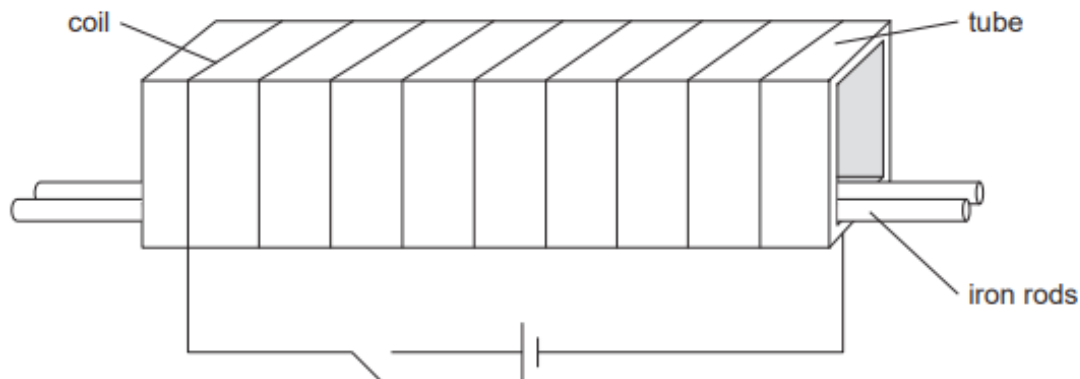


Fig. 7.1

- (a) Two iron rods are placed next to each other at the bottom of the tube. When the current is switched on, the two rods repel each other. They move to the sides of the tube.

Explain why the two iron rods repel.

.....

.....

.....

..... [2]

- (b) An iron rod and a similar copper rod are placed next to each other at the bottom of the tube. State and explain what, if anything, happens to the rods when the current is switched on.

.....

.....

.....

..... [2]

64. M/J/05/Q5

Fig. 5.1 shows a coil of wire wrapped around a plastic tube. Inside the tube are two pieces of soft iron. When the switch is closed, the compass needles point in the direction of the magnetic field produced at each position. You may ignore the magnetic field of the Earth in this question.

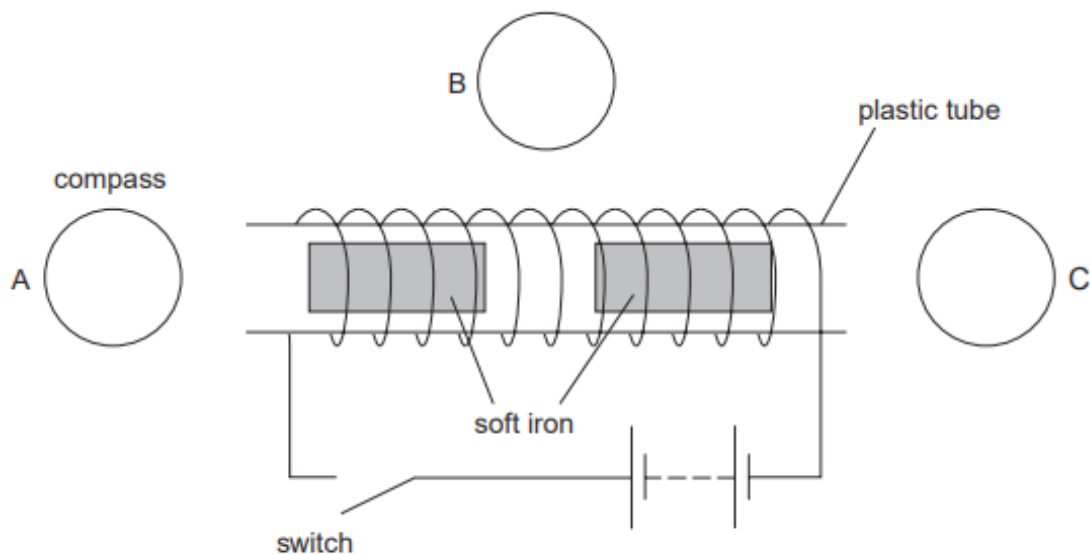


Fig. 5.1

- (a) On Fig. 5.1 mark arrows, in compasses A, B and C, to show the direction of the magnetic field at each position after the switch has been closed. [2]

- (b) When the switch is closed, the two pieces of soft iron in the tube become magnets and move.

- (i) On Fig. 5.1, mark the poles formed on each piece of soft iron. [1]

- (ii) State and explain how the pieces of iron move.

.....

 [2]

- (c) State the effect on the magnetic field of

- (i) reversing the direction of the current,

.....
 [1]

- (ii) reducing the size of the current.

.....
 [1]

65. O/N/04/Q7

Fig. 7.1 shows high voltage cables used to transmit electrical energy.

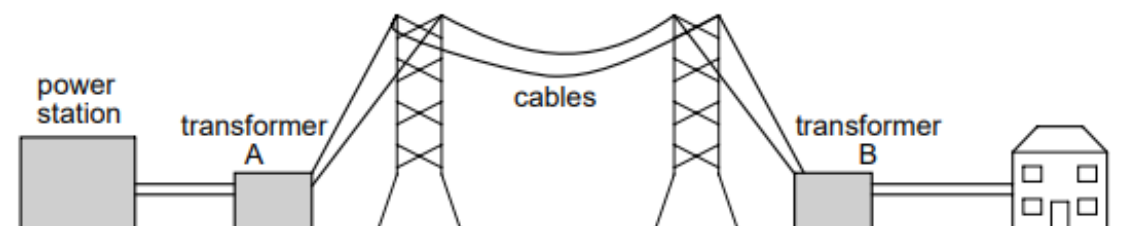


Fig. 7.1

(a) State the purpose of transformer B.

.....[1]

(b) In the space below, draw a labelled diagram to show the structure of transformer B.

[3]

(c) (i) Explain why high voltages are used to transmit electrical power.

.....
.....
.....
.....
.....
.....

- (ii) Fig. 7.2 shows how the loss of thermal energy from a cable varies with the thickness of the cable.

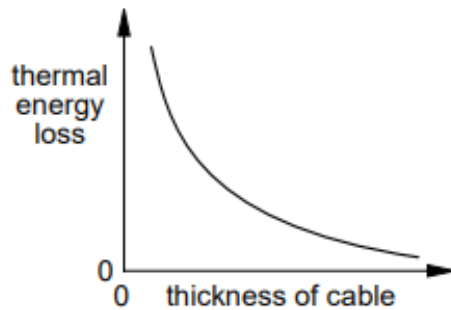


Fig. 7.2

Explain why the loss of thermal energy is less if the cable is thicker.

.....

.....

.....

.....

.....

[4]

66. M/J/04/Q5

Fig. 5.1 shows a piece of recording tape passing under the recording head of a tape recorder. An alternating current is passed through the coil. The tape is coated with a magnetic material that becomes magnetised.

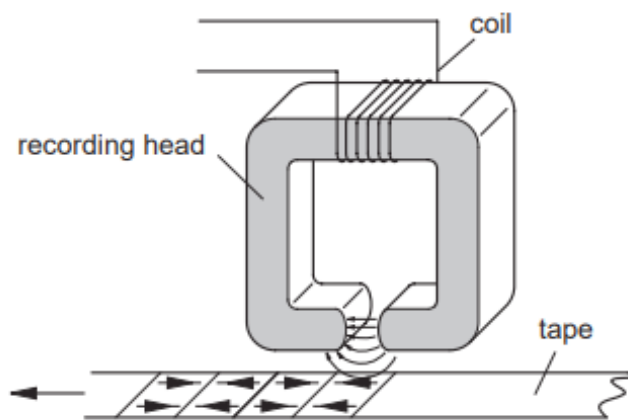


Fig. 5.1

- (a) (i)** Explain why the tape becomes magnetized.

.....

.....

.....

- (ii)** Fig. 5.1 shows that sections of the tape are magnetised in opposite directions. Explain why they become magnetised in opposite directions.

.....

.....

- (iii)** The tape is moved faster past the recording head. State how this changes the pattern on the tape.

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[3]

- (b) (i)** Explain why the coating on the tape must be of a permanent magnetic material.

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- (ii)** State the name of a permanent magnetic material.

.....

[2]

ANSWERS

1.

Question	Answer	Marks
8(a)	steel	B1
8(b)(i)	magnetic field (lines of magnet) cut (by solenoid) or magnetic field (in solenoid) changes	B1
	electromotive force / e.m.f. <u>induced</u> / current <u>induced</u>	B1
	electromotive force / e.m.f. cause current (in closed circuit)	B1
8(b)(ii)	Lenz law mentioned or to oppose the change (causing the current)	B1
	(In order to) repel magnet (upwards) or upward force on magnet or repulsion (of magnet) mentioned or to oppose the magnet or resists motion of magnet	B1
8(c)	current / it changes direction	B1
	magnitude / size of current changes or current increases / decreases	B1

2.

Question	Answer	Marks
7(a)(i)	P: (carbon) brush and Q: slip ring(s)	B1
7(a)(ii)	to connect the output terminal(s) and the (rotating) coil/brushes	B1
	without wires twisting	B1
7(b)(i)	at least one of the five positions when the output e.m.f. is zero indicated with X and no indications elsewhere	B1
7(b)(ii)	(amplitude $\equiv$) 3.6 (cm) or ($E_{\max}$ $\equiv$) 5.0×3.6 or 36 (V) seen	C1
	18 (V)	A1
7(b)(iii)	(T $\equiv$) 8.0 (m s)	B1
7(b)(iv)	(f $\equiv$) $1/T$ or $1/8.0$ or $1/(4.0 \times 2.0)$ or $1/0.0080$	C1
	130 (Hz)	A1

3.

Question	Answer	Marks
8(a)	(soft) iron	B1
8(b)	input voltage or input current is alternating	B1
	changing (magnetic) field (in primary causes changing magnetic field in secondary) (alternating emf is) <u>induced</u> (in secondary coil)	B1
	or (magnetic) field lines cut secondary	B1
8(c)	(N_s $\equiv$) $4600 \times 5 / 230$ or $V_p / V_s = N_p / N_s$ in any form	C1
	100	A1
8(d)	reverse the coils OR connect the input to the secondary / right-side coil OR secondary has more turns than <u>primary</u> / more than 4600 (with same number on primary)	B1

4.

Question	Answer	Marks
9(a)	<u>magnetic</u> field or <u>magnetic</u> flux mentioned (from magnets)	B1
	it / sides of <u>coil</u> cut (magnetic) field or (magnetic) field (strength) / flux in <u>coil</u> changes	B1
9(b)	AB on right – 6 V	B1
	coil vertical 0 V in both positions	B1
9(c)(i)	one ring (instead of two) with gaps between the two halves	C1
	each side of split ring connected to a different end of the coil and to a brush	A1
9(c)(ii)	(force on) current in a magnetic field or left-hand rule mentioned or magnetic field of coil interacts with field of magnet	B1

5.

Question	Answer	Marks
6(a)	<u>magnetic</u> field / flux mentioned	B1
	changing / alternating (magnetic) field	B1
	voltage <u>induced</u> (in secondary coil)	B1
6(b)(i)	$(V_s =) V_p \times N_s / N_p$ or $240 \times 350 / 5600$	C1
	15 V	A1
6(b)(ii)	$(I =) P / V$ or $90 / 15$	C1
	6.0 A	A1

6.

Question	Answer	Marks
6(a)	soft iron	B1
6(b)(i)	$(P =) VI$ or 220×39	C1
	$8.6 \times 10^3 \text{ W}$	A1
6(b)(ii)	$(Q =) It$ or 39×4.5 or $39 \times 4.5 \times 60$ or $39 \times 4.5 \times 60 \times 60$ or $39 \times 4.5 \times 3600$	C1
	180 or 11 000 or $39 \times 4.5 \times 60 \times 60$ or $39 \times 4.5 \times 3600$	C1
	$6.3 \times 10^5 \text{ C}$	A1
6(c)	resistance increases with temperature or resistance lower (because of (low / ambient) temperature)	B1

7.

Question	Answer	Marks
9(a)	always positive or never negative or direction does not change	B1
	constant value or size does not change	B1
9(b)(i)	$(I =) V / R$ or $12 / (20 + 28)$ or $12 / 48$	C1
	0.25 A	A1
9(b)(ii)	YZ in magnetic <u>field</u> or magnetic <u>field</u> between poles (of magnet)	M1
	(electric) current in YZ or interacts with field of current / YZ	A1
9(b)(iii)	downwards; middle, right-hand box ticked i.e. towards N ☐ towards S ☐ towards Y ☐ towards Z ☐ downwards ☒ upwards ☐	B1
9(b)(iv)	current from Z to Y page / into page and magnetic field left to right	B1
	(Fleming's) left-hand rule / other appropriate rule mentioned / described	B1
9(c)(i)	$1 / R_T = 1 / R_1 + 1 / R_2$ or $(R_T =) R_1 R_2 / (R_1 + R_2)$ or $1 / R_T = 1 / 20 + 1 / 30$ or $1 / R_T = 5 / 60$ or $1 / R_T = 1 / 12$ or $(R_T =) 20 \times 30 / (20 + 30)$	C1
	12	C1
	40 W	A1
9(c)(ii)	resistance decreases or resistance is less (with S_2 closed)	B1
	current increases and force increases	B1
9(c)(iii)	1.5	B1

8.

6(a)(i)	top magnet N pole at bottom and S pole at top	B1
	like poles repel or N and N repel	B1
6(a)(ii)	iron magnetised (by induction) / iron a temporary magnet / poles produced on iron (by magnet)	B1
	unlike poles attract or N -pole attracts S-pole or bottom pole of iron is a S pole or S pole (on iron) faces N-pole (on magnet)	B1
6(b)	at least three circles centered on wire	B1
	clockwise direction marked on at least one closed shape around wire and no arrows wrong	B1

9.

Question	Answer	Marks
5(a)	brush / carbon	B1
	split-ring or commutator	B1
5(b)(i)	faster or greater turning effect	B1
5(b)(ii)	goes backwards / turns backwards / reverses	B1
5(c)(i)	speeds up and slows down (when turned on / off)	B1
	reaches a steady speed / terminal speed / maximum speed or slows down but never to rest or cycle continuously repeats	B1
5(c)(ii)	same top / constant speed or constant speed / maximum speed for a longer time or drops to a lower speed / comes to rest	B1

10.

Question	Answer	Marks
7E(a)	magnetic field lines cut the solenoid / coils or changing magnetic field in solenoid	B1
	an electromotive force / e.m.f. is <u>induced</u>	B1
	electromotive force / e.m.f. produces the current	B1
7E(b)	negative / opposite / reversed current / deflection / reading	B1
	negative e.m.f. / voltage (induced) or magnetic field cut in opposite direction or magnetic field changes in opposite sense / decreases (owtte)	B1

11.

Question	Answer	Marks
7(a)	<u>current</u> (in coil) is (any one from) - away from A or towards B - from positive to negative - at right angles to magnetic field - makes the coil an (electro)magnet	B1
	left-hand rule stated, mentioned or used or force at right angle to field and current or top face of coil is a S-pole / bottom face a N-pole	B1
	<u>forces</u> in opposite directions (on opposite sides causing rotation) or unlike poles attract / like poles repel (only if coil is electromagnet)	B1
7(b)(i)	stronger magnetic field or (soft) iron becomes (temporary) magnet / magnetises easily	B1
	stronger / larger <u>force</u>	B1
7(b)(ii)	larger current / larger voltage / more turns on coil / wider coil / larger coil / thicker wire / stronger magnet	B1

12.

Question	Answer	Marks
6(a)	bar cuts <u>magnetic</u> field (lines / flux) or change in <u>magnetic</u> flux (in circuit with ammeter)	B1
	emf or current <u>induced</u>	B1
6(b)	negative (small)	B1
	no reading	B1
	no reading	B1
6(c)	move bar faster or wider magnet	B1
6(d)	force downwards (caused by current in magnetic field) or <u>force</u> opposes movement / motion	B1

13.

Question	Answer	Marks
9(a)	steel underlined and no other material indicated	B1
9(b)	use of (plotting) compass	B1
	place the (plotting) compass (on the paper next to the magnet) and mark a dot where the compass needle points	B1
	move compass / other end of needle next to the dot and mark a dot where the compass needle points	B1
	continue and needle points in the direction of the magnetic field	B1
9(c)(i)	(free) electrons attracted to K / positive terminal or repelled by J / negative terminal	B1
	(conventional current) from K to J or from positive or to negative or from right or to left	B1
9(c)(ii)	(Fleming) left-hand rule mentioned or equivalent rule mentioned	B1
	magnetic field related to index / pointer finger / equivalent	B1
	current related to middle finger / equivalent	B1
	thumb points down the page / equivalent and force down (the page)	B1
	or catapult field explanation: field lines clockwise when viewed from K or anticlockwise when viewed from J	B1
	extra field lines / stronger field above wire	B1
	fewer field lines / weaker field below wire	B1
	force towards weaker field	B1
9(c)(ii)	there is no force F or it disappears / becomes equal to zero	B1
	wire is magnetically screened / shielded by the iron tube	B1
9(d)	moment acts (on coil) or forces (on vertical sides) in opposite directions	B1
	currents (in vertical sides) or in opposite directions	B1
	or coil becomes / acts as a magnet or coil produces magnetic field	(B1)
	N face (of coil) attracted to S pole (of magnet) or S face (of coil) attracted to N pole (of magnet) or N face (of coil) repelled by N pole (of magnet) or S face (of coil) repelled by N pole (of magnet)	(B1)

14.

Question	Answer	Marks
6(a)	<u>iron</u> core and two coils wrapped around core	B1
	input and output clear and more turns on secondary / output coil	B1
6(b)	magnetic field mentioned	B1
	alternating current (a.c.) produces magnetic field that varies	B1
	core channels magnetic field to secondary coil and electromagnetic <u>induction</u>	B1
6(c)	less energy transferred to thermal energy in wires	B1
	(because) smaller current (for a given power)	B1

15.

Question	Answer	Marks
9(a)(i)	core becomes magnetised (due to current in coil)	B1
	iron armature attracted to core	B1
	contacts touch	B1
9(a)(ii)	steel is a permanent magnet or iron is a temporary magnet	B1
	with a steel core motor stays on or contacts stay closed (if switch S is opened)	B1
9(a)(iii)	(with relay) safer or switch S connected to low voltage or switch only handles low currents	B1
9(b)(i)	at least two lines inside coil and horizontal near centre and covering at least six wires of coil	B1
	correct shape of lines fanning out at ends of coil	B1
	at least two lines passing outside the coil from one end to the other	B1
	correct direction of field lines on at least one line and none wrong	B1
9(b)(ii)	N-pole on right hand side	B1
9(c)(i)	move magnet into / near coil or move coil	C1
	<u>move</u> magnet / coil <u>quickly</u> or <u>move</u> <u>strong</u> magnet	A1
9(c)(ii)	lines of magnetic field cut the coil or flux in coil changes	B1
	<u>induced</u> e.m.f. / voltage produced (in complete circuit)	B1

16.

Question	Answer	Marks
10(a)(i)	ammeter in series connecting switch with motor	B1
	voltmeter across motor or battery	B1
10(a)(ii)	(E =) VIt in any form numerical or algebraic or P = VI and P = E / t in any form	C1
	48 J	A1
10(a)(iii)	(E = mgh) in any form numerical or algebraic	C1
	$0.15 \times 0.8(0) \times 10$	C1
	1.2 J	A1
10(a)(iv)	ANY two of - motor / mass has kinetic energy - heat generated in motor / wires - heat generated in air due to drag / resistance <u>work done</u> to lift string	B2
10(b)(i)	mention of magnetic field (between N and S poles)	B1
	coil cuts lines of magnetic flux / magnetic field lines or flux / field in coil changes	B1
	induction	B1
10(b)(ii)	statement of Lenz's law , e.g. (induced) current flows in a direction to oppose the change producing it	B1
	coil becomes a magnet or current flowing in coil produces force in field	M1
	poles of coil attracted to opposite poles on magnet or force up on a side as side moves down	A1

17.

Question	Answer	Marks
6(a)(i)	4.0 V c.a.o.	B1
6(a)(ii)	(V =) $12 - 4.0$ or 8.0 or $4.0 / 700$ or 5.7×10^{-3}	C1
	(R =) V / I or $8.0 \times 1 / (4.0 / 700)$ or $700 \times (8.0 / 4.0)$ or $12 / (4.0 / 700)$ or 2100 (Ω)	C1
	1400 Ω	A1
6(b)	resistance of LDR decreases	B1
	current increases or p.d. across 700 Ω resistor / oscilloscope increases or p.d. across LDR decreases	M1
	trace moves up the screen	A1

18.

Question	Answer	Marks
7(a)(i)	current and <u>magnetic field</u> / <u>magnetic flux</u> (experiences a force) or the two <u>magnetic fields</u> interact	B1
7(a)(ii)	three separate approaches I / II / III	
7(a)(ii)I	B to A	B1
	left-hand rule mentioned / described	B1
	or	

7(a)(ii)II	B to A	B1
	top face of coil attracted to S-pole or top face acts as an N-pole	B1
	or	
7(a)(ii)III	B to A	B1
	more field lines under AB or fewer lines above AB	B1
7(b)	($I = $) Fx or $2 \times 9.6 \times 10^{-3} \times 0.025$ or $9.6 \times 10^{-3} \times 0.025$ or 2.4×10^{-4} (N m) or 0.024 (N cm)	C1
	4.8×10^{-4} N m or 0.048 N cm	A1
7(c)(i)	split-ring commutator or (carbon) brushes	B1
7(c)(ii)	(soft-)iron core or stronger magnet(s) or larger current or more turns or smaller resistance or greater electromotive force (e.m.f.)	B1

19.

Question	Answer	Marks
8(a)	<u>surround</u> the component with a box or use of iron mentioned	B1
	<u>surround the component with (a box of) iron</u>	B1
8(b)(i)	solenoid and d.c. power supply and switch	B1
	core	B1
	<u>iron</u> core	B1
8(b)(ii)	named or described use	B1
	what happens when electromagnet is switched on	B1
	<u>current</u> causes magnetisation or description of what happens when electromagnet is switched off	B1
8(c)(i)	ammeters deflects / shows a reading	B1
	magnetic field lines cut by solenoid or changing magnetic field in coil	B1
	e.m.f. / voltage <u>induced</u>	B1
8(c)(ii)	no deflection and no (magnetic) field lines cut or constant field	B1
8(c)(iii)	larg(er) deflection	B1
	opposite deflection	B1
	field lines cut in opposite sense or opposite change in (magnetic) field or field lines cut faster	B1

20.

Question	Answer	Marks
7(a)	current direction in copper wire correct on any part of diagram	B1
7(b)	current and <u>magnetic</u> field create a force or current (in wire) creates a <u>magnetic</u> field	B1
	left-hand rule mentioned or used or interaction of magnetic fields of magnet and current (catapult effect)	B1
7(c)	(electric) motor or any device that contains a motor	B1
	(loud)speaker or any device that contains a loudspeaker	B1

21.

Question	Answer	Marks
7(a)(i)	<u>magnetic</u> field / flux mentioned	B1
	change in field / flux (in coil) or field (lines) cut coil / wire	B1
7(a)(ii)	more turns or stronger magnet or move magnet faster	B1
7(b)(i)	 correct symbol	B1
7(b)(ii)	converts more of the <u>input energy / power</u> into light / useful energy / power or less energy / power wasted <u>with same input energy / power</u> or <u>in same time</u> or <u>same</u> input power / energy / current gives more light / less energy wasted / power wasted / heat / greater power out or <u>less</u> input power / energy gives same output power / energy	B1

22.

Question	Answer	Marks
7(a)	magnetic (material)	B1
	temporary / soft magnetic (material)	B1
7(b)(i)	it / a.c. changes direction or changes polarity / from positive to negative (continually)	B1
	it / a.c. has varying size or is sinusoidal / like a sine wave	B1
7(b)(ii)	magnetic field (in core / secondary coil) is not changing / remains constant	B1
	no (electromagnetic) <u>induction</u>	B1

23.

Question	Answer	Marks
7(a)	magnet is attracted to magnetic material or is not attracted to the non-magnetic material	B1
7(b)(i)	(direction) to the left or away <u>from region 2</u>	B1
	like poles / N-poles repel	B1
7(b)(ii)	correct direction on at least one of the eight lines and none wrong anywhere else	B1
7(b)(iii)	1 field (lines) cut the coil or field/flux (in coil) changes	B1
	2 voltage large(r) or magnetic field is strong(er) / to be in the (magnetic) field or more field (lines) cut (in a given time) / so field lines are cut or field lines close(r) together / field lines cut in small(er) time	B1

24.

Question	Answer	Marks
11(a)(i)	steel	B1 B1
11(a)(ii)	iron	
11(a)(iii)	iron B1 (i) and (iii) both correct B1 (ii) correct	
11(b)(i)	clear use (e.g. change voltage / current)	M1
	detail of operation (to transmit electricity or in a charger etc.)	A1
11(b)(ii)	1 voltage on vertical axis and time on horizontal axis	B1
	clear attempt at sinusoidal curve for at least one cycle	B1
	2 ($V_s =$) $V_p N_s / N_p$ or $220 \times 85 / 1700$	C1
	11 V	A1
11(c)(i)	1 magnetic field / flux / flux linkage mentioned	B1
	magnetic field (lines) cut solenoid or v.v. or changing (magnetic) field / flux / flux linkage (in solenoid)	B1
	induced e.m.f. / voltage	B1
	2 it / current magnetises solenoid / produces magnetic field / flux(linkage) in solenoid / <u>produces</u> a S-pole (in solenoid)	B1
	magnet repelled or experiences a force to the left	B1
11(c)(ii)	smaller reading / deflection	B1
	reading in opposite direction	B1

25.

Question	Answer	Marks
7(a)	PQ: a force towards the top of the page (second box) ticked	B1
	QR: a force towards the right of the page (bottom box) ticked	B1
7(b)	current reversed	B1
	both forces reversed	B1
7(c)	force(s) decrease or less heat generated	B1

26.

Question	Answer	Marks
7(a)(i)	current in coil (at right angles) in a magnetic field (of magnet) or left-hand rule mentioned	B1
7(a)(ii)	reverses / changes direction of current (in coil)	B1
	reverses current every half turn / when coil is vertical or reverses forces (on side AB / CD) or keeps forces in same direction for wire on one side	B1
7(b)(i)	($E =$) VIt or $2 \times 12 \times 8$ or $E = Pt$ and $P = VI$ or $E = VQ$ and $Q = It$	C1
	190 or 192 J	A1
7(b)(ii)	73% or 0.73	B1

27.

Question	Answer	Marks
7(a)(i)	arrow or other indication left to right in rod or correct in circuit labelled C	B1
7(a)(ii)	arrow from N-pole to S-pole labelled M	B1
7(a)(iii)	arrow downwards labelled F	B1
7(b)(i)	rod cuts magnetic field / flux lines or flux (in circuit) changes	B1
	induced e.m.f. / voltage / current formed	B1
7(b)(ii)	move rod faster	B1

28.

- (a) (i) X N-pole B1
 Y S-pole **and** Z N-pole B1
- (ii) they touch / move towards each other **and** opposite poles attract B1
- (b) any sensible use: starting-motor circuit; with a logic gate; B1
 nuclear power station
 corresponding explanation: current too large for dash-board switch; B1
 current too small to power device; too dangerous to reach switch

29.

- (a) current / a.c (in primary coil) creates magnetic field B1
or current / a.c magnetises iron
 changing magnetic field (in secondary) B1
- (b) it / secondary has less turns (than primary) B1
or primary has more turns (than secondary)
or (some) flux escapes
- (c) (steel is) a permanent magnet B1
or weaker fields produced
or (steel) difficult to magnetise / demagnetise
or (steel) is a hard magnetic material
- (d) passes current / charge in one direction B1
or has high resistance / is an insulator when current in
 one direction / reverse biased

30.		
(a)	(current in coil) creates magnetic field or current is at right angles to magnetic field (of permanent/cylindrical magnets)	C1
(b)	into and out of magnet or left and right or backwards and forwards	B1
	current is one way then reverses (so reverses force)	B1
(c)	$(\lambda =) v/f$ numerically or algebraic in any form 0.64 m	C1 A1
31.		
(a)	(i) diagram showing coil of wire and either - magnet or - another coil and supply (dc and switch or ac) coil of wire connected to an ammeter or voltmeter or cro or other method of detection, e.g. lamp magnet or coil moved or <u>change in current</u> mentioned if another coil used	B1 B1 B1
	(ii) ANY 2 from - move magnet (or coil) faster - larger current in primary (if transformer drawn) - more turns in coil - stronger magnet (if magnet drawn) - soft iron core	B2
	(iii) 1 direction of <u>induced</u> current/ <u>induced</u> emf opposes the change (that produces it)	B1 B1
	(iii) 2 (magnetic) flux/field/poles in coil caused by movement/(induced) current in coil statement of how opposition occurs, e.g. repulsion as magnet moves in; N pole created (by induction) at end of coil as N pole approaches	B1 B1
(b)	(i) 1 $(I =) P/V$ numerical or algebraic 15(.15) A	C1 A1
	2 $(E =) Pt$ or VIt or $500(000) \times 60 \times 60$ 1.8×10^9 J or 500 kWh	C1 A1
	(ii) low current $P = I^2R$ or $E = I^2Rt$ explained	B1 B1

32.

- (a) (i) no free electrons (in plastic) **or** all electrons are bound / structural B1
- (ii) (aluminium) is not magnet(ic) **or** cannot be magnetised B1
(iron) is a temporary / soft magnetic material **or** is not a permanent magnet B1
- (b) (i) magnetic field / flux (mentioned) B1
(magnetic) field lines cut wire / solenoid / circuit **or** changing magnetic field / flux B1
voltage/e.m.f. **induced** B1
- (ii) ($V =)IR$ **or** 0.045×1.2 **or** 0.000045×1.2 C1
 $5.4 \times 10^{-5} \text{ V}$ **or** 0.054 mV A1
($Q =)It$ **or** 0.045×0.14 **or** 0.000045×0.14 C1
 $6.3 \times 10^{-6} \text{ C}$ **or** 0.0063 mC A1
- (c) (i) larger **or** twice the current B1
(magnetic) field lines cut faster **or** (magnetic) field changes faster **or** twice the current B1
- (ii) double the current for half the time **or** larger current for less time **or** product $I \times t$ is the same B1
- (d) any **two** from:
insert S-pole (at same end)
insert (N-pole) at other end **or** from other direction
withdraw N-pole (from same end implied)
withdraw S-pole from other end **or** pass through completely B2

33.

- (a) (i) at least two straight parallel lines inside the coil B1
 at least two (complete) lines one above the coil and one below the coil B1
 third line in middle **and** evenly spaced **and** two closed loops B1
 (any crossings max. 2/3)
- (ii) current (in X) increases B1
 magnetic field becomes stronger / changes B1
 current / e.m.f. / voltage **induced** in Y / electromagnetic induction B1
 opposite deflection B1
 larger deflection B1
 magnetic field decreasing **or** quicker (rate of) change B1
- (b) (i) to increase the strength of the magnetic field B1
 to direct / concentrate the magnetic field (into the secondary coil) B1
- (ii) ($P = VI$) **or** $33\,000 \times 85$ C1
 $2.8 \times 10^6 \text{ W}$ **or** 2800 kW **or** 2.8 MW A1
- (iii) ($E = VIt$) **or** $33\,000 \times 85 \times 3600$ **or** $2.8 \times 10^6 \times 3600$ C1
 $1.0 / 1.01 / 1.008 \times 10^{10} \text{ J}$ A1

34.

- (a) (i) mention of (magnetic) field / flux (of N and S-poles) C1
 (coil/wire) cuts **magnetic** field / flux / lines A1
or magnetic flux in coil changes
- (ii) (one side of) coil cuts one way and then the other B1
or (side) moves one way and then the other / returns
or flux increases and then decreases
- (b) increase in emf for both stronger magnets and more turns B1
 no change / same frequency for both stronger magnets and more turns B1
 increase and increase for turn the coil faster B1

- 35.
- (a) (i) 1 S-pole on right of core B1
- 2 N-pole anywhere on vertical section of armature B1
and S-pole anywhere on horizontal section of armature
or
 N-pole on left of vertical section of armature **and** S-pole on right
- (ii) poles (on core) reverse / change positions B1
 (armature still) attracted (to core) B1
- (iii) (iron is a) temporary magnet B1
or (iron) easily demagnetised
or steel retains magnetism
 when current off / no battery / switch off / circuit open B1
and
 armature released / does not stay attracted / opens connections (at AB)
- (b) (i) thermistor B1
- (ii) resistance (of X) decreases B1
 current (in coil) increases **or** more voltage across coil B1
and
either relay switch closes **or** circuit (to bell) complete
- (iii) 1 $(V=) IR \text{ or } 1.5 (\times 10^{-3}) \times 2000$ C1
 $3(.0)V$ A1
- 2 $9(.0)V$ B1
- 3 $12/200 \text{ or } 0.06 (A) \text{ or } 60 (mA) \text{ seen}$ C1
or $(R_T =) 195(.12 \Omega)$
 $61(.5)mA \text{ or } 0.061(5)A \text{ or } 62 mA \text{ or } 0.062 A$ A1
- (iv) light dependent resistor **or** LDR B1
- 36.
- (a) field lines **of magnet** mentioned or magnetic flux mentioned B1
 field lines cut the coil or flux changes B1
- (b) reversed movement of magnet causes one of B1
- reversal of (induced) emf
 - reversal of (induced) current
 - field lines cut / flux change in reverse direction
- LED emits light when **current** passes in one direction B1
- (c) more current or more induced emf B1
and flux lines cut faster or faster change in flux

- 37.
- (a) one label correct and not contradicted C1
C, 1S and 1B all correct and clear and none contradicted A1
- (b) any **three** from:
magnetic field (between poles)
(coil/wire) cuts field/flux **or** field/flux cuts (coil/wire) **or** field/flux changes
(electromagnetic) induction
brushes rub against/in contact with rings B3
- (c) (half) distance across screen **or** count divisions of/measure wavelength **or** the
wavelength corresponds to one rotation B1
half distance multiplied by time base setting B1
- 38.
- (a) wire cuts field lines B1
current/e.m.f./voltage **induced** B1
- (b) larger deflection **and** to the left/opposite direction B1
- (c) no deflection/current B1
- 39.
- (a) arrows on long sides in opposite vertical directions [B1]
downwards on right **and** upwards on left **or** correct rotation shown [B1]
- (b) no (horizontal) distance between forces **or** forces through axle/pivot/axis [B1]
- (c) two halves of split ring clear and clearly connected to each end of coil [B1]
contacts/brushes labelled or described **and** connected to battery [B1]
each side of split ring touches other terminal/brush **or** current reverses in coil
or changes terminals of connection to battery [B1]
forces reverse on **sides** of coil **or** forces always in same direction on **side**
nearest a pole [B1]

- 40.
- (a) oval/circle through or near A centered on or near X
clockwise arrow on line(s) around X **and** none wrong B1
B1
- (b) fields (due to X and Y) cancel **or** X and Y fields equal and opposite B1
- (c) (i) to the left
or towards X/A/B B1
- (ii) current (in wire Y) and (magnetic) field (caused by other wire)
or two (magnetic) fields interact B1
- 41.
- (a) steel / magnadur / alnico / magnetite B1
- (b) (i) mention of cutting (lines of) magnetic field / change in (magnetic) flux M1
great(est) rate of change A1
or fast(est) cutting
or other explanation involving time
- (ii) vertical/upright B1
or turned through 90°
or normal to (magnetic) field
- 42.
- (a) (soft) iron/mu-metal B1
- (b) magnetic field **or** flux **or** flux/magnetic lines mentioned B1
changing magnetic field **or** changing flux **or** flux lines cut coil B1
induced voltage/current/e.m.f. B1
- (c) less power/energy/heat loss (**allow** no power loss/to prevent power loss) **or**
more efficient **or** thinner wire can be used (**ign.** cheaper) B1

- 43.
- (a) (i) $4.5 + 0.3$ **or** 4.8 C1
 $(I =)V/R$ **or** $12/4.8$ **or** $12/4.5$ **or** $12/0.3$ **or** $12/0.28125$ C1
2.5 A A1
- (ii) decrease resistance (of variable resistor) B1
increase current (in solenoid) B1
- (iii) 1. force on PQ/wire **or** PQ/wire moves M1
force/movement out of page/outwards/towards observer
(not upwards) A1
2. force/speed/acceleration larger B1
- (b) (i) $(P =)VI$ **or** 75×12 C1
900 W A1
- (ii) (thick wires) have low resistance B1
(thick wires) not as hot/do not melt B1
- (c) current to relay/coil/solenoid/electromagnet B1
core/relay/coil/solenoid/electromagnet magnetised B1
connections made (in motor circuit) B1
- 44.
- (a) (at least) **two** parallel horizontal lines within the cylinder B1
(at least) **two** correctly shaped lines outside the cylinder B1
- (b) (i)  (right to left) **and** on diagram (somewhere) B1
- (ii) 1. path continuously curving in same direction M1
upwards (**ignore** lines outside the shaded area) A1
2. (changes to) downwards (curve) **not** reverses/opposite direction B1
- 45.
- (a) (i) conventional current direction correct in coil/one lead B1
- (ii) at least 1 line axially through coil A
OR line above and below end of coil A B1
- at least two curved lines in ring from ends of A
to ends of B (and inside A and B) B1
- correct direction on at least one line/arrow for candidate's (i) B1

(b) (i)	(magnetic) flux/field cuts (coil B) OR field/flux changes (in coil B)	B1
	induces an e.m.f./voltage/current (in B)	B1
(ii)	(voltmeter) deflects to left/opposite (and returns to zero)	B1
	flux/field decreases/collapses/reduces OR iron loses magnetism OR change in field is in opposite direction OR to oppose flux/field change	B1
(iii)	ANY 2 lines more turns on coil B larger voltage/current (e.g. of battery)/more cells battery smaller (internal) resistance smaller resistance of wires; thicker wires; shorter wires thicker or shorter iron ring; use soft iron; coil A and B closer (on ring); more sensitive voltmeter; laminate the iron ring	B2
(c) (i)	$(P =) VI$ algebraic or numerical 384 W OR 380 W	C1 A1
(ii)	$(P =) I^2 R$ OR $(P =) V^2/R$ OR VI and V/R seen algebraic or numerical OR clear voltage of 4(.0 V) or 8(.0 V) seen	C1
	$1.6^2 \times 2.5$ OR $1.6^2 \times 5$ OR (power) 6.4 (W) seen	C1
	12.8 W OR 13 W	A1

46.		
(a)	(i) iron; soft iron; mu-metal	B1
	(ii) rod becomes (an induced) magnet opposite poles attract; N attracts S OR magnetic pole(s) on rod/at P reverses (due to induced magnetism)	B1 B1
(b)	(i) at least two circles centred on wire (no crossings) clockwise arrow on at least one circle and no arrows wrong	B1 B1
	(ii) lines closer together	B1
47.		
(a)	(i) N at top end of bar and S at bottom end	B1
	(ii) attracted to/moves towards iron core unlike poles attract	B1 B1
(b)	they disappear/bar is demagnetised/loses its poles/is weaker	B1
48.		
(a)	(i) recognisable sine/cosine curve (≥ 2.0 cycles)	B1
	(ii) larger (peak)(voltage) higher frequency/shorter period/described in words (allow shorter wavelength)	B1 B1
(b)	$(R =) V/I$ or $12/0.50$ $24\ \Omega$	C1 A1

49.

(a) upward arrow (**not** curved) on iron bar B1

(b) attraction/force not enough//weight of bar too high//friction at pivot/with copper bar B1

(c) **three** of:

electromagnet works // magnetic field created
iron bar moves/lifted up
spring pulls copper bar across/contracts
contacts break circuit // contacts open

B3

(d) more turns in coil // more iron in electromagnet // electromagnet nearer iron bar //
iron bar less weight // weaker spring
correct explanation which involves force on iron bar

M1

A1

50.

(a) all three correct: force, field, current B1

(b) (i) Fd // 20×4 ; 20×2 ; 20×0.04 ; 20×0.02 ; $20 \times 4 \times 2$; 160
80 N cm, 0.8 N m

C1

A1

(ii) more turns (on coil)
use soft iron
more current
increase AB or CD
increase BC or AD ANY 2

B1

51.

- (a) (i) d.c. current flows in one direction **or**
a.c. current flows in one direction then the other B1
- (ii) mention of magnetic field/flux B1
(magnetic) field lines // flux cuts coil // flux changes in coil B1
induction of voltage/current B1
something **relevant** reverses (e.g. field/flux cuts in one direction then the other // N pole approaches then leaves // N pole approaches and S pole approaches)
and link to a.c. B1
- (iii) **two** of: thicker wires; more turns of coil; stronger magnet; faster rotation; lower resistance (of lamp) B2
- (b) (i) to reduce heat/energy/power loss (on the power lines) B1
(higher voltage means) lower current B1
- (ii) 25:400 // 1:16 // 0.0625 B1
- (iii) reduces resistance // less power loss // costs less to run // more current // more power B1
increases weight // more support needed // more wind resistance // more ice forms // costs more to install B1
- (iv) 13 A B1
($I = P/V$) // 4.2 A // 4.17 etc A B1
must choose higher value to avoid fuse blowing // other fuses melt B1

52.

- (a) remain stationary/no effect/unaffected B1
- (b) lifted up/attracted/stick to rod (stated not implied) B1
fall down/return to dish B1
- (c) lifted up/attracted/stick to rod (stated not implied) B1
stay up/remain attracted B1

- 53.
- (a) (i) current in magnetic field **or** motor effect/LH rule **or** coil is magnet (produces) force **or** current **direction** changes **or** coil moves **or** repulsion **and** attraction
force changes direction/backwards and forwards
- (ii) air (atoms/molecules/particles) (next to cone) vibrates
compressions **and** rarefactions **or** high **and** low pressure
or vibrations passed on **or** longitudinal
- (b) the note is louder/has greater intensity (**not** changed frequency)
- 54.
- (a) both arrows point inwards
both arrows extended must pass through base of bar magnet
N marked on both needles nearest S pole
- (b) (current causes) coil/iron to become magnet/an electromagnet/creates magnetic field
iron (in coil) attracts/pulls (pivoted) iron/armature
(and L-shaped iron rotates/moves/turns) not contacts attract each other
- (c) (i) resistance decreases
- (ii) series circuit/loop with C and lamp
completely correct circuit with a battery
- 55.
- (a) meter deflects (one way)
changing magnetic field/flux in ring/coil or cutting of flux/field
induces voltage/current
- (b) ammeter returns to/remains at zero
- (c) ammeter deflects in opposite direction (then returns to zero)
field decreases/change in opposite direction **or** field/flux cuts in opposite direction

56.		
(a)	NS marked on each piece correctly	B1
(b)	NS/unlike/opposite poles attract	B1
	switch closes or soft-iron/contacts touch	B1
(c)	(i) resistance decreases	B1
	(ii) current increases clearly in coil/through thermistor	B1
	magnetic field (in coil) (and contacts close)	B1
57.		
(a)	(i) from N to S or towards right	B1
	(ii) downwards	B1
(b)	(i) rough circle around each wire (–1 any crossing lines)	B1
	correct shape around both wires or large circle around both wires	B1
	direction of field correct on any one correct line and no direction wrong	B1
	(ii) attractive force drawn on/near each wire	B1
58.		
(a)	(i) steel	B1
	(ii) rod inside (coil) with current on (at some stage)	B1
(b)	(i) (soft) iron accept Mumetal or any other soft magnetic material	B1
	(ii) all lines directly join from left to right and top line goes down and bottom line up no lines inside box and no lines cross/touch	M1 A1
59.		
(a)	at least 3 approx straight lines in middle of coil	B1
	one loop around top or bottom of coil	B1
	correct direction shown on at least one line or loop	B1
(b)	(i) 100 oscillations/cycles/waves/changes in direction (and back again) in 1 sec or time for one oscillation etc. 0.01s	B1
	(ii) coil/cone moves left to right or back and forth or oscillates/vibrates or in and out	B1
	(current in) coil produces magnetic field/pole(s)	
	or Left Hand Rule/current in magnetic field (gives force)	B1
	fields/poles (of coil) oscillate/reverse	B1
	force (exerted by magnet) oscillates/alternates (accept attracts/repels)	B1 ANY 2 B2

60.			
(a)	electromagnetic/em induction or induced current/e.m.f. (NOT magnetic/electric induction)		B1
(b)	deflects to left/opposite deflection		B1
(c)	nothing or no deflection/current/e.m.f. or needle stationary no lines of flux are cut or no change in magnetic field		B1 B1
61.			
(a)	any mention of magnetic field coil cuts lines of (magnetic) flux/field lines		C1 A1
(b)	pass current / connect coil to output / prevent wires tangling		B1
(c)	(induced) voltage or current opposes the change (producing it)		B1
(d)	large(r) coil, strong(er) magnet, iron inside coil, more turns (on coil)		B1
62.			
(a)	Y input and ground connected across resistor		B1
(b)	3 squares or 3 x 2 6V		C1 A1
(c)	line drawn at 1.5 squares		B1
63.			
(a)	rods magnetised with like poles next to each other e.g. both rods N at one end like poles repel		B1 B1
(b)	nothing happens copper is not magnetic	accept eddy currents (induced) changing flux induces current/voltage	B1 B1
64.			
(a)	arrows in A and C to right arrow in B to left or right if both A and C to left		B1 B1
(b) (i)	SNSN or NSNS		B1
(ii)	they/iron pieces attract/move together attraction of opposite poles/unlike poles/S and N	e.c.f. (i) throughout	B1 B1
(c) (i)	opposite direction/reverses/poles change		B1
(ii)	weaker (field) or (iron) demagnetises		B1

65.

(a) to step down/reduce the **voltage**

(**ignore** just step down transformer)

B1

(b) two coils (no label needed)

(**not** if primary connects secondary)

B1

output/secondary has fewer turns than input/primary clear; coils labelled

or right-hand coil has fewer turns

B1

complete (soft) iron (core) **labelled**

(**ignore** circuit symbol)

B1

(c) (i) less energy/power/heat loss/heating

(**ignore** just more efficient)

B1

current is reduced/low

(**not** if resistance changes)

B1

(ii) resistance is decrease

(resitivity is not resistance)

B1

electrical power/energy related to resistance

e.g. $P = I^2R$, P prop to R (**not** V^2/R alone)

or resistance $\propto 1/\text{area}$

(**accept** power related to R etc. given in (i))

B1

66.

(a) (i) magnetic (field) of current / coil / recording head

or head is magnetized / an electromagnet

B1

(ii) magnetism / magnetic field or current or poles on head reverses / changes direction

(**accept** "due to alternating current")

B1

(iii) each direction / one cycle longer (on tape)

B1

(b) (i) need to keep record / tape stored or played

B1

(ii) iron, steel etc

B1